Ergodic Ground-Motion Model for Response Spectra for Crustal Earthquakes in the Western United States Based on the NGA-W3 Data

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ABSTRACT

This report presents a ground-motion model (GMM) for 5%-damped pseudo-spectral acceleration (PSA) for the Western United States (WUS). The median GMM is based on converting the median Effective Amplitude Spectrum (EAS) developed by Abrahamson et al. (2025) using the NGA-W3 flatfile (November 2024 version) into a data set of PSA values. The conversion of the EAS to PSA uses a model with input parameters used in random vibration theory (RVT) but with empirically calibrated coefficients to reduce potential biases due to inconsistencies between earthquake ground-motion characteristics and statistical assumptions made in RVT-based models. The conversion covers a broad range of seismic scenarios, including scenarios that are outside the range that is well-constrained empirical data. The generated PSA dataset is uniformly sampled in terms of magnitudes, distances, and site conditions. In this approach, the extrapolation of the median GMM to large magnitudes and short distance is based on the scaling in the EAS GMM. The dataset of generated median PSA values is used to develop the base GMM for the median PSA that include the scaling with magnitude, rupture distance, depth to top of rupture (Z_{TOR}), style of faulting, and the time-averaged shear-wave velocity in the top 30 m (V_{S30}). The base model includes the V_{S30} scaling for the linear site effects. The nonlinear site effects are added to the GMM using reaults from analytical modeling of the non-linear site effects from Ilhan et al. (2025). Additional features for the hanging-wall effects and the short-period spectral shape effects (kappa proxy) are estimated from the empirical PSA data. The standard deviations terms are estimated from the empirical PSA data rather than from the variability of the converted EAS GMM. The standard deviations terms include three random effects and a within-site term. In addition to the standard random effect terms for the event (δB) and site ($\delta S2S$), the GMM includes an additional random effect for path term (δP). The path term is a new term that accounts for event-specific changes in the large-distance slope (linear R term). The PSA GMM is a regionalized model for the WUS and is ergodic within the WUS.

An evaluation of the new GMM should be completed before it is used in seismic hazard applications. The evaluation should include comparisons with GMMs from the other NGA-W3 developers once they become available as well as trial applications in probabilistic seismic hazard analyses.

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CONTENTS

ABSTRACT ii
ACKNOWLEDGMENTS iii
TABLE OF CONTENTS v
LIST OF TABLES vii
LIST OF FIGURES ix
1 INTRODUCTION 1
2 EMPIRICAL PSA DATASET 3
3 CONVERTING EAS TO PSA 11
3.1 Period-Dependent Duration Model 12
3.2 RVT-type Conversion From EAS to PSA 13
3.3 Converted PSA Dataset 14
4 FUNCTIONAL FORM FOR THE MEDIAN 29
5 REGRESSION 35
5.1 Step 1: Finite-Fault Term 35
5.2 Step 2: Base-Model Regression 36
5.3 Step 3: Additional Terms 37
5.3.1 Hanging-Wall Scaling 37
5.3.2 Tamp1.5 Scaling 37
6 RESIDUALS 51
6.1 Correction of Estimates of Random Effects 51
6.2 Random Effects 51
6.3 Within-site residuals 52
7 STANDARD DEVIATION MODEL 67
7.1 Standard Deviation Terms for Linear Site Effects 67
7.1.1 τ_E Model for Linear Site Response 67
7.1.2 τ_P Model for Linear Site Response 68
7.1.3 ϕ_{S2S} Model for Linear Site Response 68

7.1.4 ϕ_{SS} Model for Linear Site Response 69
7.1.5 Total Sigma 69
7.2 Non-linear Site Effects on Aleatory Variability 69
7.2.1 Sigma Models with Non-linear Site Response 71
7.2.2 $\phi_{amp,NL}$ Model for Non-linear Effects on ϕ_{amp} 72
8 MODEL COMPARISONS 87
8.1 Median 87
8.2 Standard Deviation 87
9 MODEL APPLICATION GUIDELINES 99
9.1 Missing Parameters 99
9.2 Use of the PSA GMM 99
REFERENCES 101

LIST OF TABLES

2.1 R_{max} model 4

2.2 Summary of Selected Subset 4

3.1 The coefficients of the 5-75% period-dependent model. 16

3.2 The coefficients of the empirical RVT-type conversion of EAS to PSA. 17

3.3 Scenarios used for generating the PSA dataset from the EAS GMM. 17

5.1 Summary of coefficients determined by global versus regional datasets 38

5.2 Coefficients for the median PSA GMM. 39

5.3 Coefficients for the median PSA GMM (cont). 40

5.4 Number of events and records by region that contribute to the HW scaling: M $>$ 6, $R_{RUP} <$ 35 km, and a net HW taper $\geq$ 0.1. 41

7.1 Coefficients for the standard deviation model based on the GMM without $\mathbf{T_{Amp1.5}}$ 74

7.2 Coefficients for the standard deviation model based on the GMM with $\mathbf{T_{Amp1.5}}$. . . 75

LIST OF FIGURES

2.1 Epicenters of the selected events by region: WUS, Japan, Europe and Middle East, and Taiwan. 5

2.2 Magnitude-distance distribution by region. (a) Western US. (b) Europe and the Middle East. (c) Japan. (d) Taiwan. 6

2.3 Magnitude-Z_{TOR} distribution by region. (a) Western US. (b) Europe and the Middle East. (c) Japan. (d) Taiwan. 7

2.4 Number of recordings for magnitude bins, R_{RUP} bins, and V_{S30} bins. (a) WUS (b) EU & ME (c) Japan and (d) Taiwan 8

2.5 (a) Number of recordings, (b) number of earthquakes, and (c) number of stations as a function of frequency for each region 9

3.1 The coefficients of the D_{5-75} period-dependent duration model. 18

3.2 Residuals of the D_{5-75} period-dependent duration model for M $\geq$ 5.5 at (a) 0.5 sec, (b) 1 sec, (c) 3 sec, and (d) 5 sec. 19

3.3 The scaling for the D_{5-75} period-dependent duration (PDD) model and for the duration normalized by the acceleration duration. (a) Distance scaling for the PDD model. (b) Distance scaling for the normalized duration. (c) V_{S30} scaling for the PDD model. (d) V_{S30} scaling for the normalized duration. (e) Magnitude scaling for the PDD model. (f) Magnitude scaling for the normalized duration. 20

3.4 The coefficients of the RVT-type model for converting EAS to PSA. 21

3.5 Distribution of the magnitude, distance V_{S30}, and Z_{TOR} for the PSA converted from the EAS GMM. 22

3.6 Distance scaling of the median PSA for T = 0.2 s and 1 s. 23

3.7 Median PSA spectrum for the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events and without the non-linear site effect, for R_{RUP}=15 km. . . . 24

3.8 Median PSA spectrum for the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events, without the non-linear site effect, for R_{RUP}= 80 km. 25

3.9 Distance scaling of the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events, without the non-linear site effect. 26

3.10 Magnitude scaling of the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events, without the non-linear site effect. 27

5.1 Examples for FF_e values based on the PSA from the converted EAS for Vs30 = 760 m/s. (a) M4 and (b) M7. 42

5.2 Model for the finite-fault term, FF(M). 43

5.3 Coefficients for the geometrical spreading (c_4) and the finite-fault term ($FF(M)$). 44

5.4 Coefficients for the magnitude scaling: c_2 and c_3 45

5.5 Coefficients for the large-distance scaling and the Z_{TOR} scaling : c_7 and c_9 46

5.6 Coefficients for the V_{S30} scaling : c_{8R} and c_{8M} 47

5.7 Coefficients for the SOF scaling and the HW scaling : c_{10} and c_{13} 48

5.8 Coefficients for the $T_{Amp1.5}$ scaling: c_{12} 49

6.1 τ, τ_P, ϕ_{S2S}, and ϕ_{SS} as a function of frequency. Pink lines represent the standard deviation from the regression. Green lines and dashed lines are computed from the random effects without and with correction. 53

6.2 Comparison of the raw residuals and corrected residuals for δB, δP, $\delta S2S$, and δWS at 0.2 sec. Upper plots are the raw residuals, middle plots are the corrected residuals, and bottom plots are the difference (corrected residuals - raw residuals) versus the number of records per event/station. 54

6.3 Comparison of the raw residuals and corrected residuals for δB, δP, $\delta S2S$, and δWS at 1 sec. Upper plots are the raw residuals, middle plots are the corrected residuals, and bottom plots are the difference (corrected residuals - raw residuals) versus the number of records per event/station. 55

6.4 (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=0.05 sec. 56

6.5 (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=0.2 sec. 57

6.6 (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=1 sec. 58

6.7 (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=3 sec. 59

6.8 δWS versus R_{RUP} in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 0.05 sec. 60

6.9 δWS versus R_{RUP} in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 0.2 sec. 61

6.10 δWS versus R_{RUP} in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 1 sec. 62

6.11 δWS versus R_{RUP} in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 3 sec. 63

6.12 δWS from the GMM with and without fNL for 0.03 sec. 64

6.13 δWS from the GMM with and without fNL for 0.3 sec. 65

7.1 Period dependence of the correlation coefficient for the δB_e and δP_e. The period-independent average is used. 76

7.2 Magnitude dependence of the τ_E for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec. 77

7.3 Coefficients of τ_E model for the GMM without and with $T_{Amp1.5}$. 78

7.4 Magnitude dependence of the τ_P for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec. 79

7.5 Coefficients of τ_P model for the GMM without and with $T_{Amp1.5}$. 80

7.6 V_{S30} dependence of the ϕ_{S2S} for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec. 81

7.7 Coefficients of ϕ_{S2S} model for the GMM without and with $T_{Amp1.5}$. 82

7.8 Magnitude dependence of the ϕ_{SS} for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec. 83

7.9 Coefficients of ϕ_{SS} model for the GMM without and with Tamp1.5. 84

7.10 Comparison of the total sigma for soil sites (V_{S30}=250 m/s and 400 m/s) for M7 and R_{RUP}=10 km. 85

8.1 Distance scaling of the median PSA of the PSA GMM with V_{S30} = 400 m/s. 89

8.2 Distance scaling of the median PSA of the PSA GMM with V_{S30} = 760 m/s. 90

8.3 Magnitude scaling of the median PSA of the PSA GMM with V_{S30} = 400 m/s. . . . 91

8.4 Magnitude scaling of the median PSA of the PSA GMM with V_{S30} = 760 m/s. . . . 92

8.5 Median PSA spectrum of the NGA-W3 PSA model and ASK14 for a strike-slip event and with the non-linear site effect for R_{RUP}=15 km. 93

8.6 Median PSA spectrum of the NGA-W3 PSA model and ASK14 for a strike-slip event and with the non-linear site effect for R_{RUP}=80 km. 94

8.7 The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with $T_{Amp1.5}$) are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M7.5, V_{S30}=400 m/s and R_{RUP} = 20 km. 95

8.8 The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with Tamp1.5) are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M7.5, V_{S30}=400 m/s and R_{RUP} = 100 km. 96

8.9 The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with $T_{Amp1.5}$) are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M4.5, V_{S30}=400 m/s and R_{RUP} = 20 km. 97

8.10 The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with $T_{Amp1.5}$ are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M4.5, V_{S30}=400 m/s and R_{RUP} = 100 km. 98

1 INTRODUCTION

In this report, we develop an ergodic ground-motion model (GMM) for pseudo-spectral acceleration (PSA) for the Western United States (WUS) by converting the Fourier Amplitude Spectrum (FAS) GMM developed by Abrahamson et al. (2025) to a PSA GMM. The effective amplitude spectrum (EAS) parameter is used for the FAS metric. The EAS uses a standard method for smoothing the FAS over frequency and for combining the two horizontal components. The EAS is converted to PSA using an empirical model with input parameters from random vibration theory (RVT). The approach differs from traditional RVT by using empirically calibrated coefficients in the model, which reduces potential bias in the conversion.

The EAS values are converted to PSA values for a large set of scenarios (magnitudes, distances, and site conditions), including scenarios that fall outside the well-constrained range of empirical data. Because we are converting the EAS GMM to PSA values, the extrapolation to large magnitudes and short distances is based on the scaling and assumptions in the developed EAS GMM (Abrahamson et al., 2025) . In addition, the PSA values are generated for regularly sampled magnitudes, distances, and site conditions, avoiding the uneven sampling of magnitudes, distances, and site conditions that exist in the empirical ground-motion data sets.

Using the generated PSA dataset, a regression is performed to develop a parametric base model PSA GMM for the median PSA, which includes the magnitude, distance, depth, style of faulting, and site condition scaling for linear site response. Because we generate a data set covering the full range of the input parameters, the regression for PSA is much simpler than the regression for the EAS. Using the base PSA GMM, we then incorporate the nonlinear site effects, as determined by the numerical modeling of site response results provided by Ilhan et al. (2025), and compute the PSA residuals from our selected subset of the NGA-W3 data. The scaling for other features of the PSA GMM are estimated from the PSA residuals. The additional features include the hanging-wall scaling and $T_{Amp1.5}$ (kappa proxy) scaling.

The PSA residuals are recomputed from the median PSA GMM, and the standard deviation terms are estimated from the PSA residuals, not from converting the variability of the EAS GMM to PSA variability.

An evaluation of the new GMM should be completed before it is used in seismic hazard applications. The evaluation should include comparisons with GMMs from the other NGA-W3 developers once they become available as well as trial applications in probabilistic seismic hazard analyses.

2 EMPIRICAL PSA DATASET

The dataset used for the PSA is the same as the EAS dataset used in Abrahamson et al. (2025), which used the November 2024 version of the NGA-W3 flatfile. The full data set consists of 67,556 recordings from 1,666 earthquakes in four regions: Western United States, Europe and the Middle East, Japan, and Taiwan. Maps of the locations of the epicenters are shown in Figure 2.1.

With the inclusion of ground-motion data from seismic stations, there is a large increase in the number of recordings at large distances, which increases the importance of addressing possible censoring of the small-amplitude ground motions in the dataset. As part of the selection of the EAS data set, Abrahamson et al. (2025) applied a magnitude-dependent maximum distance, R_{max}, to reduce the effect of censoring of small-amplitude ground motions. The R_{max} values depend on the region and the year of the earthquake, as shown in Table 2.1.

The magnitude-distance distributions of the selected datasets for the four regions are shown in Figure 2.2. The magnitude-Z_{TOR} distribution is shown in Figure 2.3. Histograms of the number of recordings by magnitude, rupture distance, and V_{S30} for each region are shown in Figure 2.4. The sampling of magnitudes, distances, and V_{S30} by region data is summarized in Table 2.2.

For each recording, we only use the PSA values at spectral periods that are within the usable period range defined by the inverse of 1.25 times the high-pass corner frequency used to filter the recording. As a result, the number of recordings in our selected dataset varies by period. Plots of the number of recordings, earthquakes, and stations for each region are shown as a function of period in Figure 2.5. There is a large decrease in the number of recordings and earthquakes with usable PSA values for the WUS region for frequencies less than 0.3 Hz.

The PSA GMM that we develop is intended for application to the WUS; however, global data is needed to constrain the scaling for the hanging-wall term. The WUS subset is used to estimate the residuals that we use to develop the standard deviation models.

Table 2.1: R_{max} model

Region	coeff	1937-2000	2001-2005	2006-2024
WUS & Taiwan	r_1	150	150	150
	r_2	120	120	120
	m_1	4.5	4.5	4.5
Europe & ME	r_1	20	20	150
	r_2	25	20	120
	m_1	5.0	5.0	4.5
Japan	r_1	150	20	20
	r_2	120	150	150
	m_1	4.5	5.0	5.0

Table 2.2: Summary of Selected Subset

Region	Subregion	Number Records	Number eqk	Magnitude Range	R_{RUP} (km) Range	V_{S30} (m/s) Range
WUS	California	42,555	665	3.0 - 7.36	0.5 - 457	106 - 2565
	Mexico	1,241	36	3.8 - 7.2	0.1 - 472	163 - 1464
	Arizona	9	4	4.0 - 4.7	60 - 171	246 - 822
	Colorado	20	7	4.0 - 5.4	18 - 212	280 - 822
	Idaho	482	61	3.7 - 6.9	9 - 389	207 - 900
	Montana	136	17	3.9 - 5.9	12 - 305	230 - 854
	Nevada	2,538	84	3.6 - 6.5	0.3 - 389	106 - 1464
	Utah	236	15	4.0 - 5.6	8 - 287	166 - 734
	Wyoming	109	8	4.0 - 5.0	13 - 205	279 - 854
	New Mexico	1	1	4.25	145	249
Europe and MidEast	Italy	4,195	79	4.0 - 6.9	0.1 - 200	90-2023
	Greece	1,639	110	4.0 - 6.9	2 - 200	91-1183
	Montenegro	11	1	7.1	4 - 118	310-585
	Macedonia	27	3	4.5 - 5.3	83 - 88	210-506
	Albania	6	1	6.5	77 - 200	383-565
	Georgia	5	1	6.2	32 - 64	393-477
	Turkey	660	15	5.0 - 7.8	0.2 - 200	131-1380
	Iran	1,177	186	4.0 - 7.4	2 - 200	215-767
Taiwan	-	5,518	21	5.8 - 7.6	0.1 - 281	121-1538
Japan	-	6,991	351	4.3 - 7.1	0.3 - 362	89-2100
	Total	67,556	1,666	3.0 - 7.8	0.1 - 472	89-2565

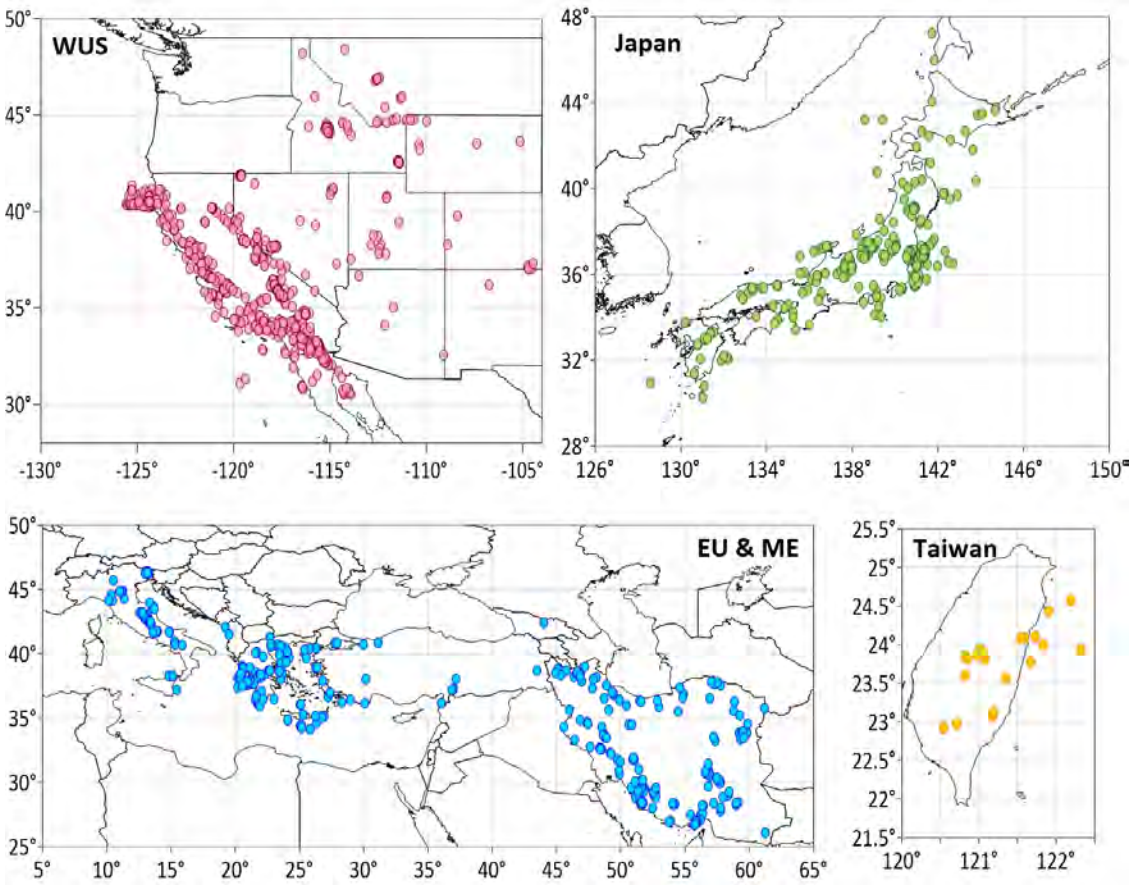


Figure 2.1: Epicenters of the selected events by region: WUS, Japan, Europe and Middle East, and Taiwan.

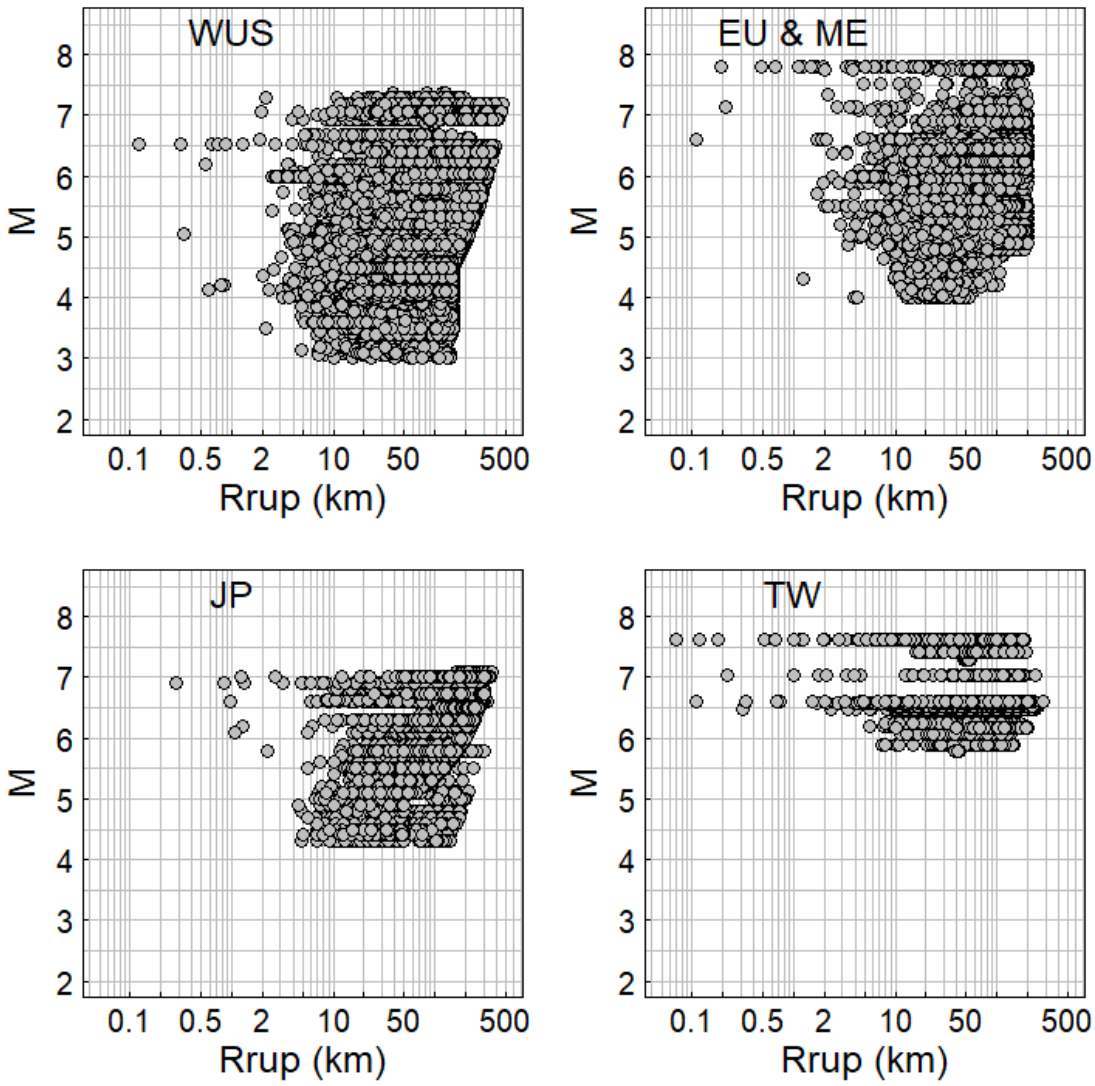


Figure 2.2: Magnitude-distance distribution by region. (a) Western US. (b) Europe and the Middle East. (c) Japan. (d) Taiwan.

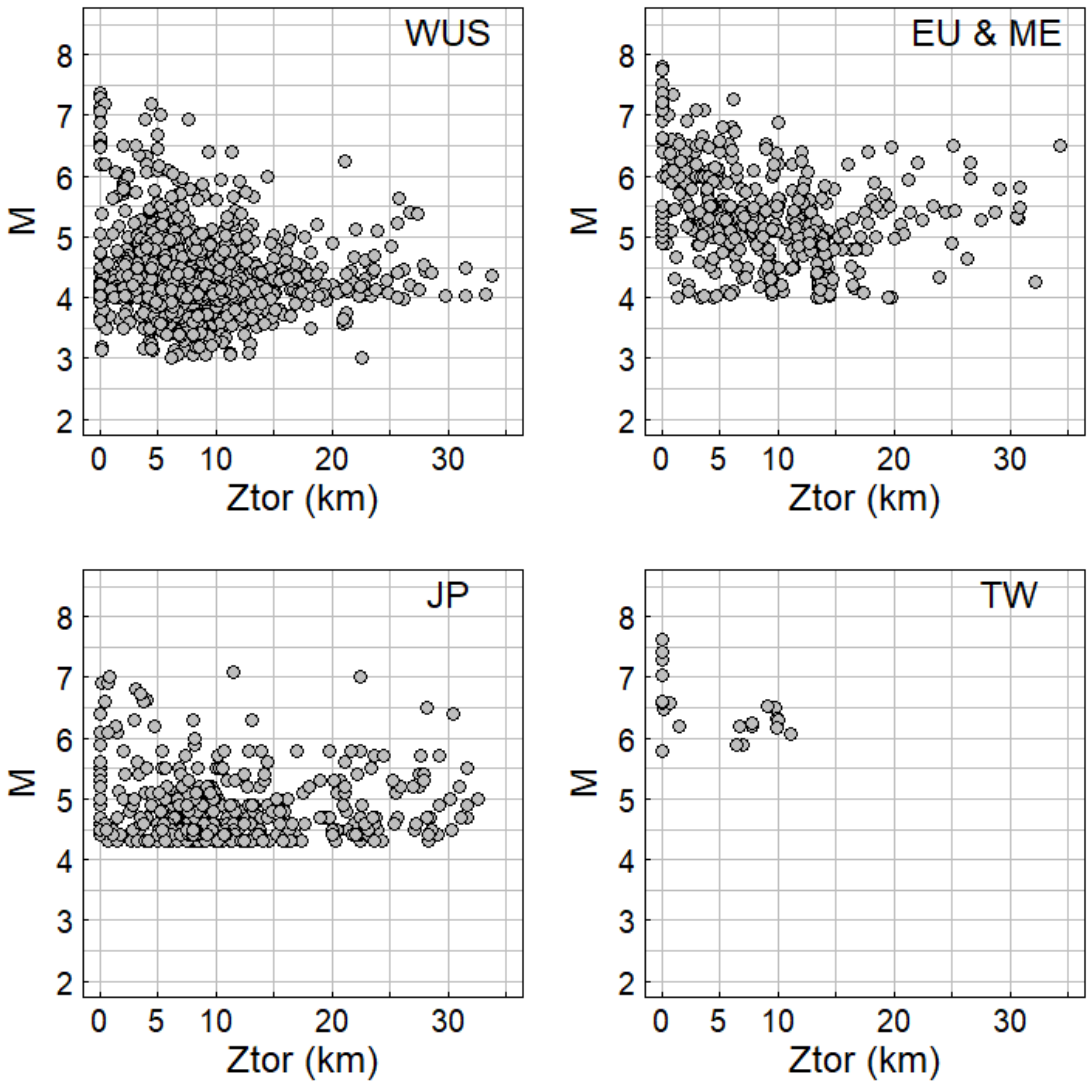


Figure 2.3: Magnitude-Z_{TOR} distribution by region. (a) Western US. (b) Europe and the Middle East. (c) Japan. (d) Taiwan.

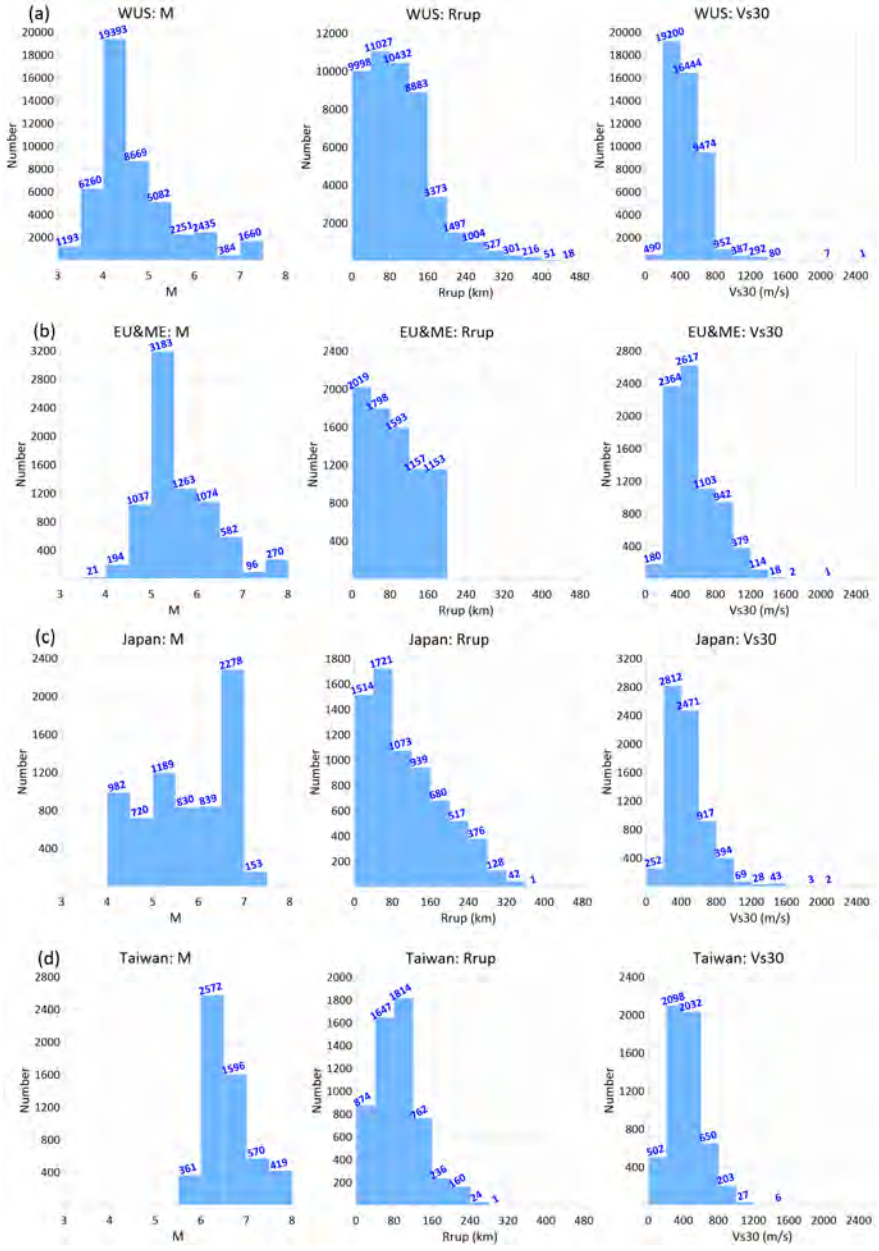


Figure 2.4: Number of recordings for magnitude bins, R_{RUP} bins, and V_{S30} bins. (a) WUS (b) EU & ME (c) Japan and (d) Taiwan

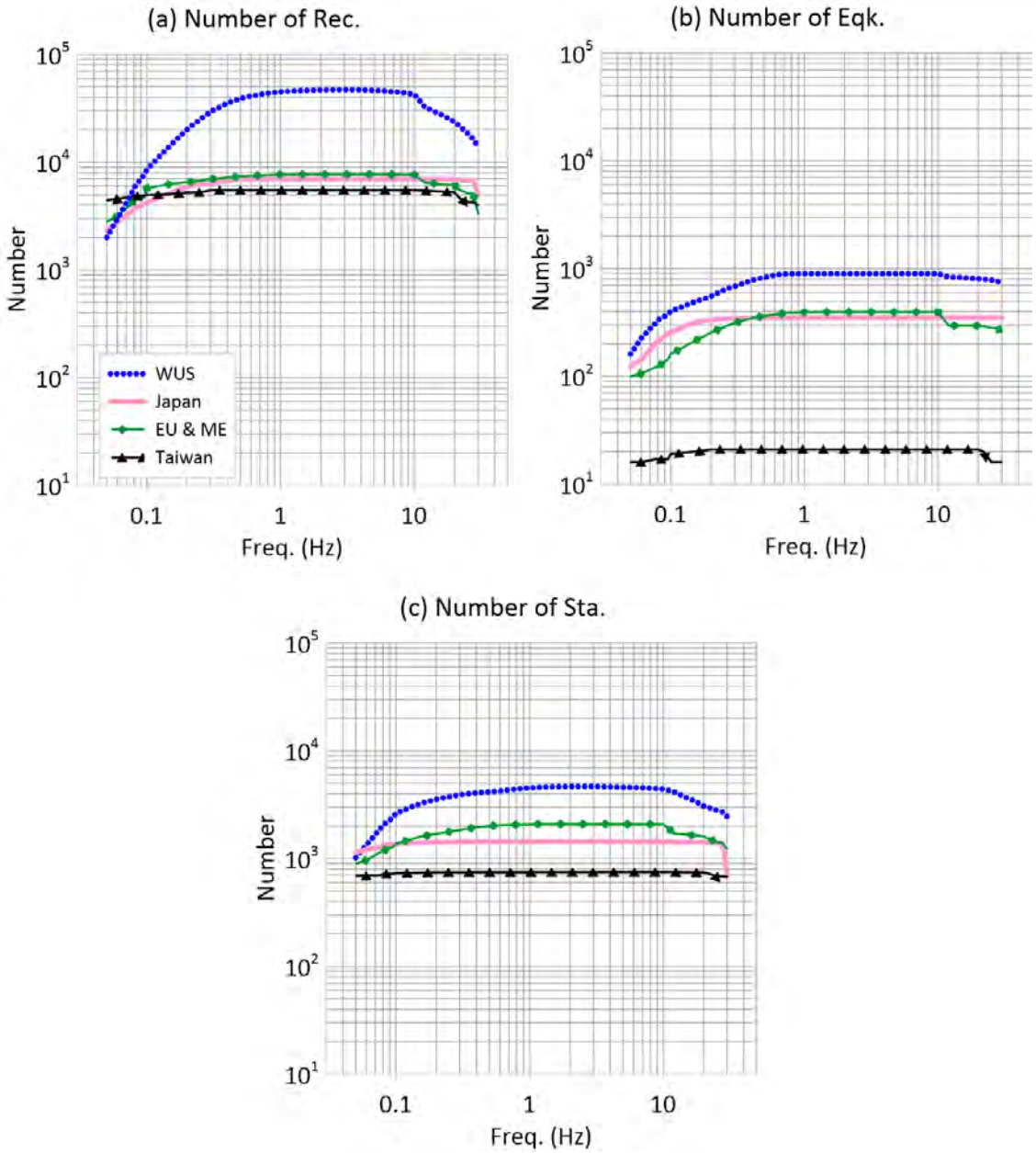


Figure 2.5: (a) Number of recordings, (b) number of earthquakes, and (c) number of stations as a function of frequency for each region

3 CONVERTING EAS TO PSA

Traditionally, Random Vibration Theory (RVT) has been used to convert the FAS of a time series to a maximum value in the time domain using Parseval's theorem and extreme value statistics. For computing response spectral values, the ground motion is convolved with the impulse response of a single-degree-of-freedom (SDOF) oscillator with damping ratio, ζ, and oscillator frequency, f_{osc}.

RVT uses the moments of the power spectrum of the oscillator's response (Y^2_{SDOF}). The kth moment, m_k, of the power spectrum of the SDOF response is defined as:

$$m_k\left(f_{osc},\zeta\right) = 2\int_0^\infty \left(2\pi f\right)^k \; Y^2_{SDOF}(f, f_{osc}, \zeta)\; df. \tag{3.1}$$

Parseval's theorem is used to compute the root-mean-square (rms) acceleration, y_{rms}, from the FAS of the time series. The rms acceleration of the SDOF response, y_{rms}, is computed as:

$$y_{rms}(f_{osc},\zeta) = \sqrt{\frac{m_0(f_{osc},\zeta)}{D_{rms}}}. \tag{3.2}$$

in which D_{rms} is the duration of the oscillator time series contributing to the rms acceleration.

RVT relates the y_{rms} to the peak value of the SDOF response in the time domain, y_{max}, by the peak factor, PF.

$$y_{max} = PF y_{rms}. \tag{3.3}$$

The estimation of the PF uses extreme value statistics. The PF depends on the number of extrema (peaks) of the time series. The more peaks in the time series, the more chances for a large value to occur. The number of extrema depends on the duration and the dominant frequency (bandwidth) of the time series. The bandwidth is quantified by $\sqrt{\frac{m_2}{m_0}}$. The number of extrema is related to the number of zero crossings, N_z, which is the product of the dominant frequency and the duration:

$$N_z \propto \sqrt{\frac{m_2}{m_0}} D_{gm}, \tag{3.4}$$

in which D_{gm} is the duration of the ground motion.

The PF depends on more than just the number of zero crossings, but we use the simplified metric that is proportional to peak factor because we include an empirical factor in our model for converting EAS to PSA that will account for scale factors on the PF.

$$PF \propto [ln(\sqrt{m_2/m_0}D_{5-75\%})]^{0.5}. \tag{3.5}$$

We use two RVT parameters for the empirical EAS-to-PSA conversion model: the rms acceleration and the $\sqrt{\frac{m_2}{m_0}}D_{gm}$.

3.1 PERIOD-DEPENDENT DURATION MODEL

Current RVT methods have used the duration of the acceleration for the D_{gm}. This assumes that the duration of the ground motion at other periods is the same as the acceleration duration; however, as shown by Abrahamson and Sung (2024), this assumption is not valid for periods greater than 0.5 sec. A period-dependent duration better represents the duration of the strong shaking for long-period ground motion than the standard approach of using the acceleration duration for all periods.

Lavrentiadis (2025b) computed the period-dependent duration for the NGA-W3 data set and a period-dependent duration GMM was developed by Sung et al. (2025). We use the Sung et al. (2025) model for the 5-75% duration in our EAS-to-PSA conversion. A brief summary of the period-dependent duration model is given here.

The period-dependent duration model is developed as an adjustment to the traditional acceleration duration.

$$D_{5-75}(T) = f_1(R, V_S30) + f_2(D_{5-75-acc}), \tag{3.6}$$

in which $D_{5-75-acc}$ is the time interval between reaching 5% and 75% of the Arias intensity of the accelerogram.

The $f_1(R, V_S30)$ term reflects the differences in the distance and site scaling for the acceleration duration, $D_{5-75-acc}$, and the period-dependent duration, $D_{5-75}(T)$. There is no magnitude dependence in the f_1 term because it is captured in the magnitude scaling of the $D_{5-75-acc}$ values.

The residuals for the duration are consistent with a power-normal distribution (Pinilla-Ramos et al., 2023). Therefore, the period-dependent duration model uses a power-normal transformation with exponent n and is given by:

$$[D_{5-75}(T)]^n = [D_{Path}(T) + D_{Site}(T) + c_5 D_{5-75-acc}(T)]^n + \delta WS, \tag{3.7}$$

in which

$$D_{Site}(T) = \begin{cases} 0 & \text{for} \leq 3 \text{ km} \\ ln\left(\frac{Min(V_{S30},1000)}{2000}\right)\frac{c_4(T)}{7}(R_{RUP}-3) & \text{for 3 km} < R_{RUP} \leq 10 \text{ km} \\ ln\left(\frac{Min(V_{S30},1000)}{2000}\right)c_4(T) & \text{for } R_{RUP} > 10 \text{ km,} \end{cases} \tag{3.8}$$

$$D_{Path}(T) = \begin{cases} 0 & \text{for } R_{RUP} \leq 3 \text{ km} \\ \frac{c_{71}(T)}{7}(R_{RUP}-3) & \text{for 3 km} < R_{RUP} \leq 10 \text{ km} \\ c_{71}(T) + \frac{c_{72}(T)-c_{71}(T)}{90}(R_{RUP}-10) & \text{for 10 km} < R_{RUP} \leq 100 \text{ km} \\ c_{72}(T)R_{RUP} & \text{for } R_{RUP} > 100 \text{ km.} \end{cases} \tag{3.9}$$

The coefficients for the period-dependent duration model are shown in Table 3.1 and are plotted in Figure 3.1. Examples of the residuals of the period-dependent duration model for M $\geq$ 5.5 are shown in Figure 3.2. The residuals for the WUS data set are centered on zero, as shown by the binned averages (pink points). In addition to the WUS data, we show the residuals from the 2023 M7.8 Turkey earthquake. This event has the best sampling of the near-fault ground motions from a large-magnitude earthquake. At short distances, there are large negative residuals from the Turkey event (smaller durations) that reflect shorter durations of the near-fault ground motions for this event. For the period-dependent duration model, we do not modify the duration model to fit these short-distance, large-magnitude durations. This event warrants further evaluation of the near-fault durations.

Examples of the magnitude, distance, and V_{S30} scaling of the period-dependent duration are shown in Figure 3.3. This figure shows a significant increase in the duration for periods greater than 0.5 sec, consistent with the results of Abrahamson and Sung (2024).

3.2 RVT-TYPE CONVERSION FROM EAS TO PSA

We develop an empirically calibrated model for converting EAS to PSA that is based on two parameters used in RVT: the rms acceleration and the number of zero crossings. For developing the conversion model, the moments of the EAS are computed using the median EAS values from the GMM, not from the EAS of the recordings. This approach simplifies the application of the conversion for the suite of scenarios by avoiding the need to include the variability of the $EAS(f)$, and its inter-frequency correlation in the computation of the moments. The key is that the application of the EAS-to-PSA conversion will be consistent with the development of the conversion model. A limitation of this approach is that we can only use the conversion for the median values, and not the standard deviations.

We use the $D_{5-75}(T)$ period-dependent ground-motion duration described above, rather than the duration for acceleration. The initial functional form used for conversion from EAS to PSA is given by:

$$
\begin{aligned}
ln(PSA(T)) =& a_1(M, R_{RUP}, T) + a_2(T) ln\left(\frac{m_0(T)}{D_{5-75}(T)}\right) \\
&+ a_3(T)\sqrt{ln\left(\sqrt{\frac{m_2(T)}{m_0(T)}} D_{5-75}(T)\right)}.
\end{aligned}
\tag{3.10}
$$

There can be a strong tradeoff between a_1 and a_3. Therefore, we fix the $a_3(T)$ term to unity, which assumes that the peak factor is proportional to the square root of the natural logarithm of the number of zero crossings.

An initial regression showed a bias for very short distances ($R_{RUP} < 3$ km). There is sparse ground-motion data in the short-distance range, but it is an important part of the GMM for applications. Therefore, we add an additional short-distance term, $f_R(M, R_{RUP})(T)$, to remove the bias. The additional term is given by:

$$
f_R(M, R_{RUP})(T) = \begin{cases} d_1(M) & \text{for } R_{RUP} \leq 1 \text{ km} \\ d_1(M) - 0.5 d_1(R_{RUP} - 1) & \text{for 1 km} < R_{RUP} \leq 3 \text{ km} \\ 0 & \text{for } R_{RUP} > 3 \text{ km,} \end{cases}
\tag{3.11}
$$

in which

$$
d_1 = 0.1(max(M, 5)) - 0.7. \tag{3.12}
$$

The final equation for converting the EAS to PSA is given by:

$$
\begin{aligned}
ln(PSA(T)) =& a_1(M, R_{RUP}, T) + a_2(T) ln\left(\frac{m_0(T)}{D_{5-75}(T)}\right) \\
&+ \sqrt{ln\left(\sqrt{\frac{m_2(T)}{m_0(T)}} D_{5-75}(T)\right)} + f_R(M, R_{RUP})(T).
\end{aligned}
\tag{3.13}
$$

The $a_1(T)$ and $a_2(T)$ terms are estimated by regression. The $a_1(T)$ term depends on magnitude and distance. Instead of using a functional form for the magnitude and distance dependence of $a_1(T)$, we compute $a_1(T)$ for four magnitude bins and three distance bins and use linear interpolation between the bins. The coefficients for these bins are listed in Table 3.2 and are plotted in Figure 3.4.

3.3 CONVERTED PSA DATASET

The median EAS values are converted to PSA for the large suite of scenarios listed in Table 3.3. There are 92,700 scenarios with PSA values generated for 23 spectral periods between 0.01 and 10

sec. The sampling of magnitudes, rupture distances, V_{S30}, and Z_{TOR} is shown in Figure 3.5. The range of Z_{TOR} values for each magnitude is based on the range of values in the WUS data set for each style of faulting.

Examples of the distance scaling for T=0.2 sec and T=1 sec are shown in Figure 3.6. These PSA values represent the median PSA without aleatory variability. The ranges of PSA values for each distance reflect the range of Z_{TOR} values. For the large-magnitude events, there is a smaller range of PSA values due to the smaller range of possible Z_{TOR} values for the WUS crustal thickness.

The response spectra for the generated PSA values are compared to the ASK14 median values in Figures 3.7 and 3.8 for distances of 15 and 80 km, respectively. The distance scaling is compared in Figure 3.9, and the magnitude scaling is compared in Figure 3.10. The main differences are in the short-period range for the small and moderate magnitudes (M<6.5), with the generated PSA values smaller than the ASK14 GMM, and in the long period range for the large magnitudes, with the generated PSA values larger than the ASK14 GMM for the 15 km case.

Table 3.1: The coefficients of the 5-75% period-dependent model.

Period(s)	c_4	c_5	c_{71}	c_{72}
0.02	-0.001	1.000	-0.009	-0.010
0.05	0.000	1.000	-0.009	-0.008
0.075	0.000	0.990	-0.010	-0.008
0.1	0.000	0.985	-0.010	-0.009
0.15	0.000	0.980	-0.010	-0.010
0.2	-0.171	0.960	-0.010	-0.011
0.3	-0.400	0.936	-0.006	-0.010
0.4	-0.545	0.923	0.004	-0.009
0.5	-0.672	0.913	0.015	-0.008
0.75	-0.991	0.903	0.037	-0.005
1	-1.154	0.903	0.057	-0.001
1.5	-1.159	0.902	0.096	0.006
2	-1.161	0.902	0.117	0.013
3	-1.163	0.901	0.132	0.032
4	-1.164	0.901	0.122	0.039
5	-1.166	0.900	0.110	0.037
7.5	-1.169	0.900	0.102	0.021
10	-1.189	0.900	0.093	0.012

Table 3.2: The coefficients of the empirical RVT-type conversion of EAS to PSA.

Period(s)	a11 M<7.2 R<20	a_{12} M>7.2 R>80	a_{13} 6<M<7.2 R<20	a_{14} 6<M<7.2 R>80	a_{15} 5<M<6 R<20	a_{16} 5<M<6 R>80	a_{17} M<3.5 R<20	a_{18} M<3.5 R>80	a_2
0.01	0.136	0.256	0.270	0.291	0.205	0.205	0.760	0.668	0.502
0.02	0.136	0.256	0.270	0.291	0.205	0.205	0.760	0.668	0.502
0.025	0.140	0.268	0.252	0.294	0.214	0.214	0.779	0.675	0.503
0.03	0.130	0.263	0.233	0.283	0.205	0.205	0.755	0.659	0.503
0.05	0.079	0.239	0.139	0.220	0.156	0.156	0.582	0.560	0.500
0.075	-0.014	0.165	0.019	0.117	0.060	0.060	0.370	0.419	0.493
0.1	-0.073	0.120	-0.055	0.051	-0.006	-0.006	0.225	0.325	0.489
0.15	-0.038	0.170	-0.058	0.059	0.028	0.028	0.076	0.242	0.492
0.2	-0.034	0.164	-0.081	0.020	0.046	0.046	-0.071	0.105	0.493
0.25	-0.124	0.052	-0.204	-0.106	-0.031	-0.031	-0.236	-0.079	0.489
0.3	-0.133	0.027	-0.223	-0.137	-0.029	-0.029	-0.267	-0.122	0.490
0.4	-0.184	-0.054	-0.275	-0.219	-0.041	-0.041	-0.330	-0.216	0.487
0.5	-0.257	-0.211	-0.386	-0.359	-0.149	-0.149	-0.433	-0.339	0.484
0.75	-0.486	-0.520	-0.602	-0.621	-0.367	-0.367	-0.631	-0.590	0.478
1	-0.733	-0.820	-0.843	-0.884	-0.605	-0.605	-0.880	-0.909	0.471
1.5	-0.784	-0.920	-0.891	-0.946	-0.646	-0.646	-0.946	-1.102	0.468
2	-0.840	-1.015	-0.946	-1.018	-0.700	-0.700	-1.017	-1.327	0.461
2.5	-0.784	-1.005	-0.892	-0.982	-0.667	-0.667	-0.958	-1.369	0.460
3	-0.749	-0.983	-0.862	-0.963	-0.644	-0.644	-0.932	-1.398	0.460
5	-0.602	-0.816	-0.746	-0.883	-0.535	-0.535	-0.865	-1.504	0.460
6	-0.667	-0.850	-0.826	-0.970	-0.608	-0.608	-0.970	-1.640	0.460
7.5	-0.690	-0.834	-0.864	-1.016	-0.638	-0.638	-1.038	-1.737	0.460
10	-0.641	-0.730	-0.836	-0.996	-0.596	-0.596	-1.050	-1.775	0.460

Table 3.3: Scenarios used for generating the PSA dataset from the EAS GMM.

Parameters	N	Values
Magnitudes	14	M3 to M8.0, step 0.5
R_{RUP} (km)	23	1, 3, 5, 7, 10, 12, 15, 20, 24, 30, 35, 40, 50, 60 70, 85, 100, 125, 150, 180, 200, 250, 300
ZTOR (km)	12	0, 0.5, 1, 2.5, 5, 7.5, 10, 12.5, 15, 17.5, 20, 25
VS30 (m/s)	8	160, 200, 280, 400, 550, 760, 900, 1000
SOF	3	NML, SS, RV

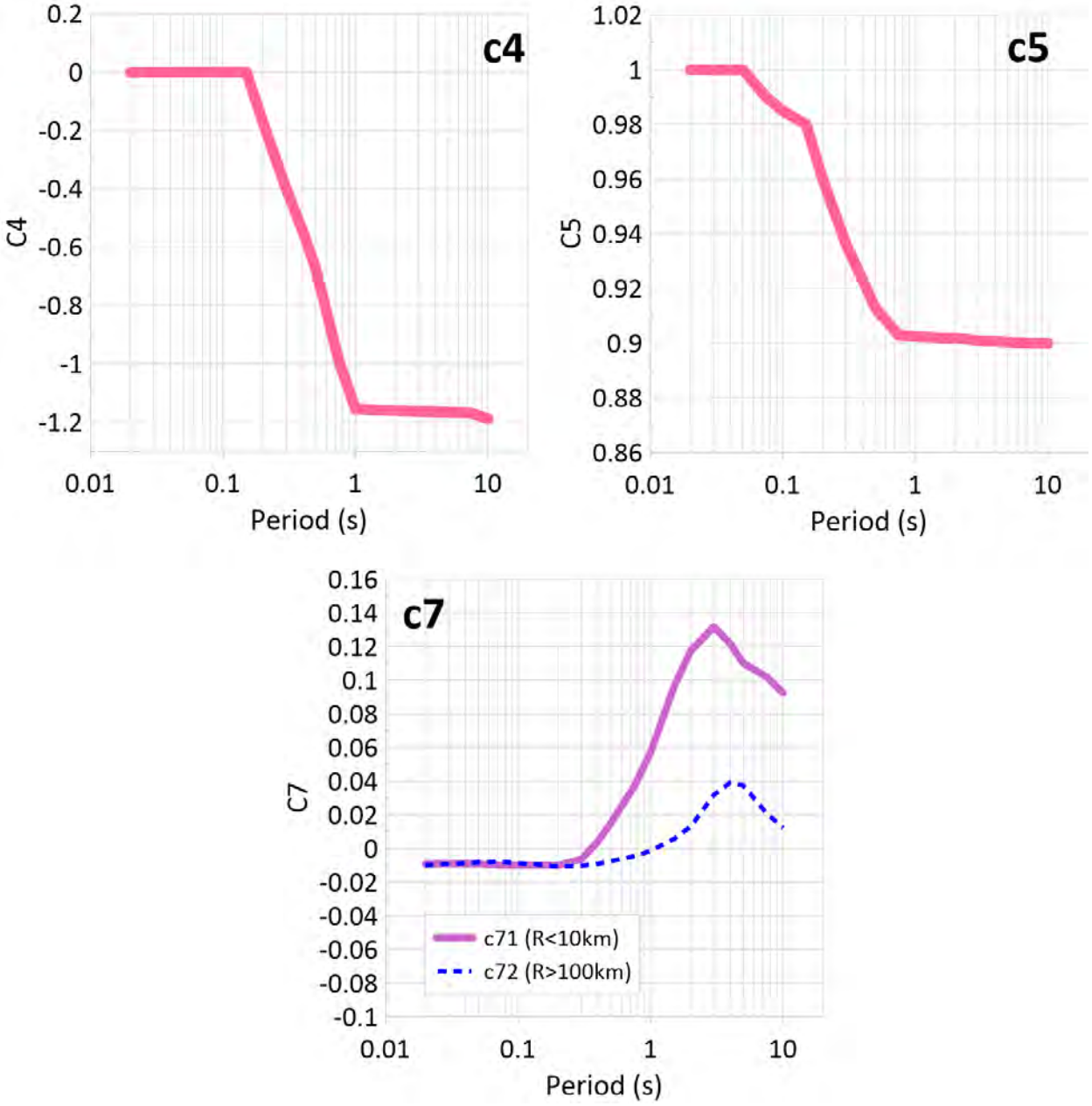


Figure 3.1: The coefficients of the D_{5-75} period-dependent duration model.

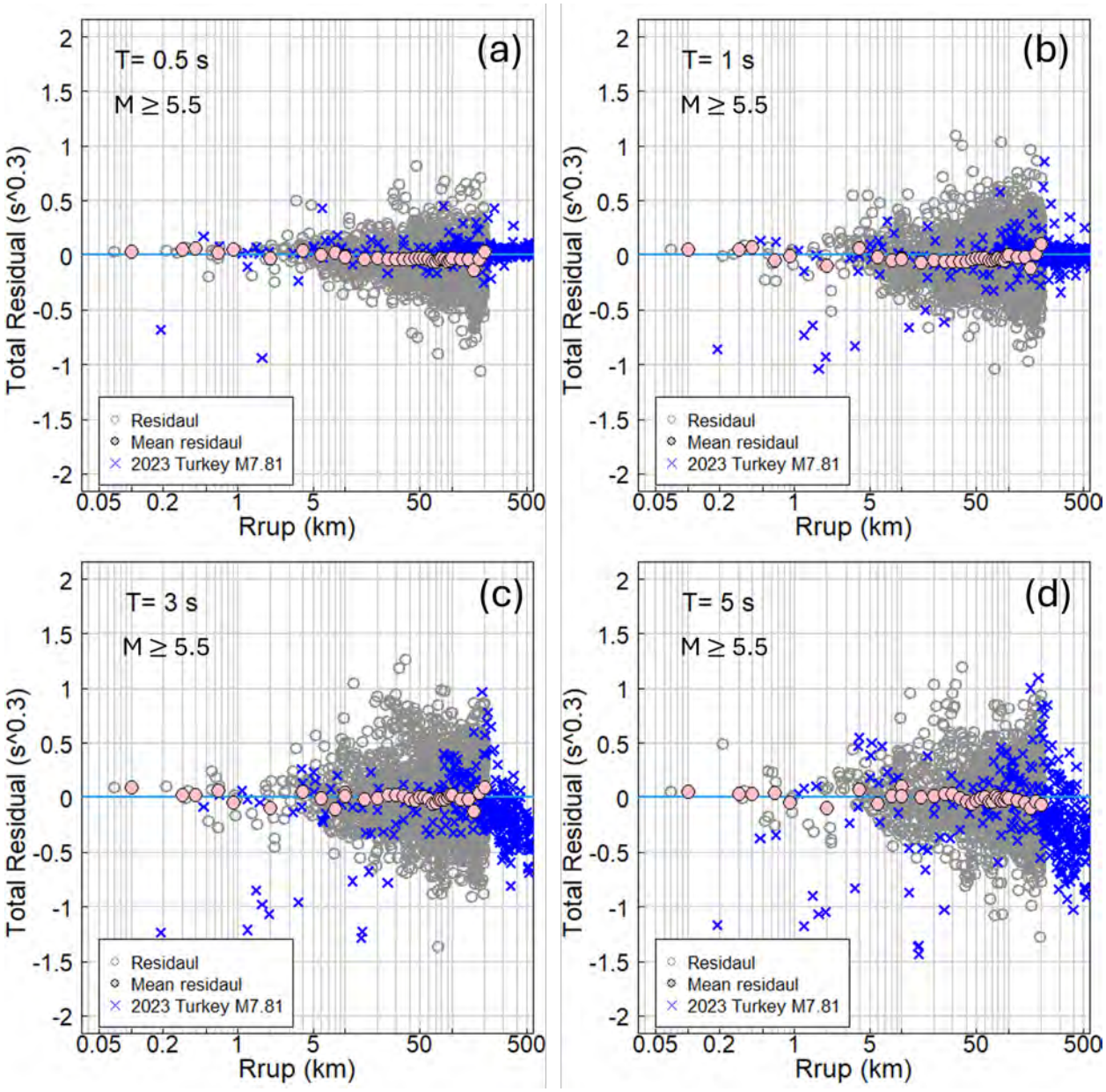


Figure 3.2: Residuals of the D_{5-75} period-dependent duration model for M $\geq$ 5.5 at (a) 0.5 sec, (b) 1 sec, (c) 3 sec, and (d) 5 sec.

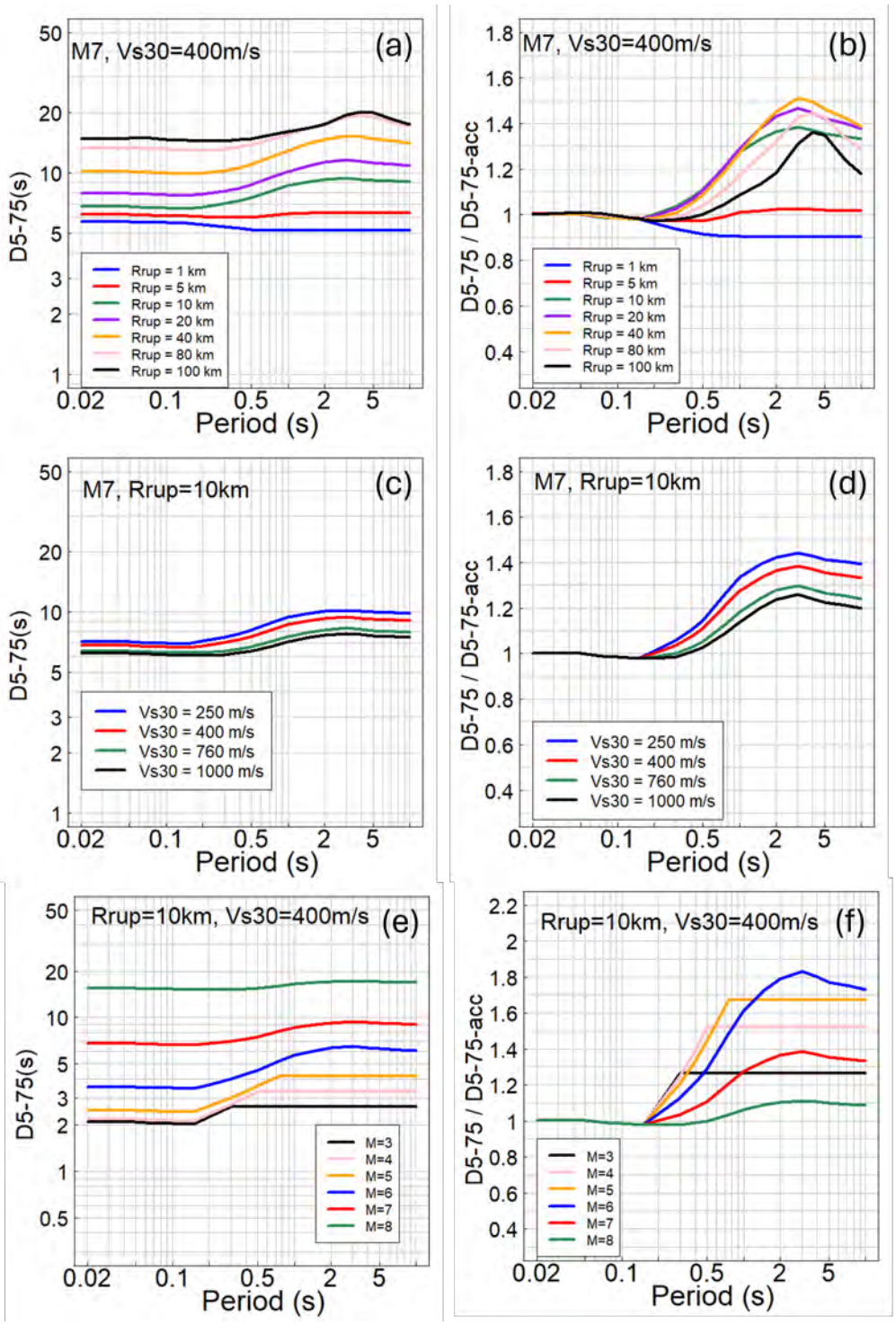


Figure 3.3: The scaling for the D_{5-75} period-dependent duration (PDD) model and for the duration normalized by the acceleration duration. (a) Distance scaling for the PDD model. (b) Distance scaling for the normalized duration. (c) V_{S30} scaling for the PDD model. (d) V_{S30} scaling for the normalized duration. (e) Magnitude scaling for the PDD model. (f) Magnitude scaling for the normalized duration.

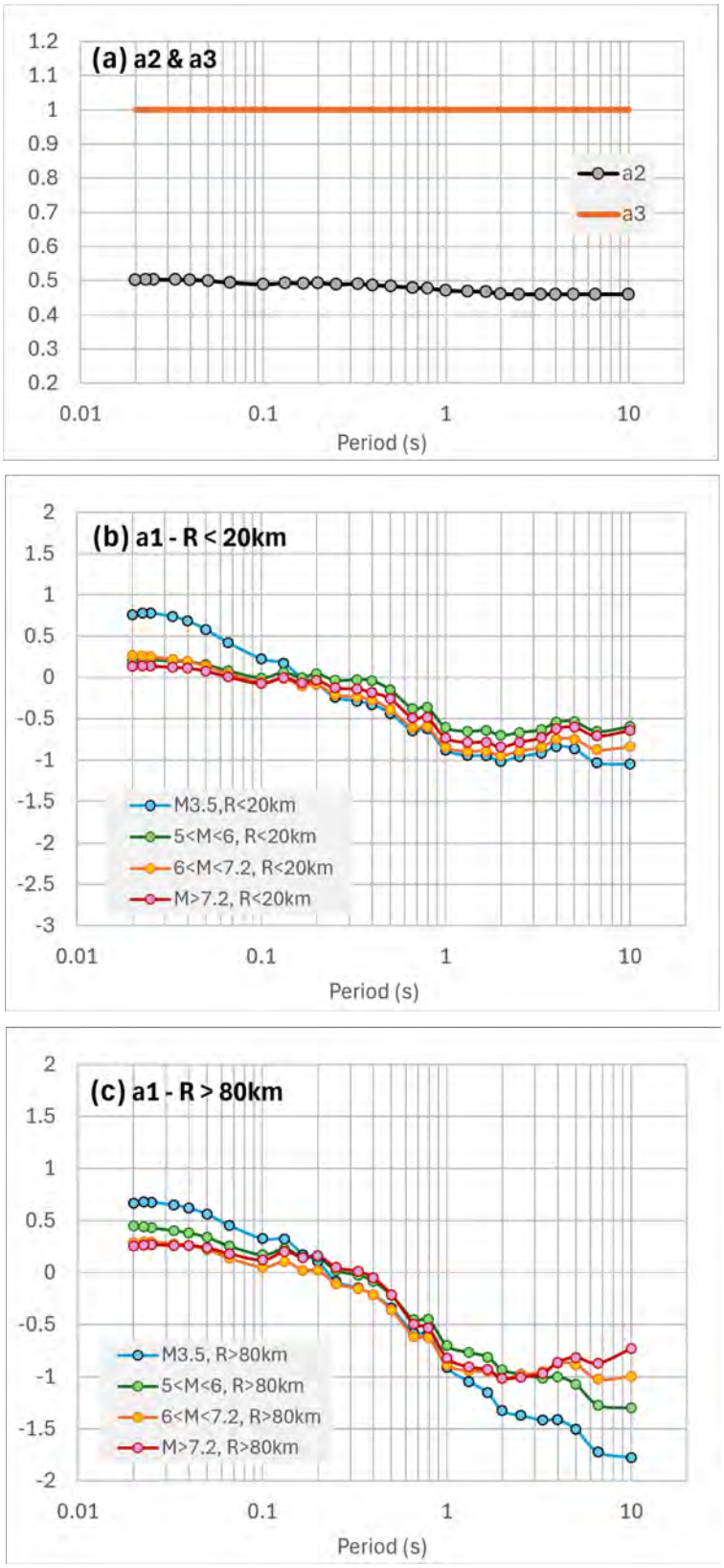


Figure 3.4: The coefficients of the RVT-type model for converting EAS to PSA.

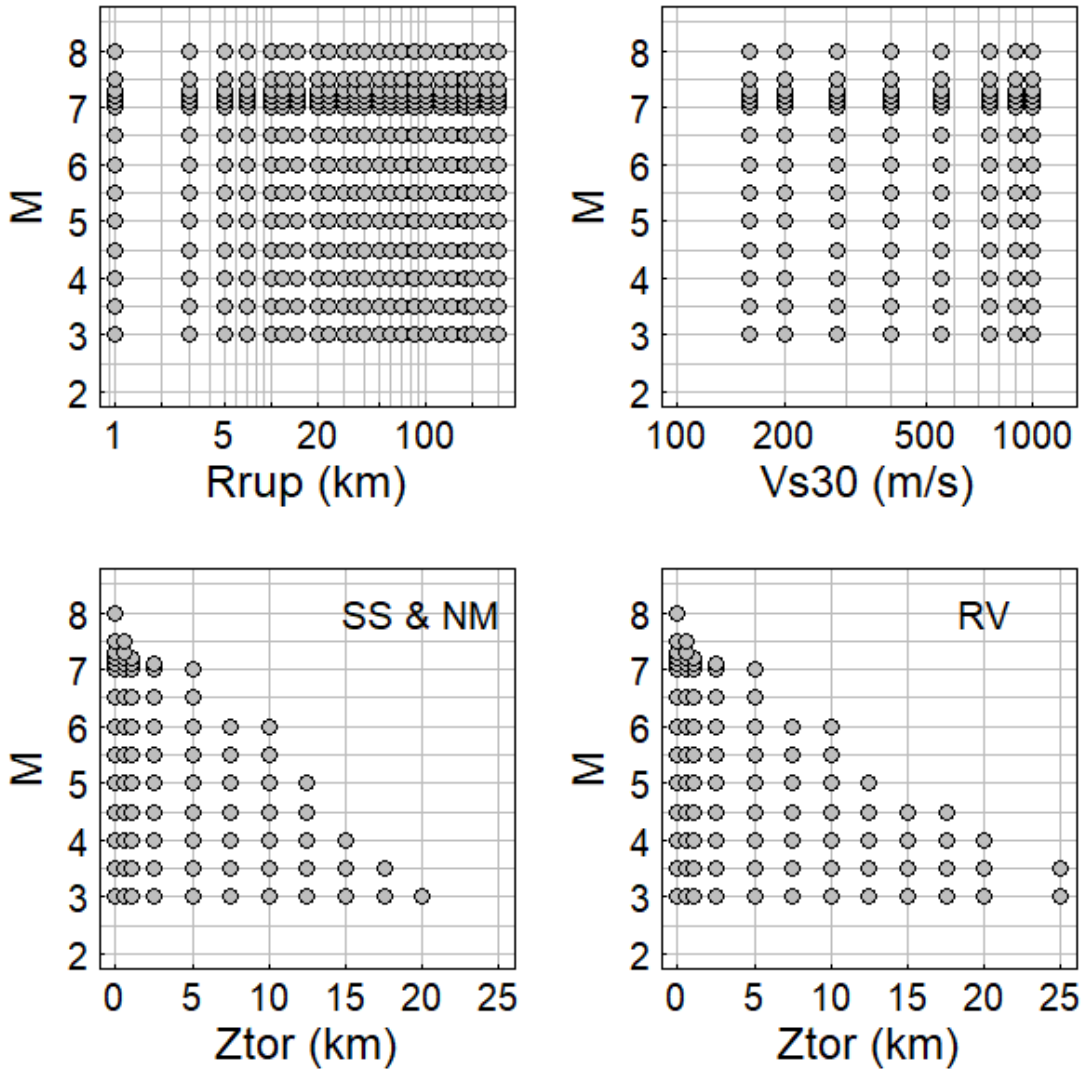


Figure 3.5: Distribution of the magnitude, distance V_{S30}, and Z_{TOR} for the PSA converted from the EAS GMM.

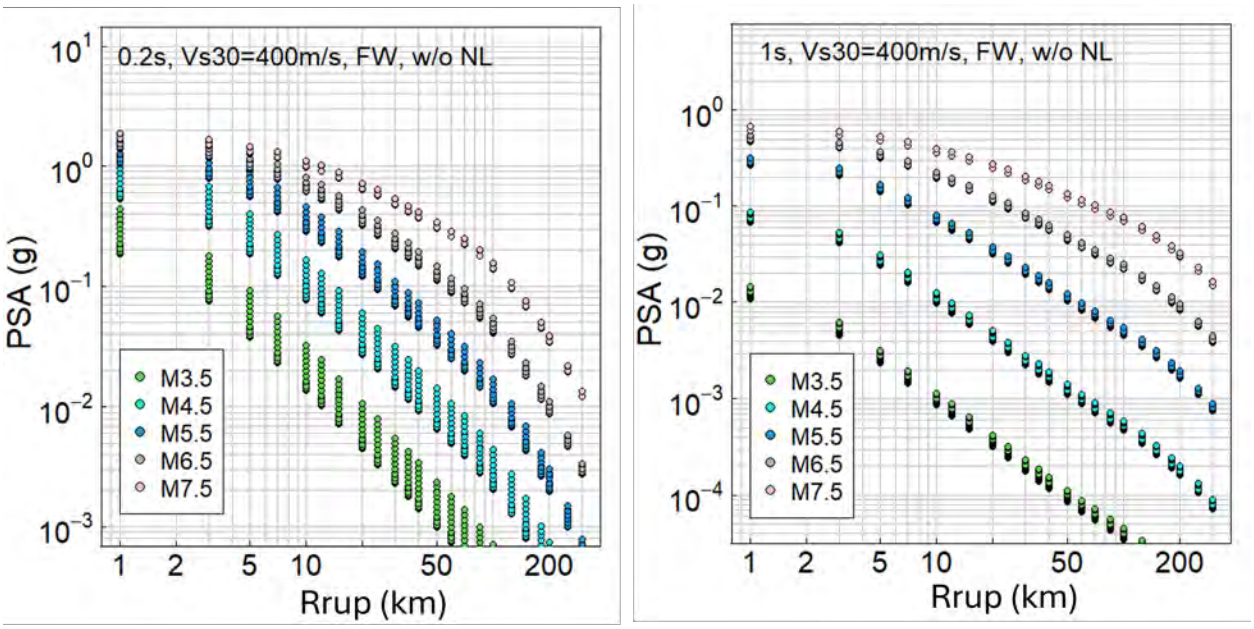


Figure 3.6: Distance scaling of the median PSA for T = 0.2 s and 1 s.

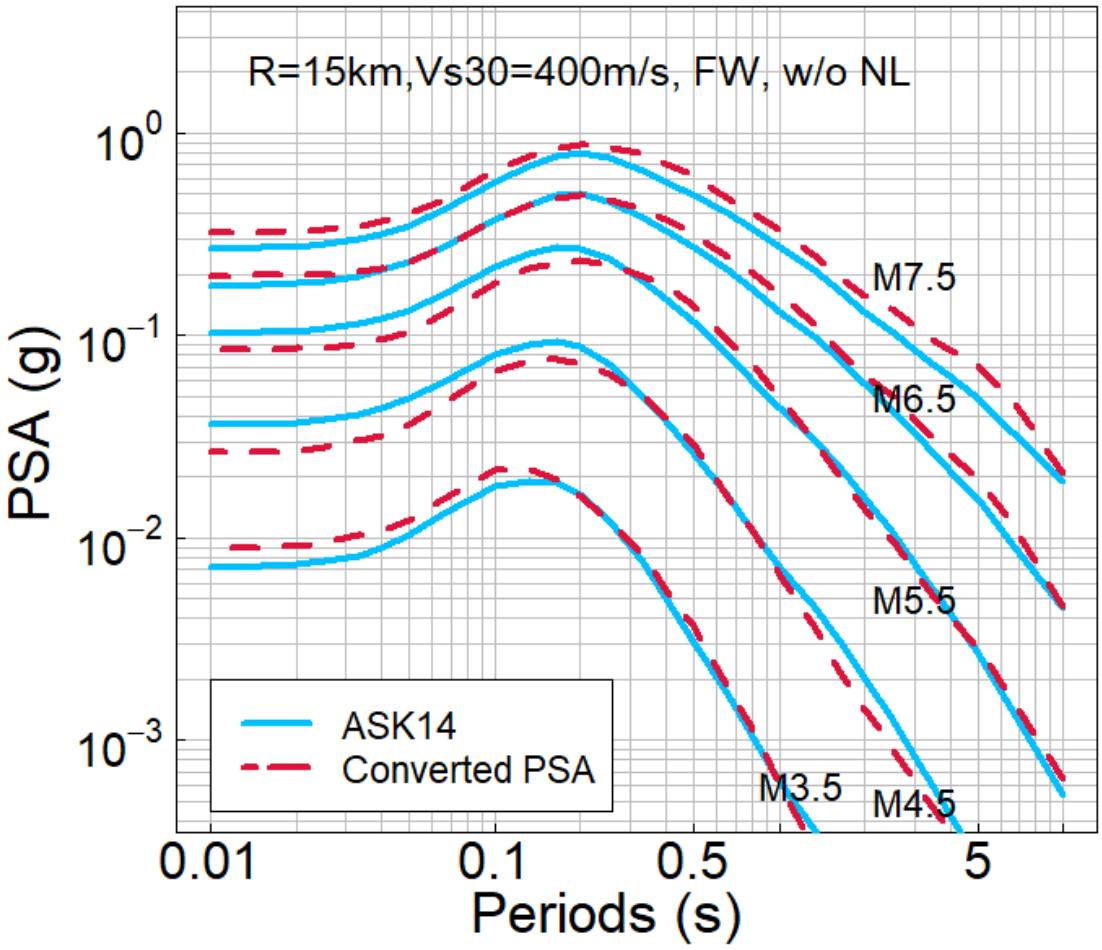


Figure 3.7: Median PSA spectrum for the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events and without the non-linear site effect, for R_{RUP}=15 km.

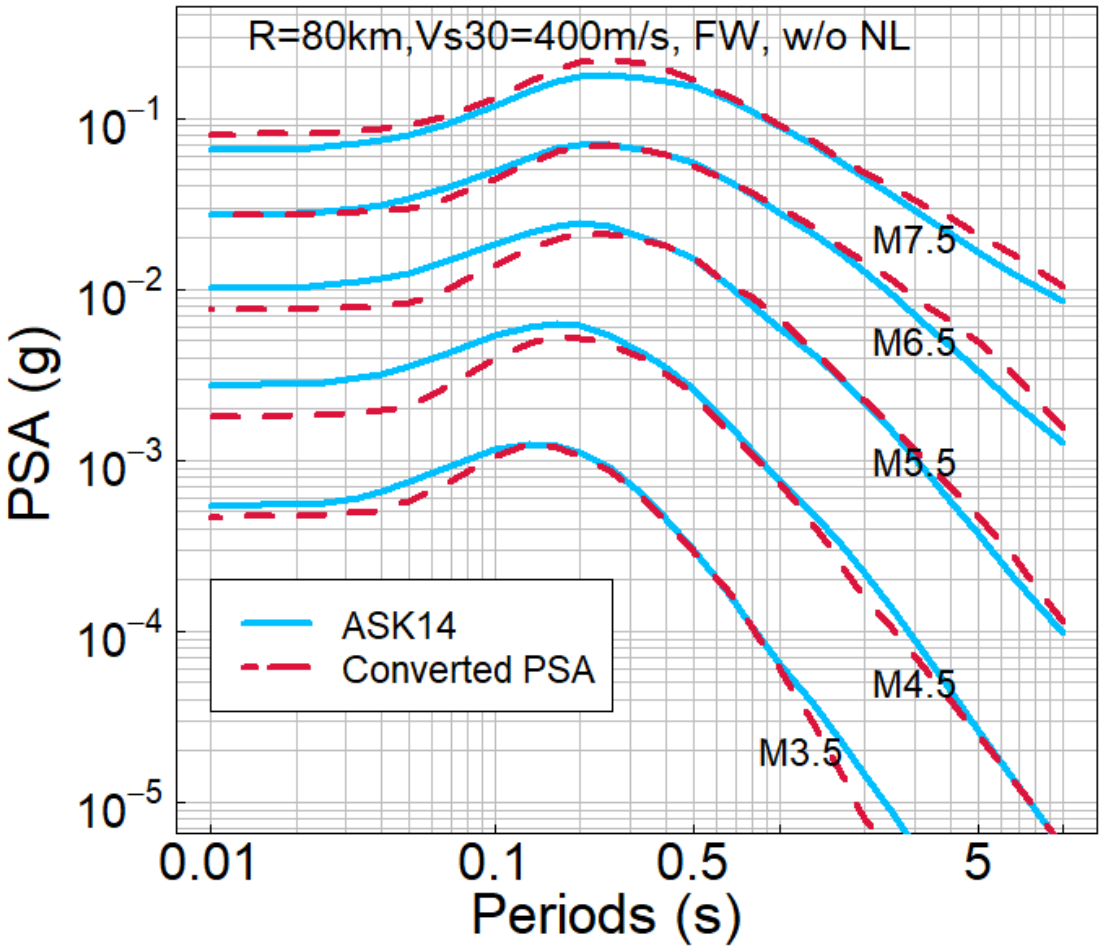


Figure 3.8: Median PSA spectrum for the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events, without the non-linear site effect, for R_{RUP}= 80 km.

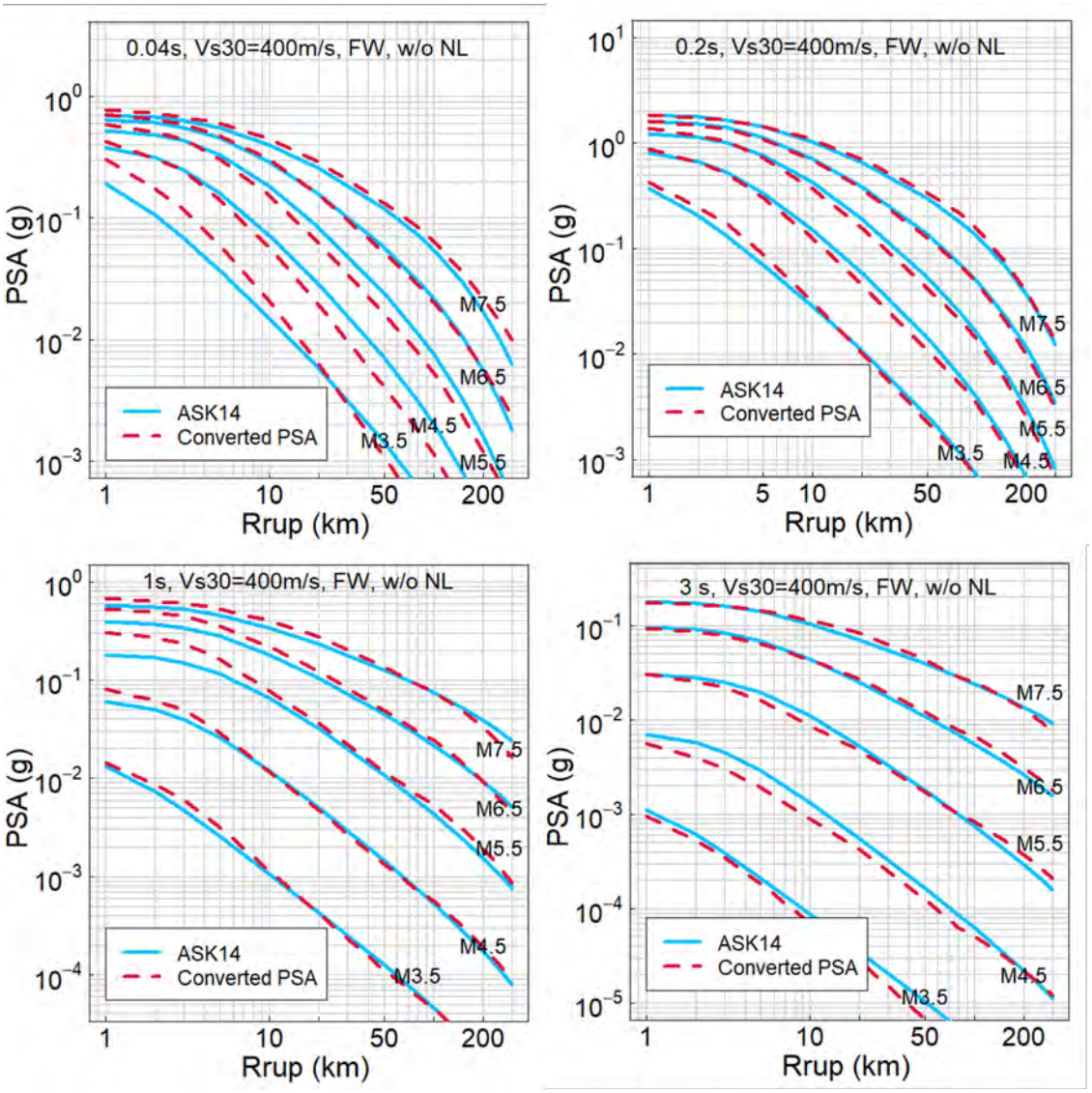


Figure 3.9: Distance scaling of the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events, without the non-linear site effect.

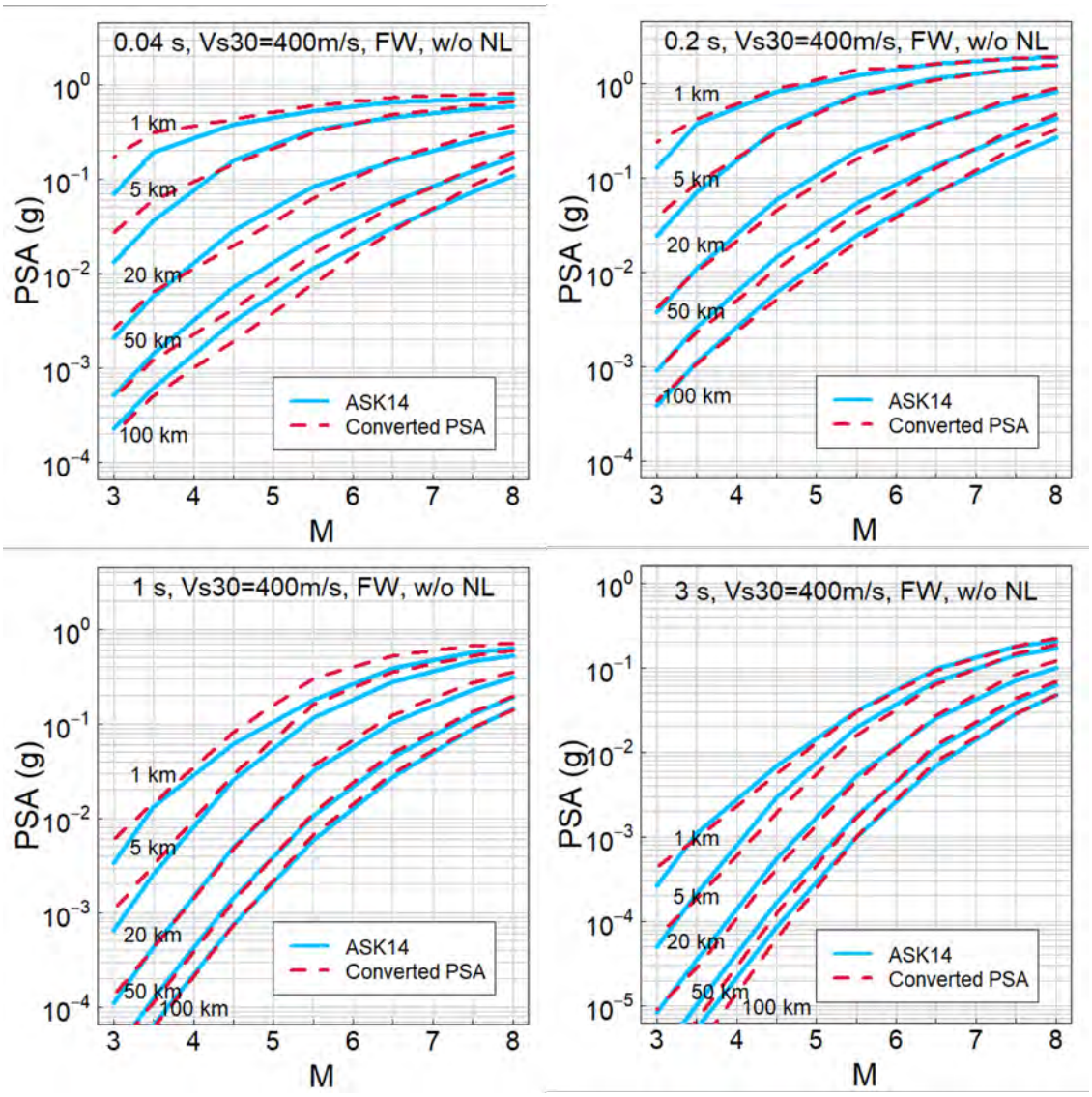


Figure 3.10: Magnitude scaling of the PSA from the EAS GMM and for the ASK14 GMM for strike-slip events, without the non-linear site effect.

4 FUNCTIONAL FORM FOR THE MEDIAN

The functional form of the PSA GMM is similar to the functional form used for the EAS GMM (Abrahamson et al., 2025). The general form of the PSA GMM is given by:

$$\begin{aligned} ln(PSA(T)) =& c_1 + f_m(M) + f_p(M, R_{RUP}) + f_{sl}(V_{S30}) + f_{snl}(V_{S30}, PSA_{800}(T)) + f_z(Z_{TOR}) \\ &+ f_{HW}(M, R_x, Z_{TOR}, dip) + f_{SOF}(F_{RV}, F_{NML}) + f_{SSD}(M) + f_{Tamp1.5}(\delta_{Tamp1.5}) \\ &+ \delta B_e + \left(\frac{R_{rup} - 50}{100}\right) \delta P_e + \delta S2S_s + \delta WS_{es}. \end{aligned} \tag{4.1}$$

Each term is described below.

Magnitude scaling

The functional form for the magnitude scaling is given by:

$$f_m(M) = c_2(M - 6) + \left(\frac{c_2 - c_3}{c_n}\right) ln(1 + \exp(c_n(c_m - M))), \tag{4.2}$$

in which M is the moment magnitude. This form transitions from linear scaling at small magnitudes with a slope of c_3 to linear scaling at large magnitudes with a slope of c_2. The transition between the small-magnitude scaling and the large-magnitude scaling is centered at c_m, and the width of the transition is controlled by the c_n term.

Path scaling

In the ASK14 GMM, the path term includes a magnitude-dependent geometrical spreading term. For this GMM, we use a constant geometrical spreading term (c_4) to be consistent with the modeling of the geometrical spreading and finite-fault terms in the EAS GMM. The geometrical spreading term transitions to a fixed value of -0.5 (surface wave geometrical spreading) at large distances ($R_{RUP} > 50$ km).

The large-distance scaling of the EAS, related to inelastic attenuation, is frequency dependent. At large distances, the PSA values at short periods are affected by a wide range of frequencies. The frequency content is magnitude dependent, with greater low-frequency content for large-magnitude events. For the EAS, the effect of the anelastic attenuation is independent of magnitude, but for the PSA, the large-distance scaling of the short-period PSA can be dependent on magnitude because the greater long-period content of the EAS for larger magnitudes will contribute to the short-period PSA values at large distances. Therefore, we allow for a magnitude-dependent coefficient, $c_7(M)$, to scale the linear R term.

The functional form of the path term is given by:

$$f_p(M, R_{RUP}) = c_4\, ln\left(R_{RUP} + FF(M)\right) + (-0.5 - c_4) ln R_1 + c_7(M) R_{RUP}, \tag{4.3}$$

in which $R_1 = \sqrt{R_{RUP}^2 + 50^2}$.

Finite-fault term

In the Chiou and Youngs (2014) PSA GMM, the finite-fault effects are modeled by a magnitude-dependent finite-fault term, FF(M), that uses a hyperbolic function. The EAS GMM (Abrahamson et al., 2025) replaced the hyperbolic function with an exponential function of magnitude. We use this same exponential function for the FF(M) term in the PSA GMMs.

$$FF(M) = exp\left(\frac{b_2 M_1}{1 + b_3 M_1}\right), \tag{4.4}$$

in which M_1 is defined as $M_1 = \max(M - 3.0, 0)$.

Site scaling

The site term includes the V_{S30} scaling. The basin-depth scaling is not included in our ergodic GMM for three reasons. First, in the November 2024 version of the NGA-W3 flatfile, there were issues with the reliability of the $Z_{1.0}$ estimates. Second, the basin effects depend on the ray paths entering the basin, which is easier to model with a non-ergodic path term than as a constant factor based on $Z_{1.0}$. Third, there is a strong correlation between V_{S30} and $Z_{1.0}$ in California (Pinilla-Ramos et al., 2022), implying that the inclusion of these two parameters in the model provides marginal improvements to the accuracy of the GMM.. Therefore, the basin effects were not included in our ergodic GMM, but will be included in the non-ergodic GMM that is being developed in a separate study.

The V_{S30} scaling is separated into a linear term, f_{sl}, and a non-linear term, f_{snl}. The functional form of the linear site term is given by:

$$f_{sl}(V_{S30}) = c_8(M, R_{RUP})\, ln\left(\frac{Min(V_{S30}, 1000)}{800}\right). \tag{4.5}$$

At large V_{S30} values ($V_{S30} > 1000$ m/s), the correlation of the V_{S30} and the deeper $V_S(z)$ profile that controls site amplification is weaker than for soil sites. Therefore, we limit the scaling for the linear site term to V_{S30} values less than 1000 m/s.

For the EAS, the linear site term applies to all scenarios, but for PSA, the amplification for linear site amplification can depend on the spectral shape. At short periods, the peak in the response spectrum will be dependent on magnitude. Similarly, at large distances, the period of the peak in the response spectrum will be dependent on distance. The scaling for the linear site is expected to depend on both magnitude and distance. Therefore, we allow for a magnitude and distance dependence in the c_8 coefficients.

The non-linear site term is fixed to the analytical-based site response model by Ilhan et al. (2025) and is given by:

$$f_{snl}(V_{S30}, PSA_{800}(T) = \begin{cases} f_2(f, V_{S30} = 200)\ ln\left(\frac{PSA_{800}(T)+f_3(f)}{f_3}\right) & \text{for } V_{S30} \leq 200 \text{ m/s} \\ f_2(f, V_{S30})\ ln\left(\frac{PSA_{800}(T)+f_3(f)}{f_3(f)}\right) & \text{for 200 m/s} \leq V_{S30} \leq 800 \text{ m/s} \\ 0 & \text{for } V_{S30} > 800\text{m/s}, \end{cases} , \tag{4.6}$$

in which $PSA_{800}(T)$ is the PSA(T) for V_{S30} = 800 m/s. The V_{S30} dependence of f_2 is given by:

$$f_2(V_{S30}, T) = f_4(T)[exp(f_5(T)(Min(800, V_{S30}) - 360) - exp(f_5(T)400)]. \tag{4.7}$$

The f_3 coefficient controls the threshold ground-motion level for the transition from linear to non-linear site amplification. The f_4 and f_5 are period-dependent coefficients given by Ilhan et al. (2025).

Depth scaling

The depth scaling is based on the scaling in the ASK14 GMM and is given by:

$$f_z(Z_{TOR}) = c_9 Min(Z_{TOR}, 20). \tag{4.8}$$

The scaling is limited to the Z_{TOR} values less than 20 km, which reflects the limits of the well-sampled depths for events in the WUS (see Figures 2.3). In previous GMMs, the Z_{TOR} scaling was strongest at short periods (e.g., Abrahamson et al., 2014), the Z_{TOR} has been interpreted as a proxy for stress drop.

SOF scaling

The style-of-faulting (SOF) scaling adopts a simple period-dependent constant for normal and reverse earthquakes, given by:

$$f_{SOF}(F_{RV}, F_{NML}) = c_{10n}F_{NML} + c_{10r}F_{NML}, \tag{4.9}$$

in which F_{NML}=1 for normal mechanism earthquakes ($-30 \geq Rake \geq -150$) and 0 otherwise; F_{RV}=1 for reverse mechanism earthquakes ($30 \leq$ Rake ≤ 150) and 0 otherwise.

Hanging-wall scaling

The functional form for the Hanging-wall (HW) scaling uses the simulation-based model given by Donahue and Abrahamson (2014).

$$f_{HW}(M, R_x, Z_{TOR}, dip) = c_{13}T_{HW1}(dip)T_{HW2}(M)T_{HW3}(R_x)T_{HW4}(Z_{TOR})T_{HW5}(Ry_0). \tag{4.10}$$

The c_{13} term is estimated from the regression, whereas the five other terms, T_{HW1} - T_{HW5}, are constrained by the results of Donahue and Abrahamson (2014) with the modification to the R_x scaling at large distances implemented by Abrahamson et al. (2014). The five taper terms for the HW model are given by:

$$T_{HW1}(\text{dip}) = \begin{cases} \frac{60}{45} & \text{for dip} \leq 30 \\ \frac{90-\text{dip}}{45} & \text{for dip} > 30 \end{cases} \tag{4.11}$$

$$T_{HW2}(M) = \begin{cases} 1 + a_2(M - 6.5) & \text{for M} > 6.5 \\ 1 + a_2(M - 6.5) - (1 - a_2)(M - 6.5)^2 & \text{for 5.5} < \text{M} < 6.5 \\ 0 & \text{for M} \leq 5.5 \end{cases} \tag{4.12}$$

$$T_{HW3}(R_x) = \begin{cases} h_1 + h_2(R_x/R_1) + h_3(R_x/R_1)^2 & \text{for } R_x \leq R_1 \\ 1 - \frac{R_x - R_1}{R_2 - R_1} & \text{for } R_1 < R_x \leq R_2 \\ 0 & \text{for } R_x > R_2 \end{cases} \tag{4.13}$$

$$T_{HW4}(Z_{TOR}) = \begin{cases} 1 - 0.01Z_{TOR}^2 & \text{for } Z_{TOR} \leq 10 \text{ km} \\ 0 & \text{for } Z_{TOR} > 10 \text{ km} \end{cases} \tag{4.14}$$

$$T_{HW5}(Ry_0) = \begin{cases} 1 & \text{for } Ry_0 \leq Ry_1 \\ 1 - 0.2\ (Ry_0 - Ry_1) & \text{for } Ry_1 < Ry_0 < Ry_1 + 5 \\ 0 & \text{for } Ry_0 \geq Ry_1 + 5, \end{cases} \tag{4.15}$$

in which

$$
\begin{aligned}
Ry_1 &= R_x\, tan(20) \\
a_2 &= 0.2 \\
h_1 &= 0.25 \\
h_2 &= 1.5 \\
h_3 &= -0.75 \\
R_1 &= W cos(dip) \\
R_2 &= 3R_1.
\end{aligned}
\tag{4.16}
$$

Tamp1.5 scaling

$T_{amp1.5}$ is defined as the shortest period at which the normalized spectrum, $PSA(T)/PGA$, reaches 1.5 (Largent et al. (2025)). The $\delta_{Tamp1.5}$ is the residual (in LN units) of a site-specific estimate of $T_{amp1.5}$ relative to a $T_{amp1.5}$ GMM (e.g., Largent et al., 2025). The $\delta T_{amp1.5}$ is related to the difference in the short-period spectral content of the ground motion compared to the median short-period spectral content for a given magnitude, distance, and site condition. A negative $\delta T_{amp1.5}$ indicates that the site-specific spectral content is richer in high-frequency contents, similar to a lower than average κ_0 value. Therefore, we use the $\delta T_{amp1.5}$ as a proxy for differences between the site-specific κ_0 and the median κ_0 for this given site condition.

Because we treat $\delta T_{amp1.5}$ as a proxy for κ_0, we limit the ground-motion scaling due to the $\delta T_{amp1.5}$ parameter to short-periods on rock sites ($V_{S30} > 600$ m/s). The scaling is then tapered to zero for V_{S30} = 450 m/s. The $\delta T_{amp1.5}$ scaling is given by:

$$
f_{Tamp1.5} = \begin{cases} c_{12}(f)\delta_{Tamp1.5} & \text{for } V_{S30} > 600 \text{ m/s} \\ \frac{ln(V_{S30}/450)}{ln(600/450)} c_{12}(f)\delta_{Tamp1.5} & \text{for } 450 \leq V_{S30} \leq 600 \text{ m/s} \\ 0 & \text{for } V_{S30} < 450 \text{ m/s,} \end{cases}
\tag{4.17}
$$

Static stress drop scaling

A parameter that is a proxy for the static-stress-drop, given by $M - 4 - log_{10}(RupArea)$, was included in the EAS GMM. An evaluation of the scaling of the δB with this static-stress-drop proxy found a very weak correlation, even at long periods (e.g., T=10 sec). At very long periods, the ground motion should be related to the average displacement on the rupture. The very weak correlation suggests that there may be issues with the estimates of the rupture areas or with a tradeoff in the scaling of the δB with other parameters in the GMM. Therefore, we do not include this term in the current PSA model. We will revisit the static-stress-drop scaling as part of the evaluation of the PSA GMM.

5 REGRESSION

The regression is conducted in three main steps.

- Step 1: Estimate the coefficients for the finite-fault term using the median PSA values from the converted EAS median GMM.
- Step 2: Estimate the coefficients for the base GMM using the median PSA values converted from the EAS median GMM.
- Step 3: Estimate the coefficients for the HW scaling and $T_{amp1.5}$ scaling using the empirical PSA data

For most of the terms, we do not smooth the period-dependence of the coefficients. We plan to smooth the coefficients after completing the comparisons of our GMM with the other NGA-W3 GMMs and making any revisions to our GMM based on the feedback from these comparisons.

In the development of the EAS GMM, the global data was used to constrain the coefficients related to the large-magnitude and short-distance scaling. The region-specific scaling was based on the WUS subset. The use of the global and regional datasets is summarized in 5.1.

5.1 STEP 1: FINITE-FAULT TERM

The functional form for the finite-fault term, $FF(M)$, given by equation (4.4) has two coefficients: b_2 and b_3. To estimate the b_2 and b_3 coefficients, we first fit the distance scaling of the PSA values converted from the EAS values separately for each magnitude to estimate an event-specific finite-fault term, denoted FF_e:

$$f_e(R_{RUP}) = c_4\, ln\,(R_{RUP} + FF_e) + (-0.5 - c_4) ln R_1 + c_7(M) R_{RUP}. \tag{5.1}$$

Figure 5.1 shows examples of the FF_e values obtained from regression analysis for a small-magnitude event (M4) and a large-magnitude event (M7).

The resulting magnitude dependence of the FF_e values for a range of spectral periods is shown in Figure 5.2. The event-specific $FF(M)$ estimation for our GMM has a much stronger

magnitude scaling than the CY14 GMM because our $FF(M)$ model sets the finite-fault term to 1 km for a M3 event, whereas the CY14 GMM did not focus on the small-magntiude part of the $FF(M)$ model. The $FF(M)$ for the PSA has slightly stronger magnitude scaling than the $FF(M)$ for EAS GMM. The period-dependent coefficients, b_2 and b_3, are plotted in Figure 5.3 and listed in Table 5.2.

5.2 STEP 2: BASE-MODEL REGRESSION

The period-dependent finite-fault term, FF(M), is held fixed, and the regression is conducted for the base model GMM, without the nonlinear site term, the hanging-wall term, and the $T_{amp1.5}$ term. The base model is given by:

$$\begin{aligned} ln(PSA) =& c_1 + f_m(M) + f_p(M, R_{RUP}) + f_{sl}(V_{S30}) + f_z(Z_{TOR}) + f_{SOF}(F_{RV}, F_{NML}) \\ &+ \delta B_e + \left(\frac{R_{rup} - 50}{100}\right)\delta P_e + \delta S2S_s + \delta WS_{es}. \end{aligned} \tag{5.2}$$

We use the converted PSA dataset to build the base model for the magnitude, distance, Z_{TOR}, style-of-faulting, and linear V_{S30} scaling.

The initial results showed that there was over-saturation for small magnitudes (M4) at short distances (1 km). We imposed a constraint that there is no over-saturation (i.e., the median PSA monotonically increases with magnitude at all distances). Without this constraint, at short periods, the median PSA from M3-M4 events will exceed the median PSA from M5 events at R_{RUP}=1 km. A rupture distance of 1 km for M3-M4 events may be an unlikely scenario, but we impose the no over-saturation constraint on the extrapolation of the GMM to small magnitudes and very short distances.

An additional term, $f_{R2}(M, R_{RUP})(T)$, is incorporated into the base model to avoid over-saturation at small magnitudes:

$$f_{R2}(M, R_{RUP}) = \begin{cases} d_2(M) & \text{for } R_{RUP} \leq 0.1 \text{ km} \\ d_2(M) - 1.9d_2(R_{RUP} - 1) & \text{for } 0.1 \text{ km} < R_{RUP} \leq 2 \text{ km} \\ 0 & \text{for } R_{RUP} > 2 \text{ km}, \end{cases} \tag{5.3}$$

in which

$$d_2(M) = \begin{cases} \text{-0.35} & \text{for } M \leq 6 \\ \text{-0.175 } (M - 8) & \text{for } 6 < M \leq 8 \\ 0 & \text{for } M > 8 \end{cases} \tag{5.4}$$

With this constraint, our modified base GMM is given by:

$$\begin{aligned} ln(PSA) =& c_1 + f_m(M) + f_p(M, R_{RUP}) + f_{sl}(V_{S30}) + f_z(Z_{TOR}) + f_{SOF}(F_{RV}, F_{NML}) \\ &+ f_{R2}(M, R_{RUP}) + \delta B_e + \left(\frac{R_{rup} - 50}{100}\right)\delta P_e + \delta S2S_s + \delta WS_{es}. \end{aligned} \tag{5.5}$$

The coefficients for the base model are shown in Figures 5.3 - 5.8 and are listed in Tables 5.2 and 5.3. The period-dependence of the coefficients is smoothed for the SOF and HW scaling. The other coefficients have not been smoothed. The coefficients will be smoothed after completing the evaluation of the current GMM by comparing the scaling with that for GMMs from other developers in the NGA-W3 project.

5.3 STEP 3: ADDITIONAL TERMS

With the base model coefficients fixed, we add the fixed non-linear site term, f_{snl}, to the GMM and compute the PSA residuals from the empirical PSA data set (not from the converted EAS GMM). Using these residuals, we first estimate the HW scaling and then estimate the $T_{amp1.5}$ scaling.

5.3.1 Hanging-Wall Scaling

To have enough data to constrain the HW effects, the HW scaling is estimated using the residuals of a subset of the global dataset. The sampling of sites with potential for significant HW effects is given in Table 5.4, which lists events with M >6 and with the net HW taper (product of the five HW tapers in equation 4.10) greater than 0.1.

For this evaluation of the HW coefficient, c_{13}, the residuals are computed relative to the WUS PSA GMM because we have not yet developed a global PSA GMM.

The HW coefficient, c_{13}, is shown in Figure 5.7. The c_{13} coefficient from the expanded dataset is similar to the value from ASK14, indicating that using the residuals from the global dataset computed relative to the WUS PSA GMM leads to reasonable results. The c_{13} coefficients for the HW term are listed in Table 5.3.

The HW coefficients for the PSA data are similar to the HW coefficients for the EAS GMM, but there is slightly weaker scaling at short periods.

5.3.2 Tamp1.5 Scaling

We compute the residuals for the WUS data, including the HW scaling term. The $T_{amp1.5}$ scaling is estimated from the new residuals.

The period dependence of the coefficients for the $T_{amp1.5}$ scaling is shown in Figure 5.8. The $T_{amp1.5}$ is intended to be a proxy for κ_0, which only affects the short-period range of the response spectrum. Therefore, the coefficients are set to zero for periods greater than 0.2 sec. The strongest scaling is at a period of 0.07 sec, which reflects the shift in the peak of the PSA spectrum to lower periods for low-kappa sites. The coefficients for $T_{Amp1.5}$ are listed in Table 5.3.

Table 5.1: Summary of coefficients determined by global versus regional datasets

parameters	Global	WUS	Scaling
c1	-	o	constant term, from PSA data
c2	o	-	large magnitude scaling from converted EAS GMM
c3	-	o	small magnitude scaling from converted EAS GMM
cm	o	-	midpoint of the magnitude transition, from converted EAS GMM
cn	o	-	width of the magnitude transition, from converted EAS GMM
c4	o	-	geometric spreading, scaling from converted EAS GMM
b1	o	-	finite-fault effect, scaling from converted EAS GMM
b2	o	-	finite-fault effect, scaling from converted EAS GMM
b3	o	-	finite-fault effect, scaling from converted EAS GMM
c7	-	o	linear R, scaling from converted EAS GMM
c8	-	o	V_{S30}, scaling from converted EAS GMM
c9	o	-	Z_{TOR}, scaling from converted EAS GMM
c10n	-	o	SOF scaling from PSA data
c10r	-	o	SOF scaling from PSA data
c12	-	o	Tamp1.5 scaling from PSA data
c13	o	-	Hanging-wall scaling from PSA data

Table 5.2: Coefficients for the median PSA GMM.

Period(s)	c_1	c_2	c_3	c_4	b_2	b_3	cm_st0	cn_st0	c_{70}	c_{72}
0.01	-2.178	1.280	1.616	-1.700	0.577	-0.042	5.207	16.500	-0.0129	-0.0086
0.02	-2.153	1.280	1.606	-1.700	0.573	-0.042	5.207	16.500	-0.0130	-0.0087
0.025	-2.139	1.280	1.582	-1.700	0.566	-0.043	5.207	16.500	-0.0134	-0.0087
0.03	-2.102	1.280	1.569	-1.700	0.560	-0.044	5.207	16.500	-0.0136	-0.0089
0.05	-1.852	1.280	1.553	-1.700	0.552	-0.043	5.215	16.500	-0.0144	-0.0094
0.075	-1.388	1.280	1.563	-1.700	0.581	-0.035	5.220	14.582	-0.0149	-0.0105
0.1	-1.114	1.300	1.593	-1.700	0.626	-0.026	5.223	12.649	-0.0149	-0.0112
0.15	-0.868	1.350	1.728	-1.709	0.668	-0.024	5.228	10.311	-0.0140	-0.0118
0.2	-0.844	1.400	1.864	-1.679	0.675	-0.028	5.231	8.944	-0.0130	-0.0116
0.25	-0.965	1.400	1.973	-1.666	0.671	-0.031	5.233	7.999	-0.0118	-0.0109
0.3	-1.111	1.400	2.034	-1.666	0.662	-0.034	5.234	7.287	-0.0112	-0.0103
0.4	-1.439	1.400	2.180	-1.659	0.640	-0.038	5.237	6.324	-0.0100	-0.0092
0.5	-1.715	1.400	2.264	-1.654	0.629	-0.038	5.247	5.657	-0.0089	-0.0083
0.75	-2.126	1.400	2.364	-1.646	0.630	-0.038	5.592	4.615	-0.0074	-0.0071
1	-2.591	1.400	2.456	-1.627	0.632	-0.026	5.770	4.000	-0.0064	-0.0057
1.5	-3.100	1.401	2.573	-1.590	0.636	-0.018	5.863	3.260	-0.0056	-0.0042
2	-3.336	1.400	2.707	-1.557	0.683	-0.012	5.894	2.828	-0.0055	-0.0038
2.5	-3.467	1.401	2.737	-1.507	0.691	-0.012	5.925	2.530	-0.0055	-0.0035
3	-3.624	1.428	2.759	-1.478	0.700	-0.010	5.936	2.304	-0.0058	-0.0036
5	-4.016	1.482	2.884	-1.430	0.722	-0.011	6.038	1.789	-0.0063	-0.0030
6	-4.214	1.534	2.883	-1.425	0.730	-0.006	6.129	1.629	-0.0061	-0.0023
7.5	-4.469	1.544	2.904	-1.423	0.722	-0.005	6.209	1.454	-0.0059	-0.0015
10	-4.996	1.559	2.901	-1.420	0.710	-0.008	6.280	1.265	-0.0061	-0.0005

Table 5.3: Coefficients for the median PSA GMM (cont).

Period (s)	c_{80}	c_{82}	c_{8b}	c_{8c}	c_{9}	c_{10n}	c_{10r}	c_{13}	c_{12}	$c_{1,adj-Tamp1.5}$
0.01	-0.629	-0.848	0.016	-0.021	0.0413	-0.193	0.023	0.561	-0.247	0.009
0.02	-0.605	-0.822	0.026	-0.022	0.0417	-0.193	0.023	0.561	-0.307	0.023
0.025	-0.576	-0.794	0.034	-0.024	0.0421	-0.193	0.023	0.561	-0.377	0.047
0.03	-0.533	-0.751	0.038	-0.025	0.0428	-0.193	0.023	0.561	-0.488	0.074
0.05	-0.395	-0.605	0.039	-0.028	0.0449	-0.193	0.023	0.561	-0.725	0.078
0.075	-0.304	-0.452	0.037	-0.030	0.0475	-0.193	0.023	0.561	-0.747	0.087
0.1	-0.352	-0.449	0.027	-0.036	0.0478	-0.193	0.023	0.561	-0.513	0.021
0.15	-0.457	-0.557	0.015	-0.040	0.0437	-0.193	0.023	0.561	-0.155	-0.003
0.2	-0.568	-0.678	0.006	-0.044	0.0392	-0.193	0.023	0.561	0.000	-0.019
0.25	-0.704	-0.765	0.002	-0.043	0.0352	-0.193	0.023	0.561	0.000	-0.039
0.3	-0.776	-0.847	0.001	-0.039	0.0326	-0.193	0.023	0.561	0.000	0.000
0.4	-0.924	-0.977	0.003	-0.031	0.0264	-0.193	0.023	0.561	0.000	0.000
0.5	-1.015	-1.058	0.013	-0.015	0.0217	-0.193	0.023	0.561	0.000	0.000
0.75	-1.079	-1.115	0.026	0.005	0.0152	-0.193	0.023	0.562	0.000	0.000
1	-1.163	-1.120	0.041	0.029	0.0103	-0.193	0.023	0.565	0.000	0.000
1.5	-1.168	-1.141	0.055	0.054	0.0042	-0.193	0.023	0.556	0.000	0.000
2	-1.101	-1.121	0.064	0.075	-0.0031	-0.193	0.023	0.514	0.000	0.000
2.5	-1.041	-1.061	0.058	0.084	-0.0080	-0.193	0.023	0.433	0.000	0.000
3	-0.999	-1.024	0.045	0.090	-0.0097	-0.193	0.023	0.288	0.000	0.000
5	-0.857	-0.881	0.032	0.089	-0.0174	-0.193	0.023	0.082	0.000	0.000
6	-0.817	-0.816	0.023	0.079	-0.0176	-0.193	0.023	0.031	0.000	0.000
7.5	-0.762	-0.736	0.016	0.060	-0.0181	-0.193	0.023	0.044	0.000	0.000
10	-0.684	-0.616	0.010	0.033	-0.017	-0.193	0.023	0.040	0.000	0.000

Table 5.4: Number of events and records by region that contribute to the HW scaling: M $>$ 6, $R_{RUP} <$ 35 km, and a net HW taper $\geq$ 0.1.

Region	Number of events	Number of records
CA	8	43
Mexico	2	7
Turkey	7	24
Italy	8	26
Greece	4	4
Montenegro	1	4
Iran	2	3
Taiwan	9	125
Japan	7	53

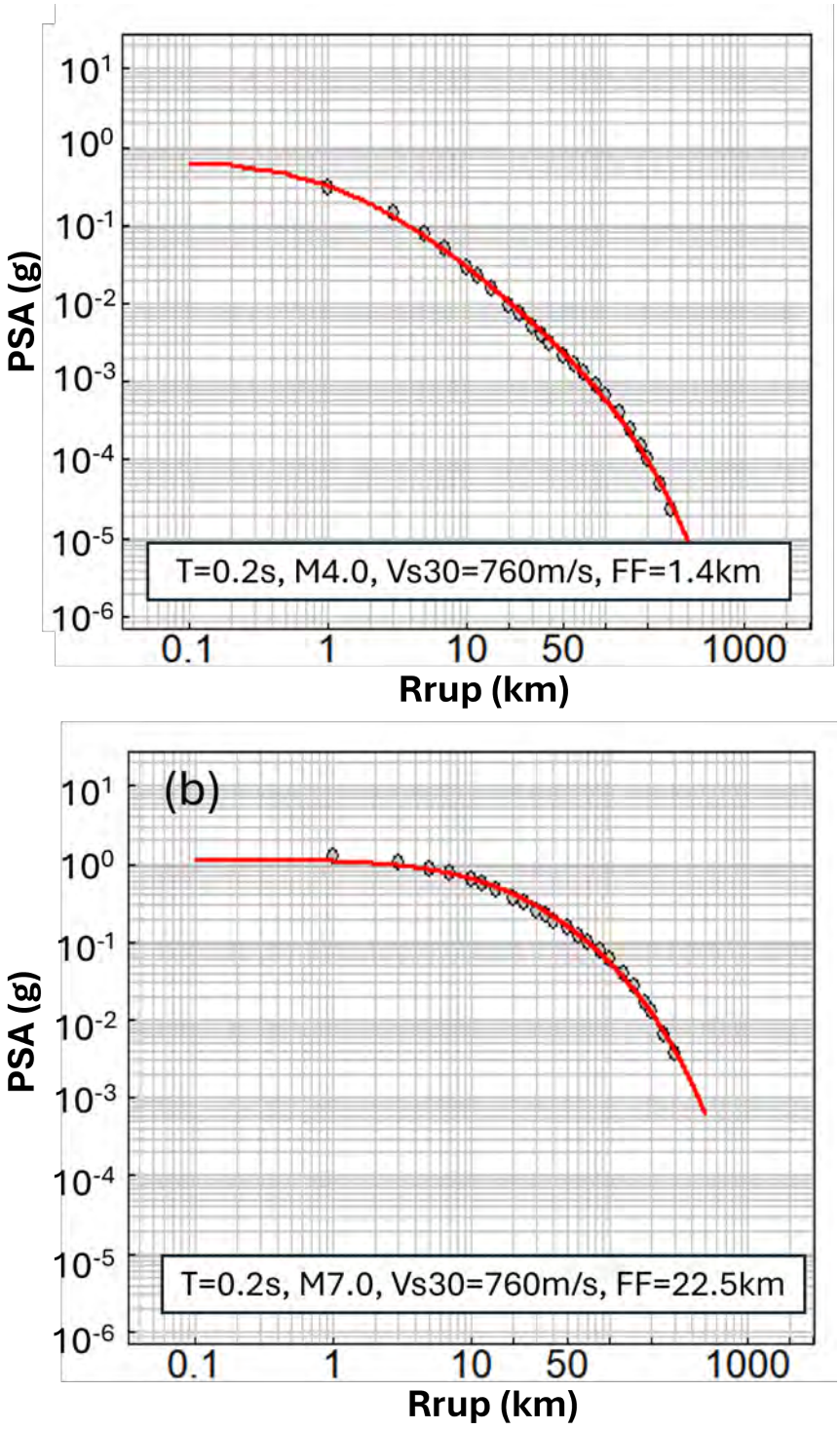


Figure 5.1: Examples for FF_e values based on the PSA from the converted EAS for Vs30 = 760 m/s. (a) M4 and (b) M7.

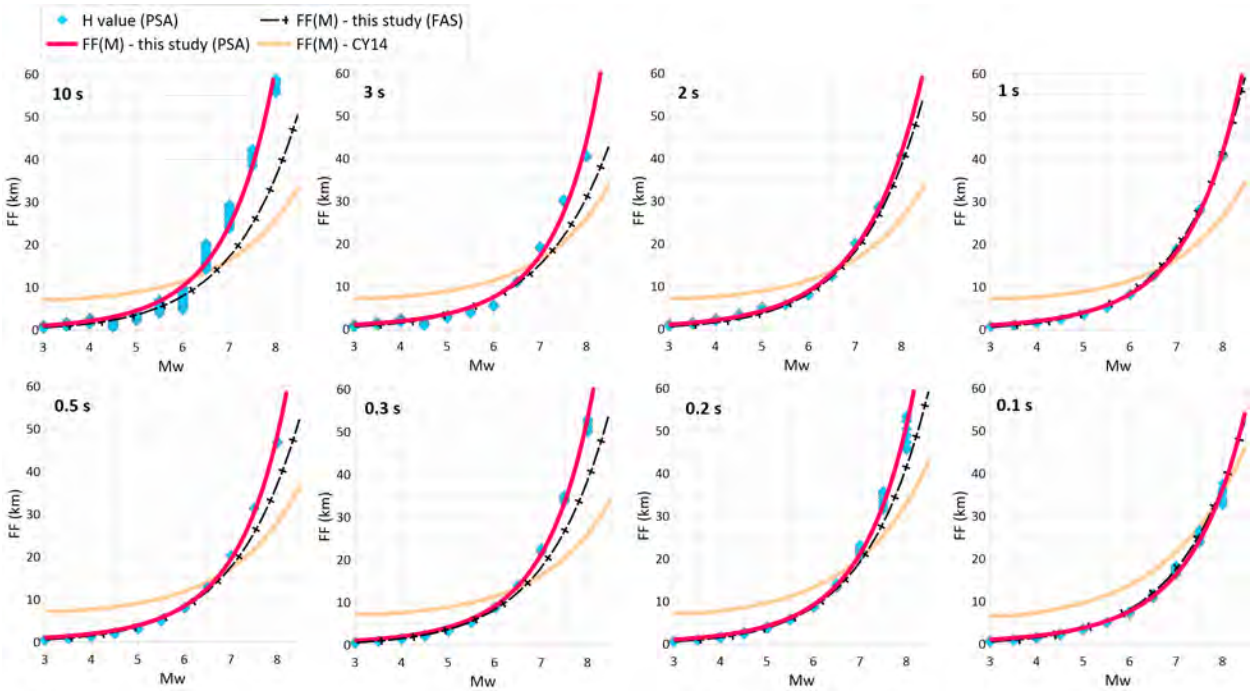


Figure 5.2: Model for the finite-fault term, FF(M).

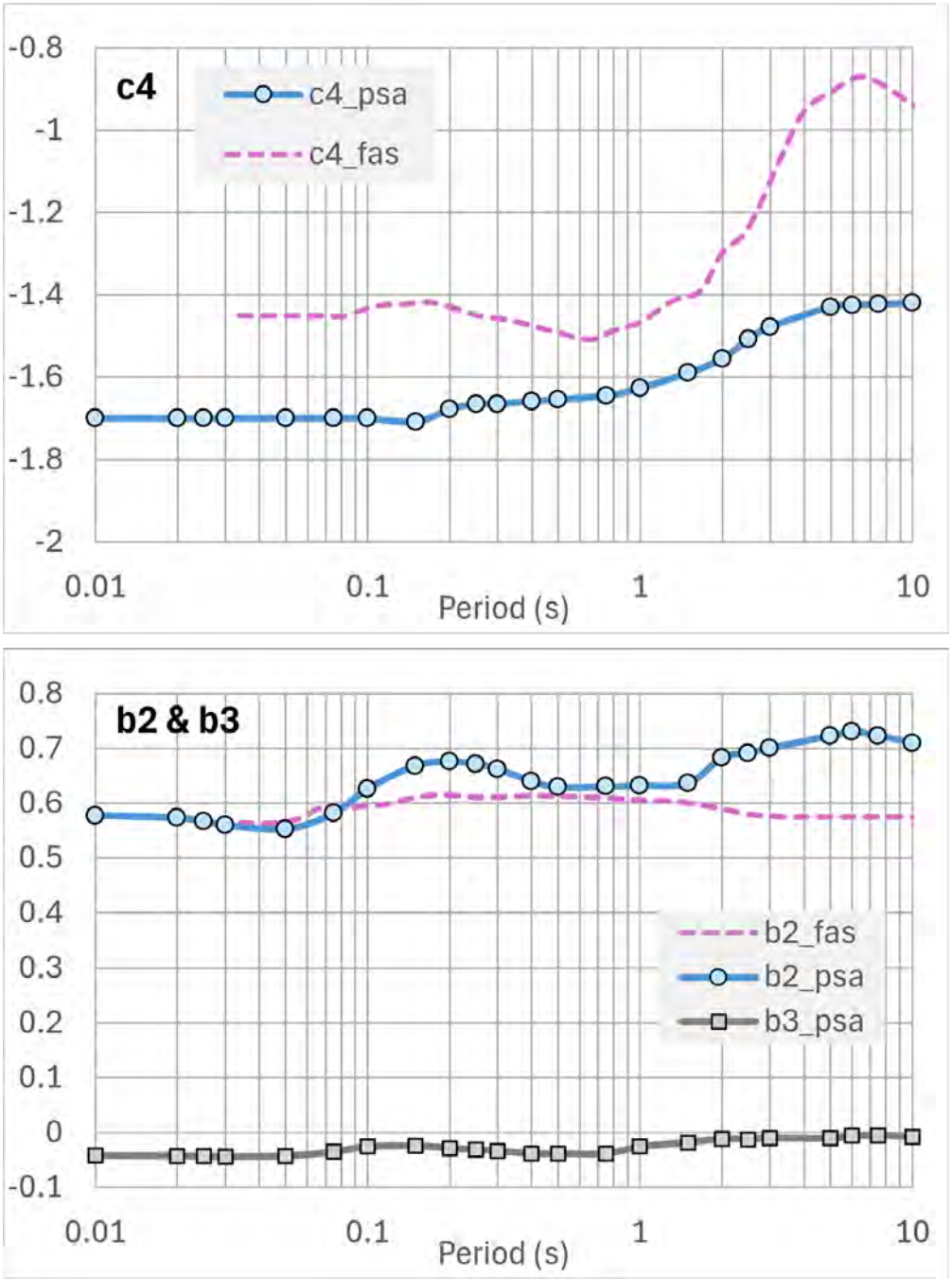


Figure 5.3: Coefficients for the geometrical spreading (c_4) and the finite-fault term ($FF(M)$).

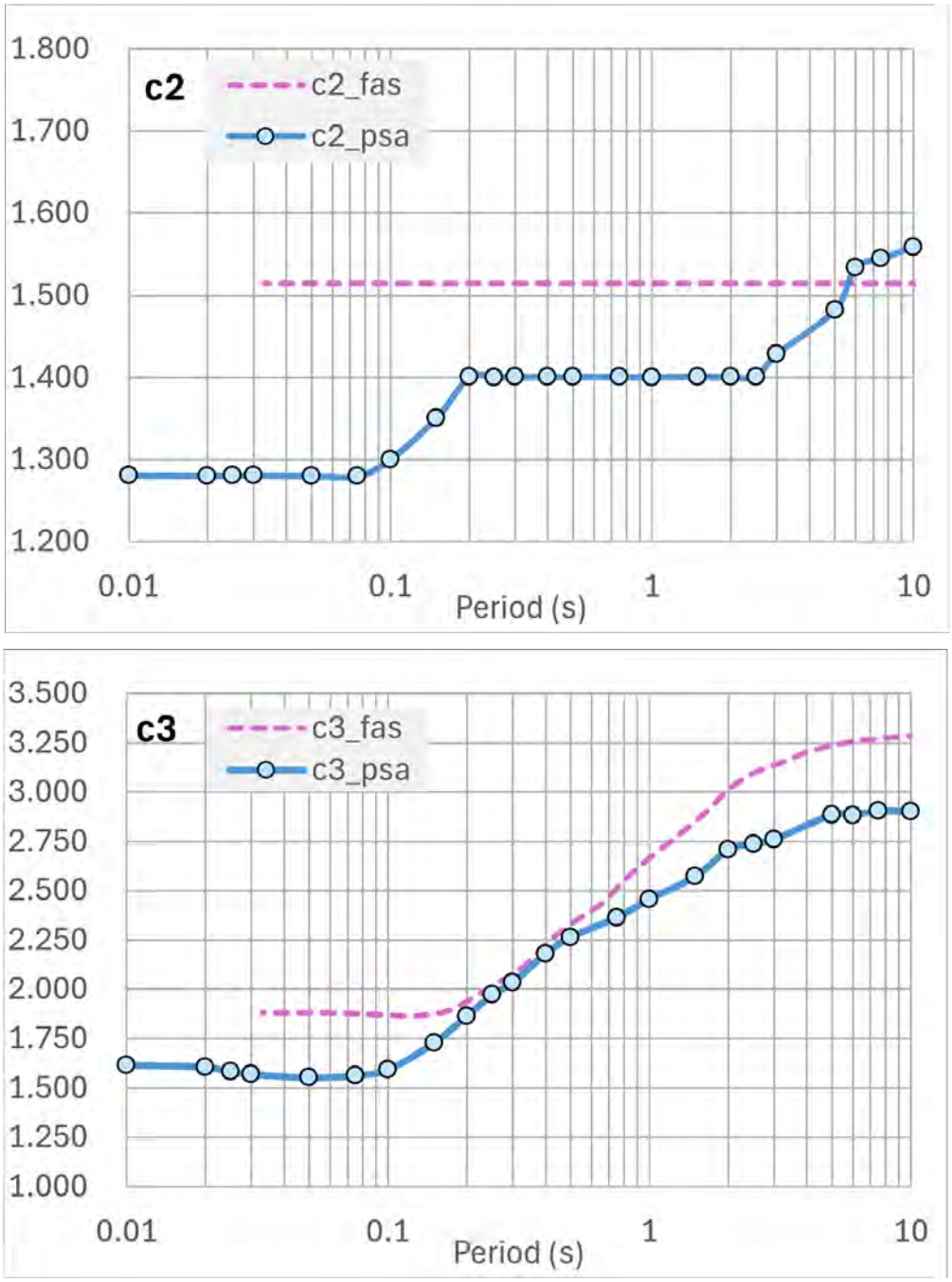


Figure 5.4: Coefficients for the magnitude scaling: c_2 and c_3

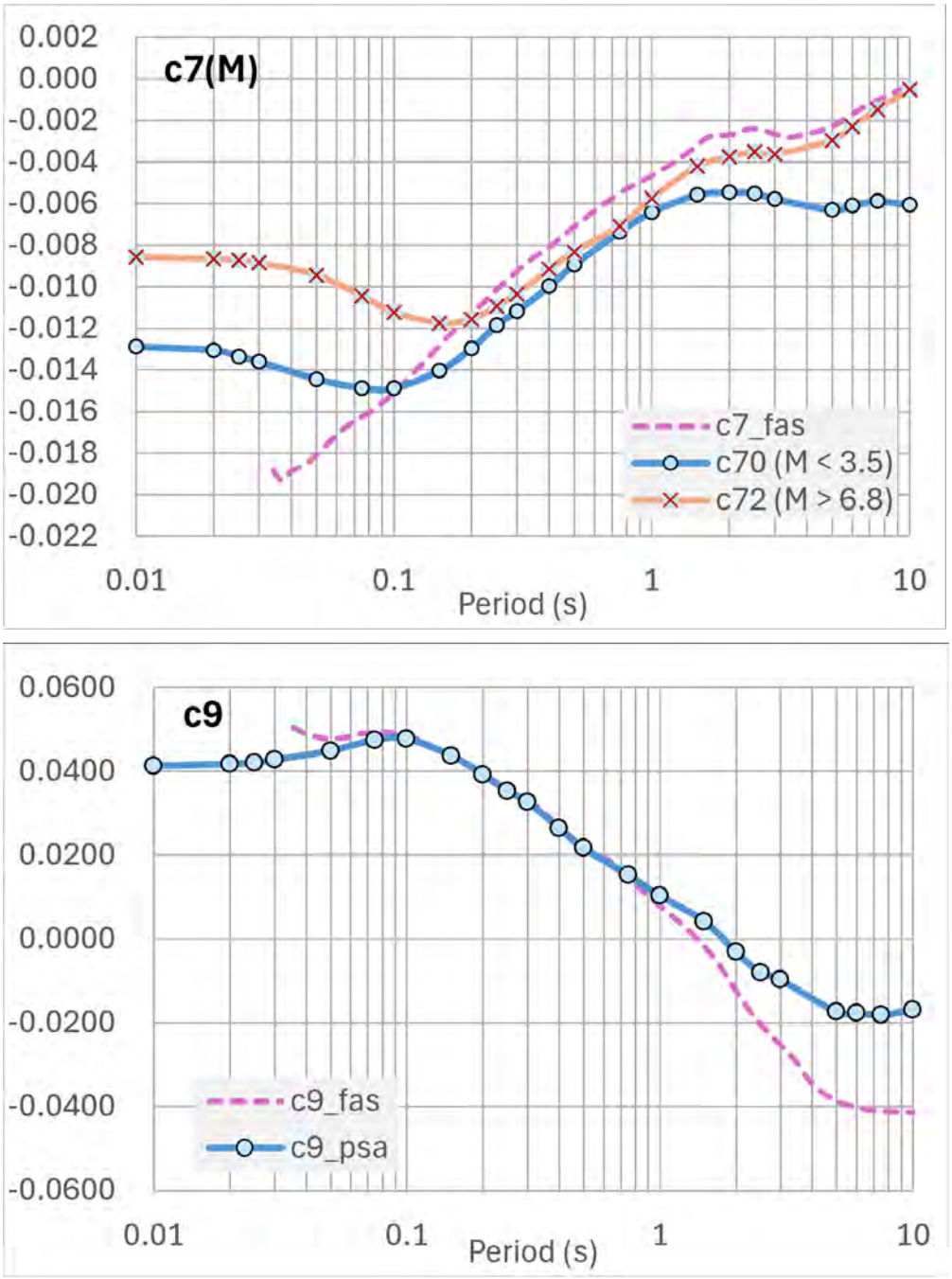


Figure 5.5: Coefficients for the large-distance scaling and the Z_{TOR} scaling : c_7 and c_9

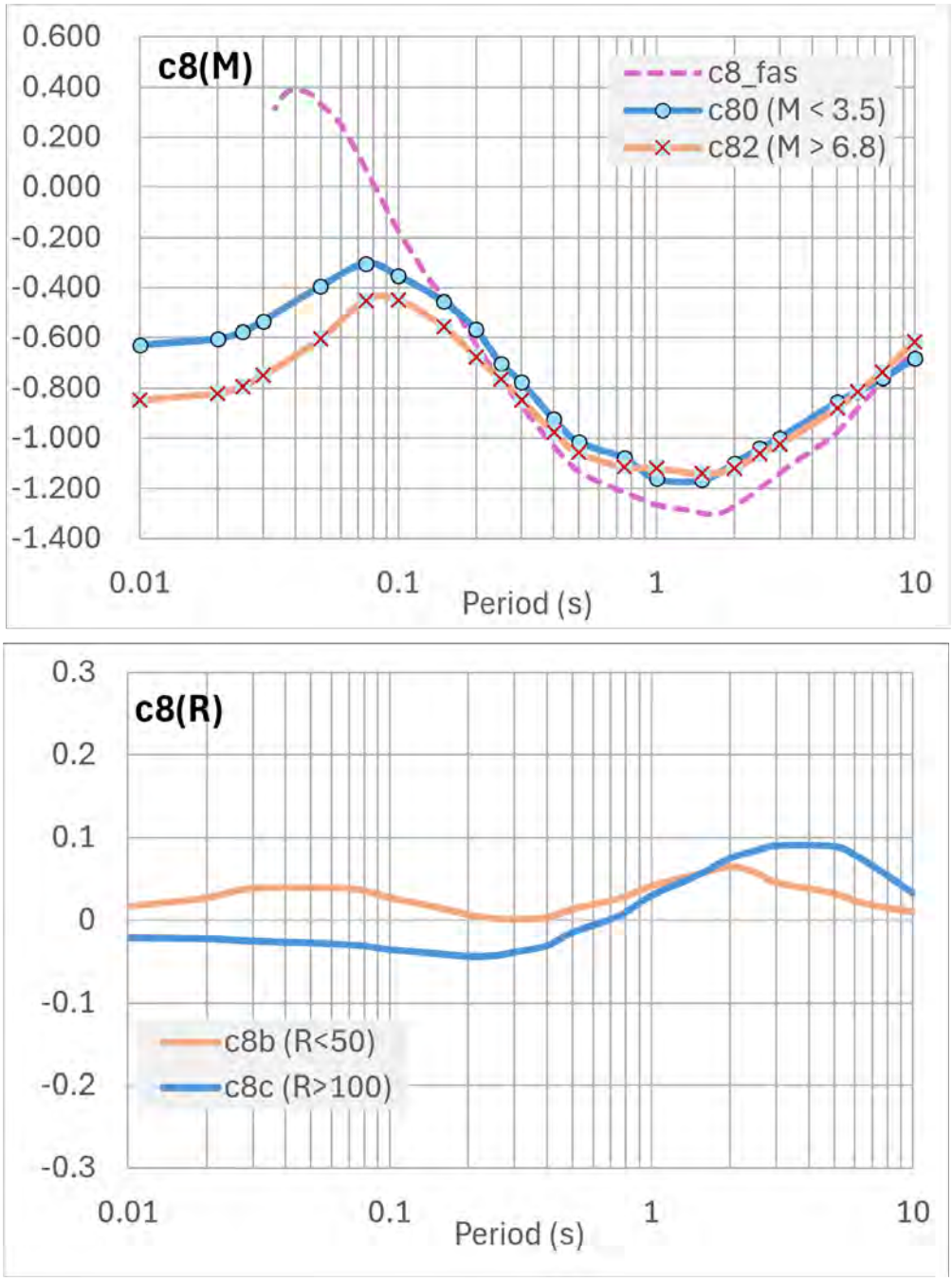


Figure 5.6: Coefficients for the V_{S30} scaling : c_{8R} and c_{8M}

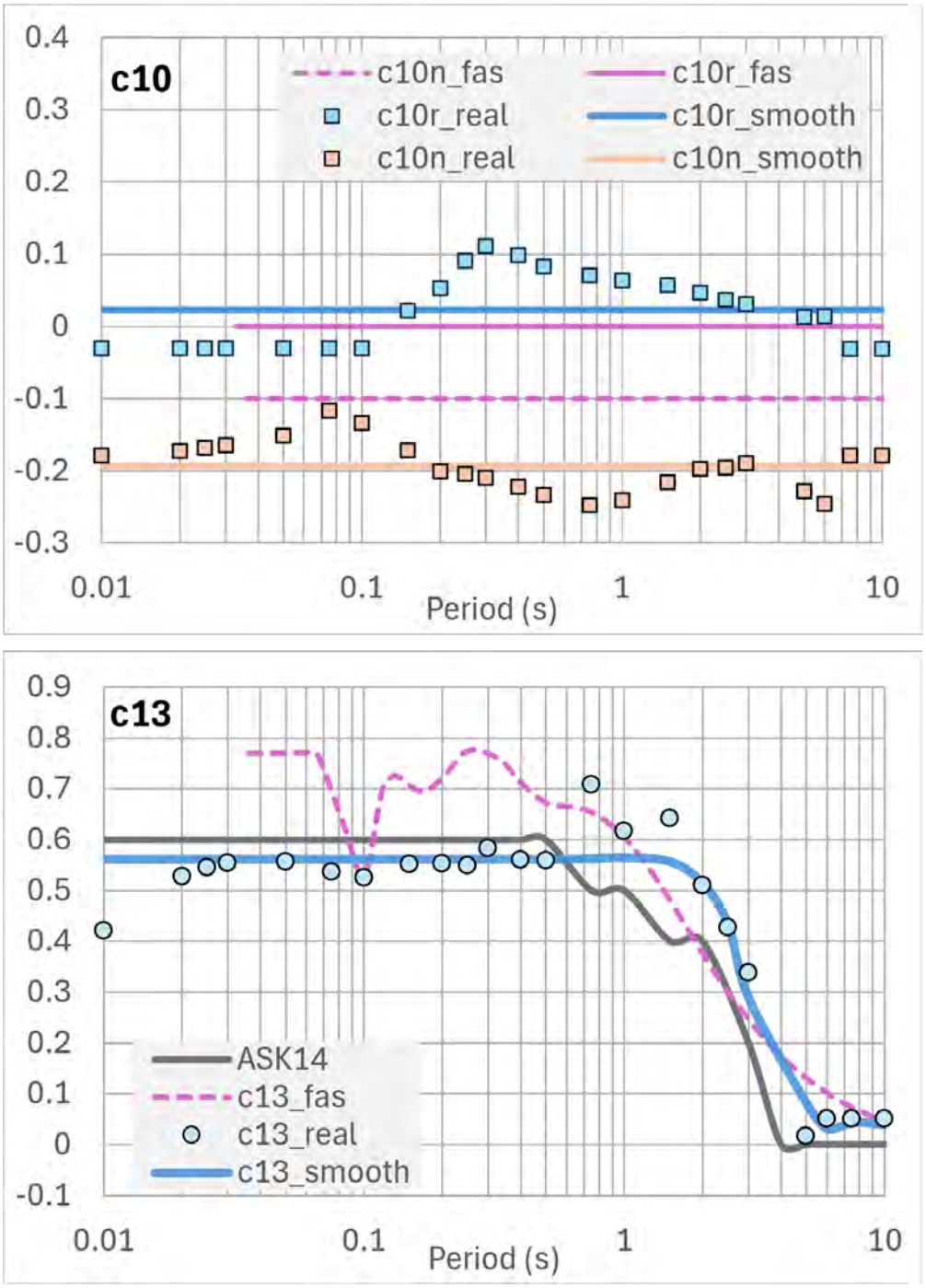


Figure 5.7: Coefficients for the SOF scaling and the HW scaling : c_{10} and c_{13}

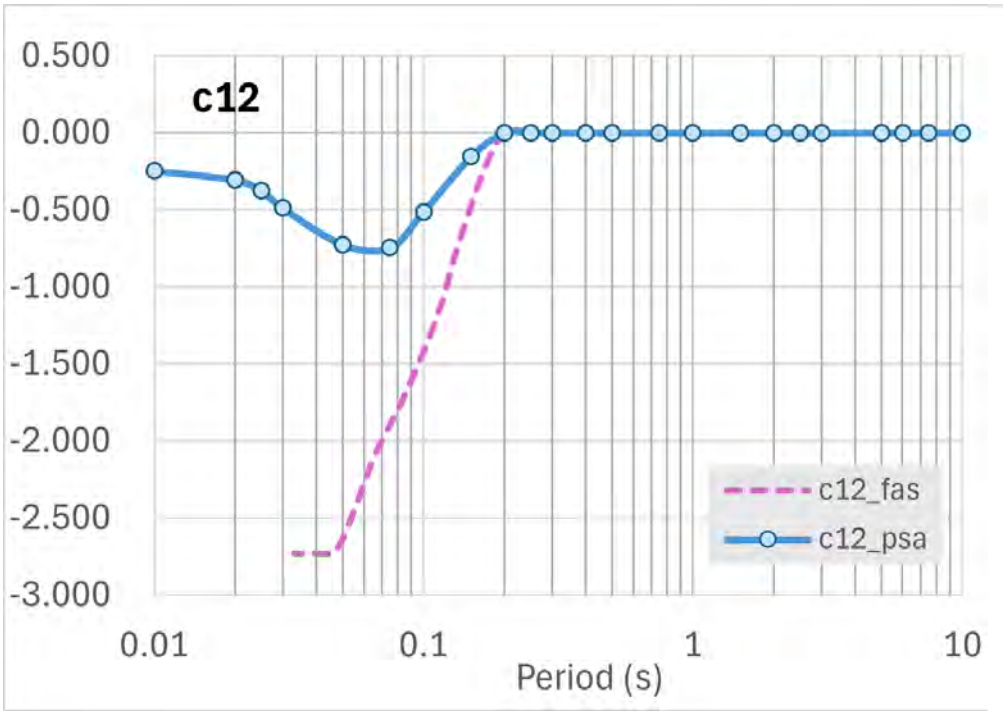


Figure 5.8: Coefficients for the $T_{Amp1.5}$ scaling: c_{12}

6 RESIDUALS

Our PSA GMM includes a new term, $T_{Amp1.5}$. Because this parameter is not standard and may not be available for applications, we compute two sets of residuals: one set with the $T_{Amp1.5}$ term and one set without the $T_{Amp1.5}$ term. Including the $T_{Amp1.5}$ term leads to a small reduction in the standard deviation terms at short periods.

6.1 CORRECTION OF ESTIMATES OF RANDOM EFFECTS

If there is a small number of recordings per event or recordings per site, then the standard deviation of the estimates of the random effects will tend to be smaller than the standard deviation computed as part of the regression (Kuehn et al., 2025). As an example, the standard deviations of the uncorrected random effects are compared to the standard deviations from the regression in Figure 6.1. The standard deviations computed from the point estimates of the random effects are systematically smaller than the standard deviations of the random effects from the regression.

A correction to the random effects developed by Lavrentiadis (2025a) is applied to the point estimates of the random effects. This correction does not change the mean random effect, but it increases the variability of the random effects for point estimates with a small number of recordings per event or site. As an example, the uncorrected and corrected estimates of the three random effects (δB, δP, and $\delta S2S$) for T=0.2 sec and T=1 sec are shown in Figures 6.2 and 6.3, respectively. The size of the correction is shown in the lower row of Figures 6.2 and 6.3. The corrections span a larger range for events and sites with fewer recordings. With the correction, the standard deviations of the corrected event terms are consistent with the standard deviations from the regression, as shown in Figure 6.1.

6.2 RANDOM EFFECTS

The corrected random effects are used to evaluate the magnitude, distance, and V_{S30} dependence of the standard deviations of the random effects.

Event terms

The event terms, δB_e, for periods of 0.05, 0.2, 1.0, and 3.0 sec are shown as functions of magnitude and Z_{TOR} in the top two frames of Figures 6.4 - 6.7. For each period, the binned mean values are shown by the red circles. The δB_e values do not show any systematic trends with magnitude or Z_{TOR} for these four spectral periods, but there is less variability for the larger magnitudes.

Path terms

The path terms, δP_e, are shown as a function of magnitude in the third frames of Figures 6.4 - 6.7 Similar to the event terms, the path terms do not show systematic trends with magnitude.

Site terms

The site terms, $\delta S2S_s$, are shown as a function of V_{S30} in the bottom frames of Figures 6.4 - 6.7 The site terms do not show a trend with V_{S30}. There is an indication of larger variability of the short-period $\delta S2S$ for V_{S30} values between 600 and 700 m/s, but this is not seen for the V_{S30} values greater than 700 m/s.

6.3 WITHIN-SITE RESIDUALS

The within-site residuals, δWS, are shown as a function of rupture distance, for separate magnitude bins in Figures 6.8 - 6.11 for periods of 0.05, 0.2, 1.0, and 3.0 sec. Overall, the mean residuals do not show trends with distance. At short distances ($<$ 2 km) and small magnitudes, there is a positive residual for the short periods. This is related to the over-saturation of the small-magnitude events that was seen in the initial regression, but we constrained the GMM to not allow over-saturation for the small magnitudes.

The non-linear site response scaling based on analytical modeling is evaluated by plotting the within-site residuals as a function of PSA(T) for V_{S30}=800 m/s for soil sites. The residuals are shown in Figures 6.12 and 6.13 for spectral periods of 0.03 and 0.3 sec, respectively. For V_{S30} values between 200-250 m/s and 250-300 m/s, including the non-linear site model reduces the over-prediction for large $PSA_{800}(T)$ values.

For the softer sites with $V_{S30} < 200$ m/s, there are positive residuals for T=0.03 sec at moderate input motions (PSA(T=0.03) of 0.03 g to 0.1 g), indicating that the non-linear site effects from the analytical modeling may be too strong for sites with $V_{S30} < 200$ m/s.

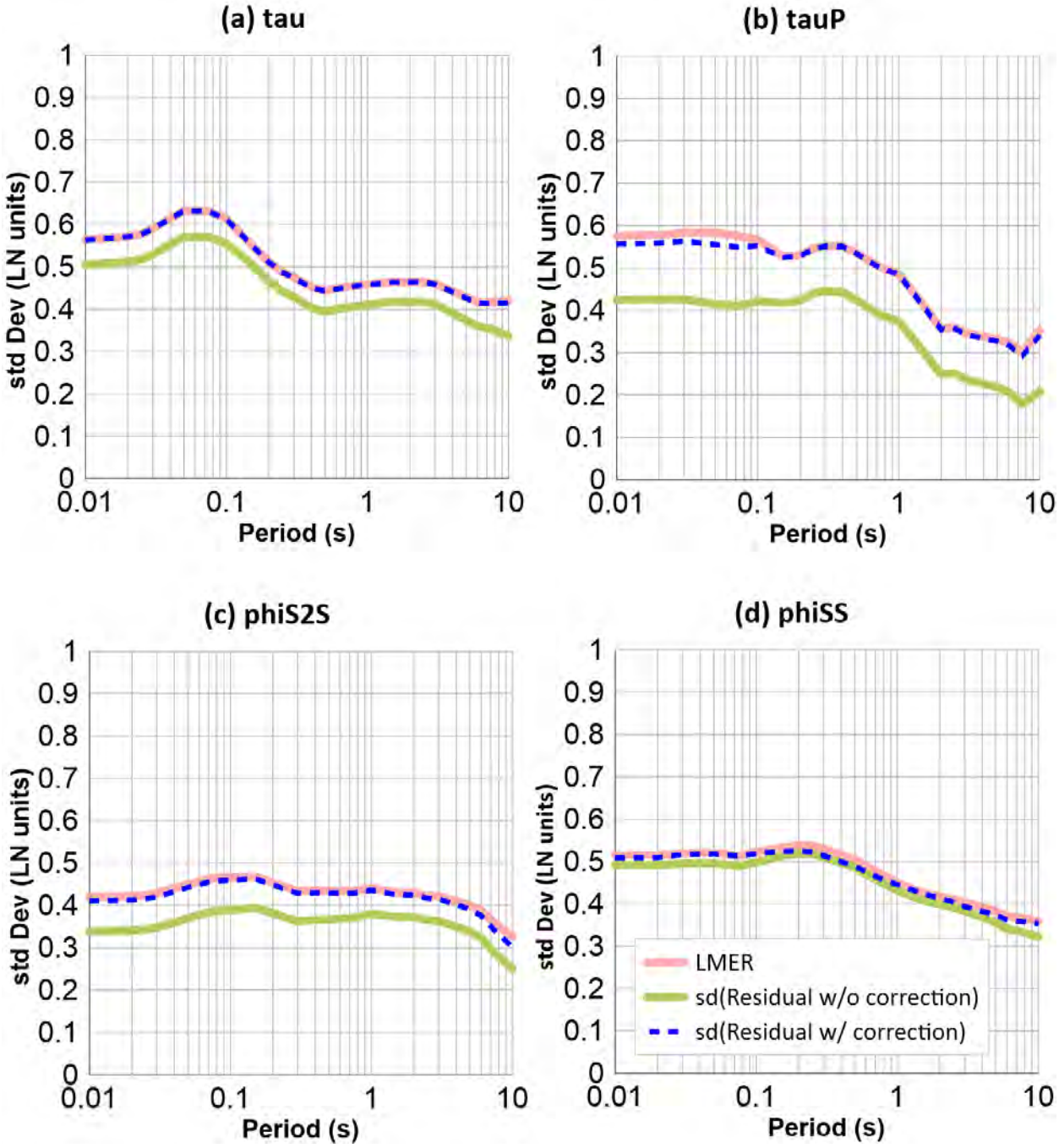


Figure 6.1: τ, τ_P, ϕ_{S2S}, and ϕ_{SS} **as a function of frequency. Pink lines represent the standard deviation from the regression. Green lines and dashed lines are computed from the random effects without and with correction.**

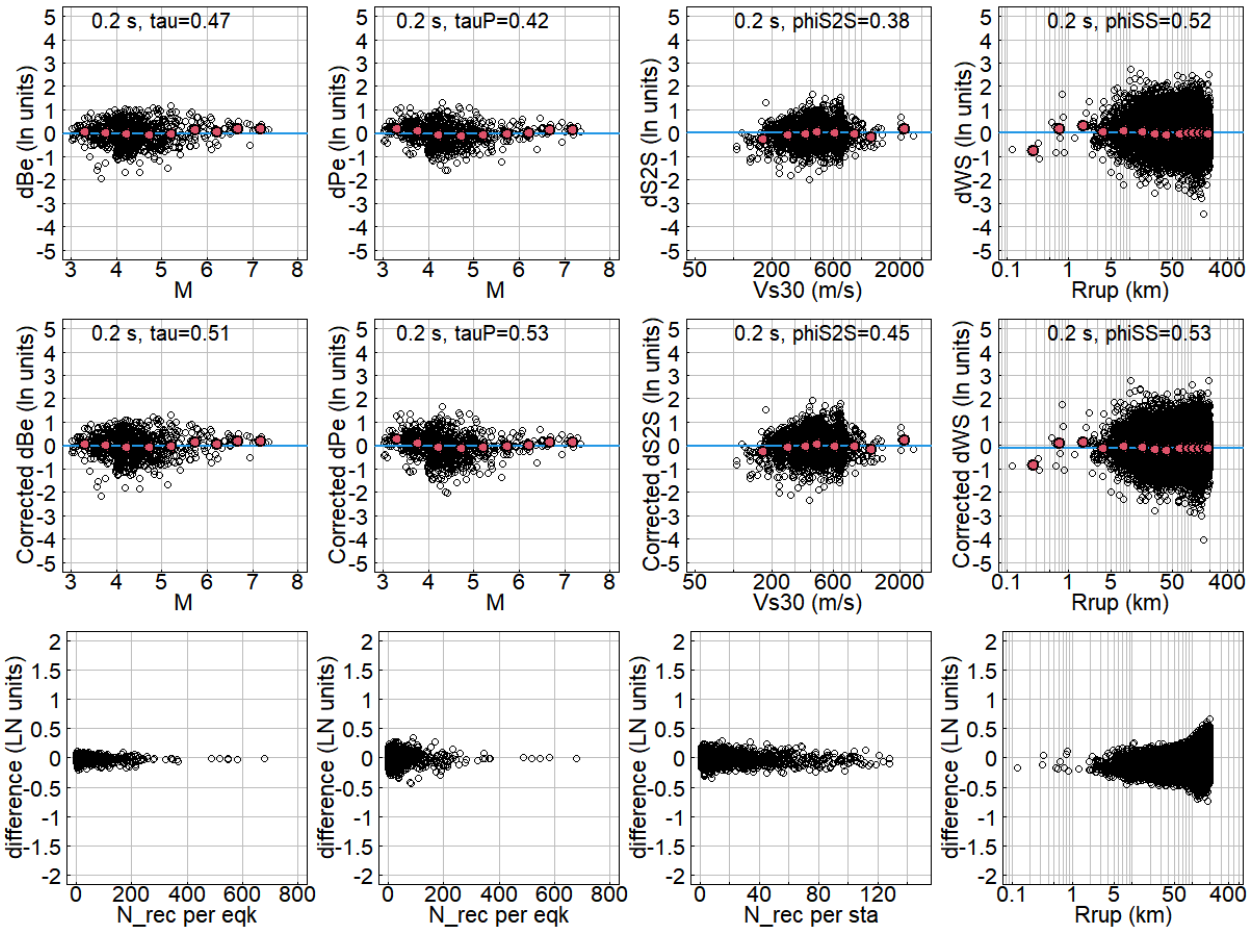


Figure 6.2: Comparison of the raw residuals and corrected residuals for δB, δP, $\delta S2S$, and δWS at 0.2 sec. Upper plots are the raw residuals, middle plots are the corrected residuals, and bottom plots are the difference (corrected residuals - raw residuals) versus the number of records per event/station.

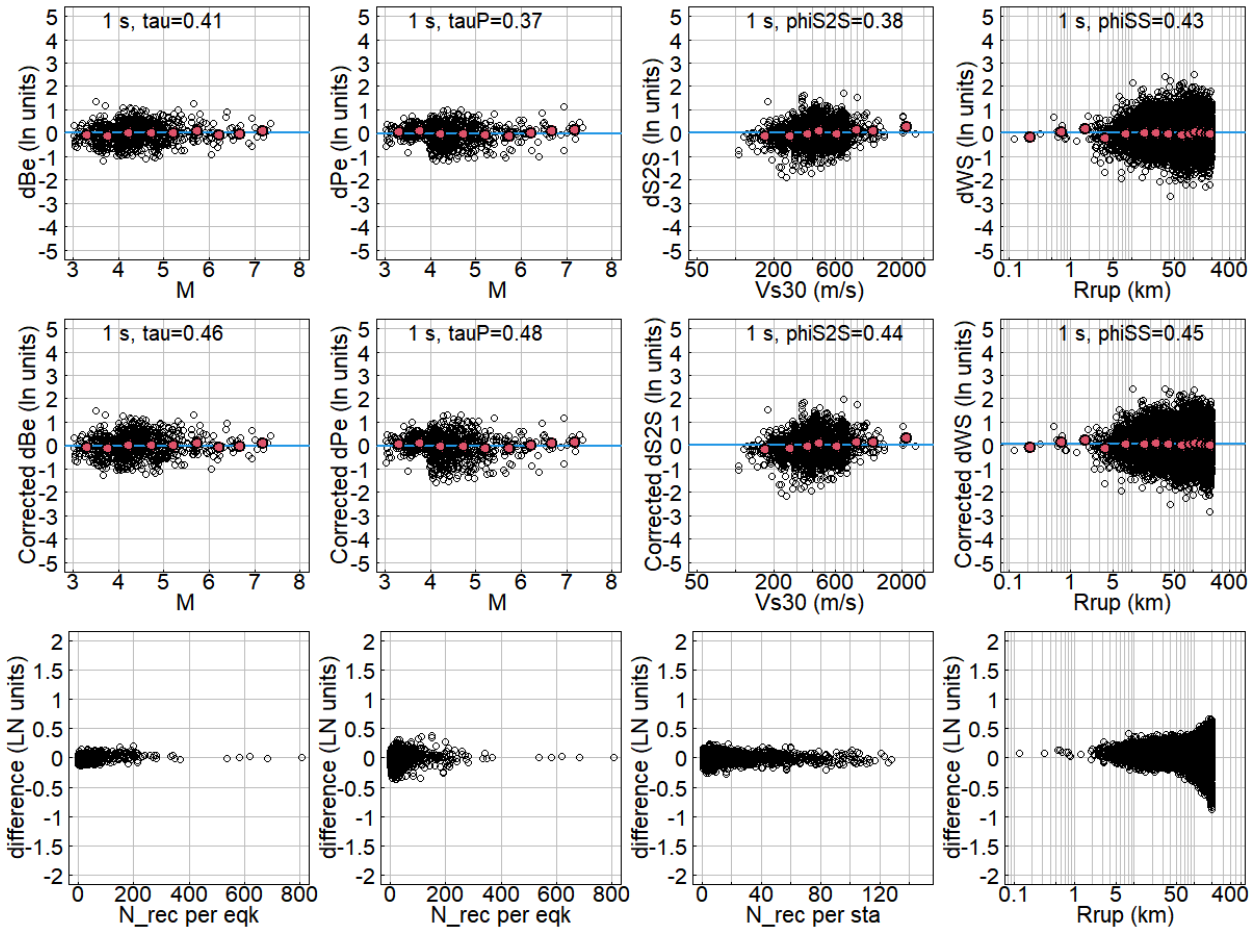


Figure 6.3: Comparison of the raw residuals and corrected residuals for δB, δP, $\delta S2S$, and δWS at 1 sec. Upper plots are the raw residuals, middle plots are the corrected residuals, and bottom plots are the difference (corrected residuals - raw residuals) versus the number of records per event/station.

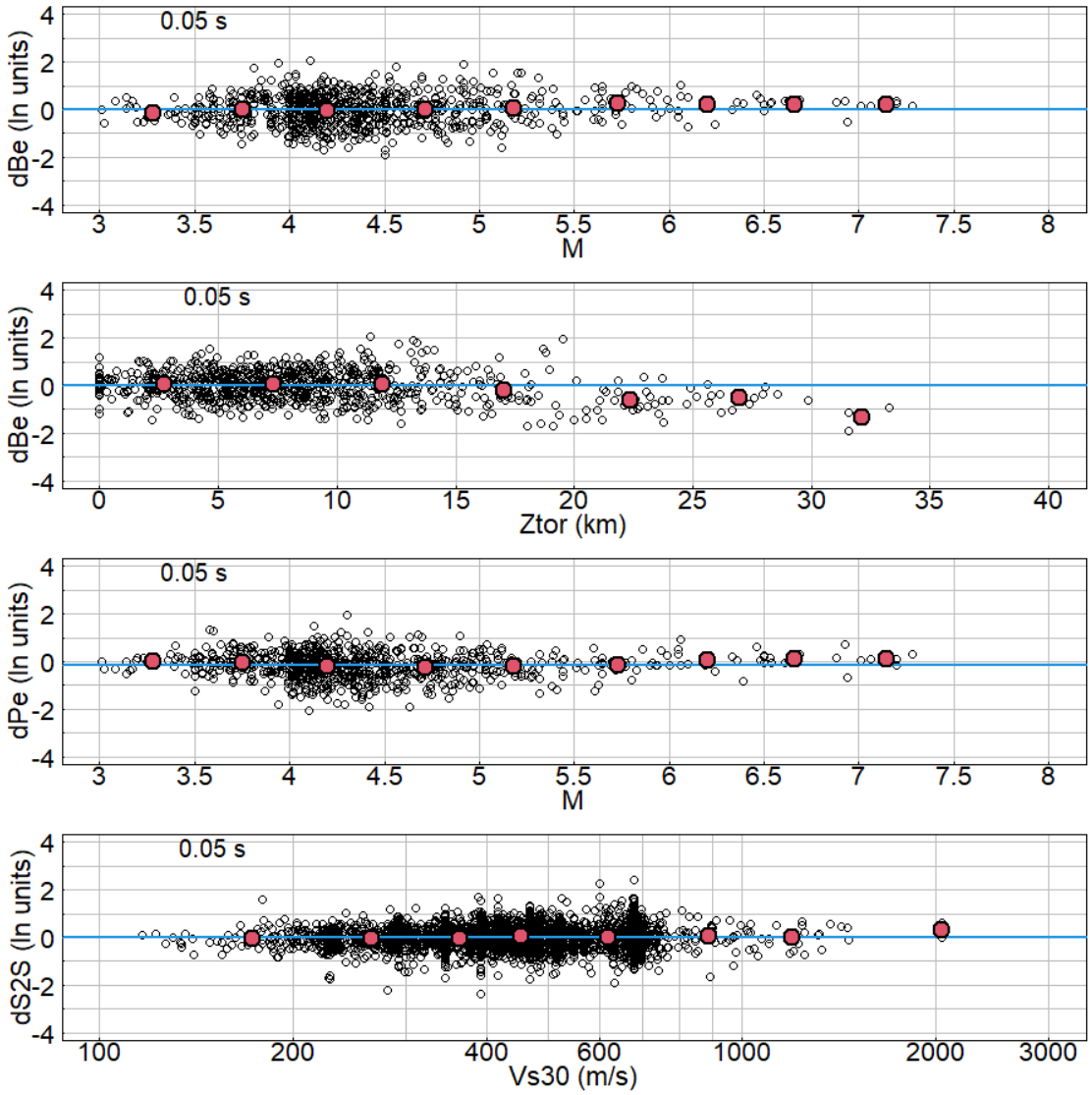


Figure 6.4: (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=0.05 sec.

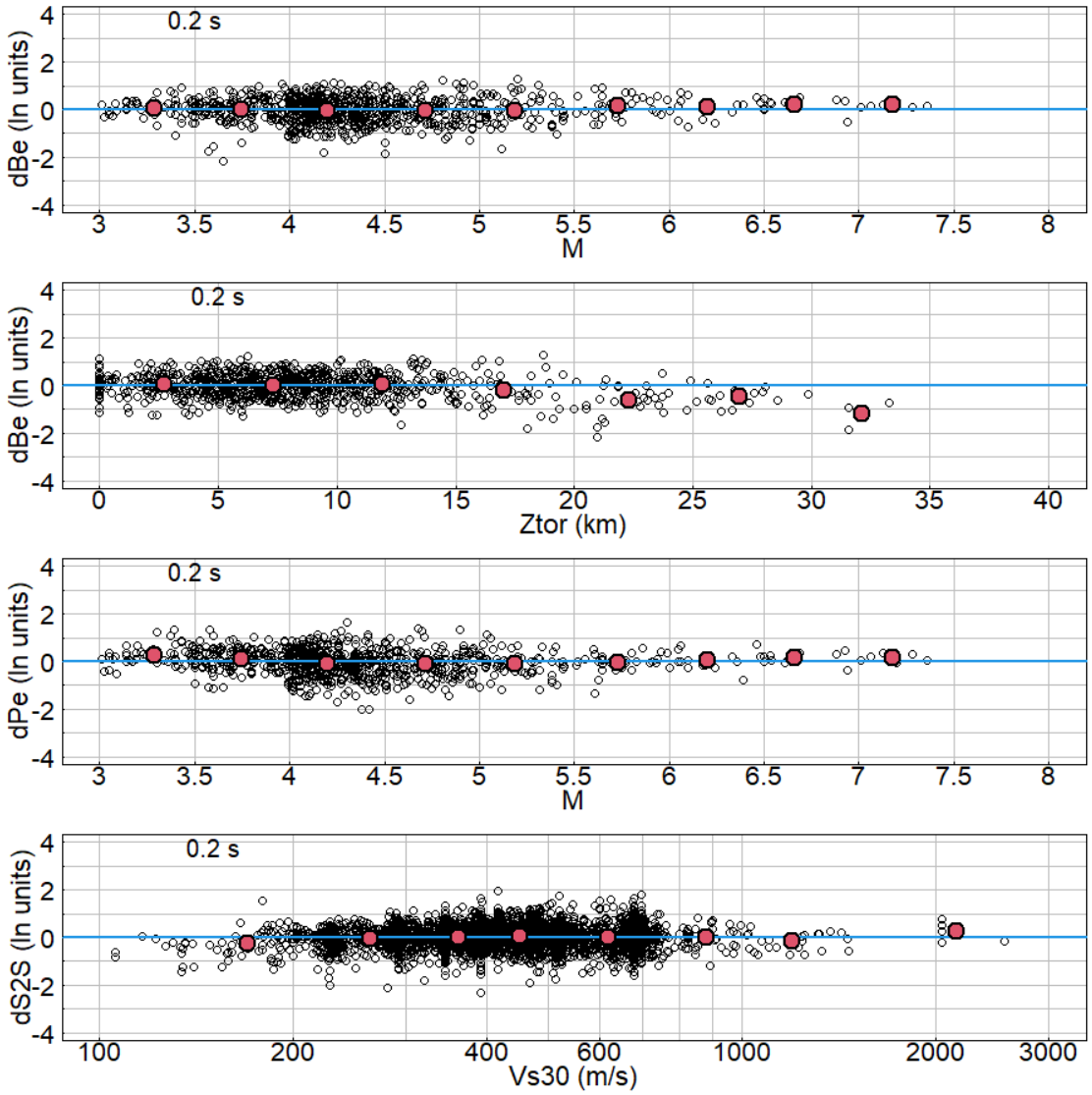


Figure 6.5: (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=0.2 sec.

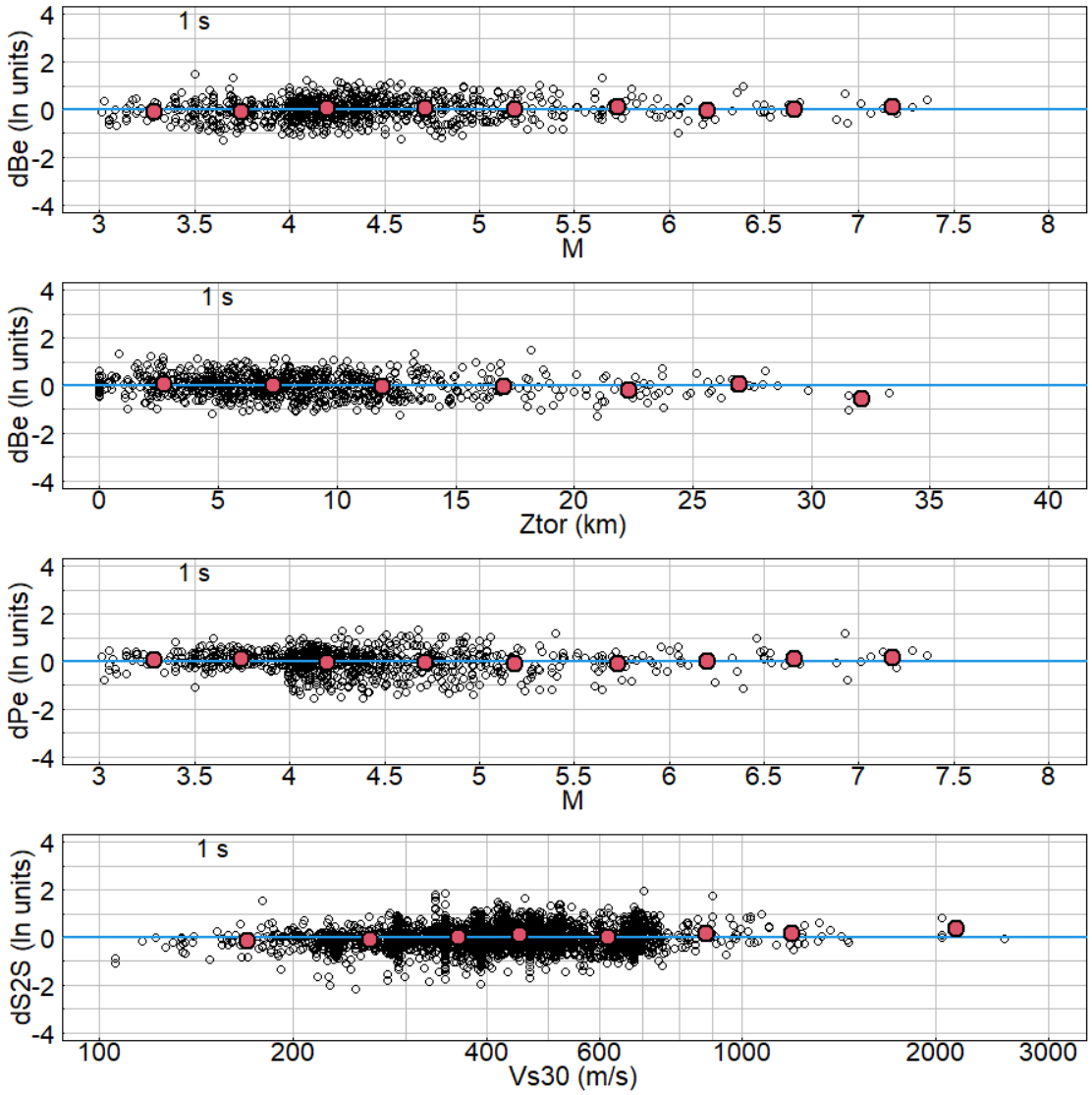


Figure 6.6: (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=1 sec.

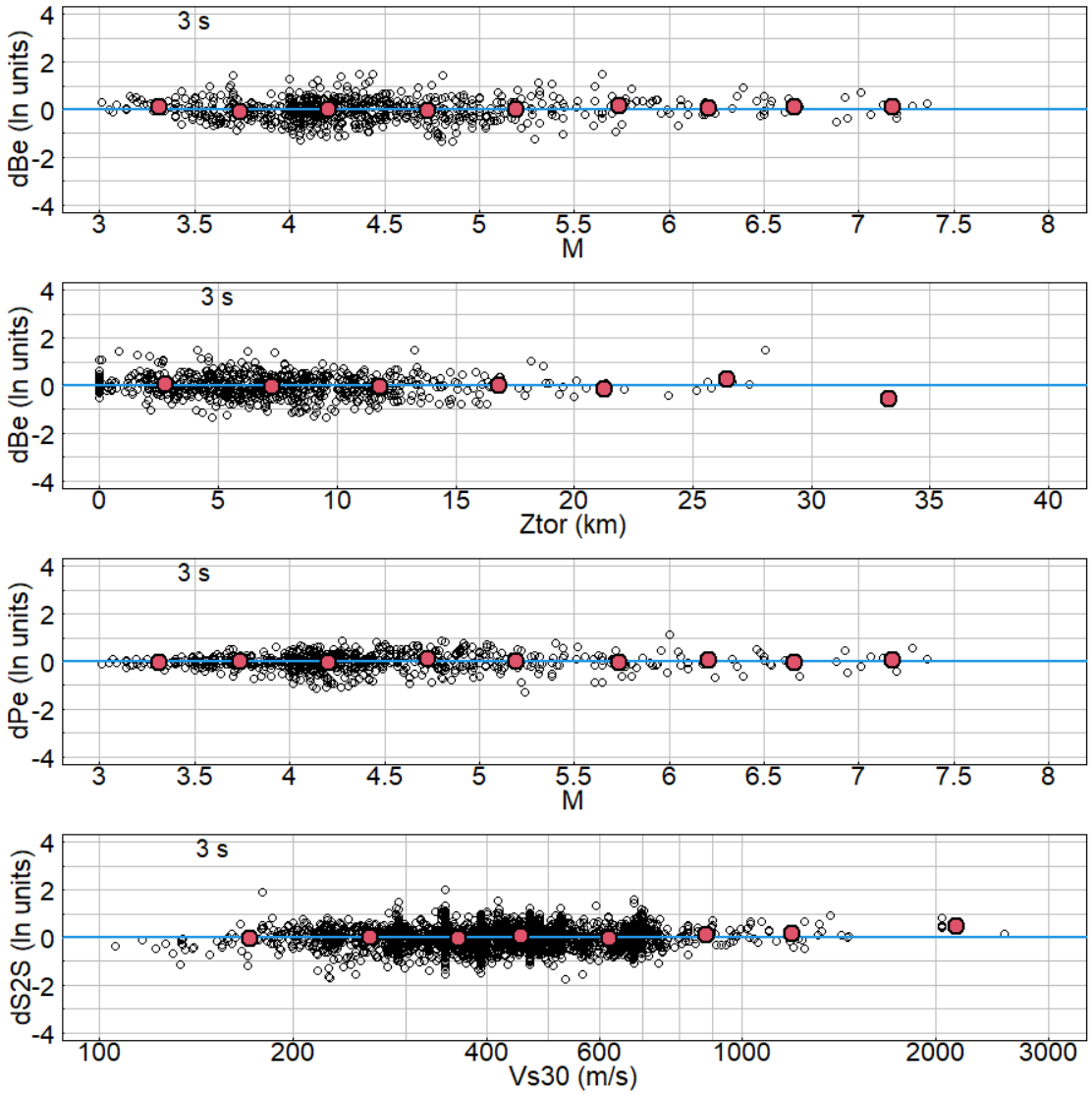


Figure 6.7: (a) δB versus M, (b) δB versus Z_{TOR}, (c) δP versus M, and (d) $\delta S2S$ versus V_{S30} for T=3 sec.

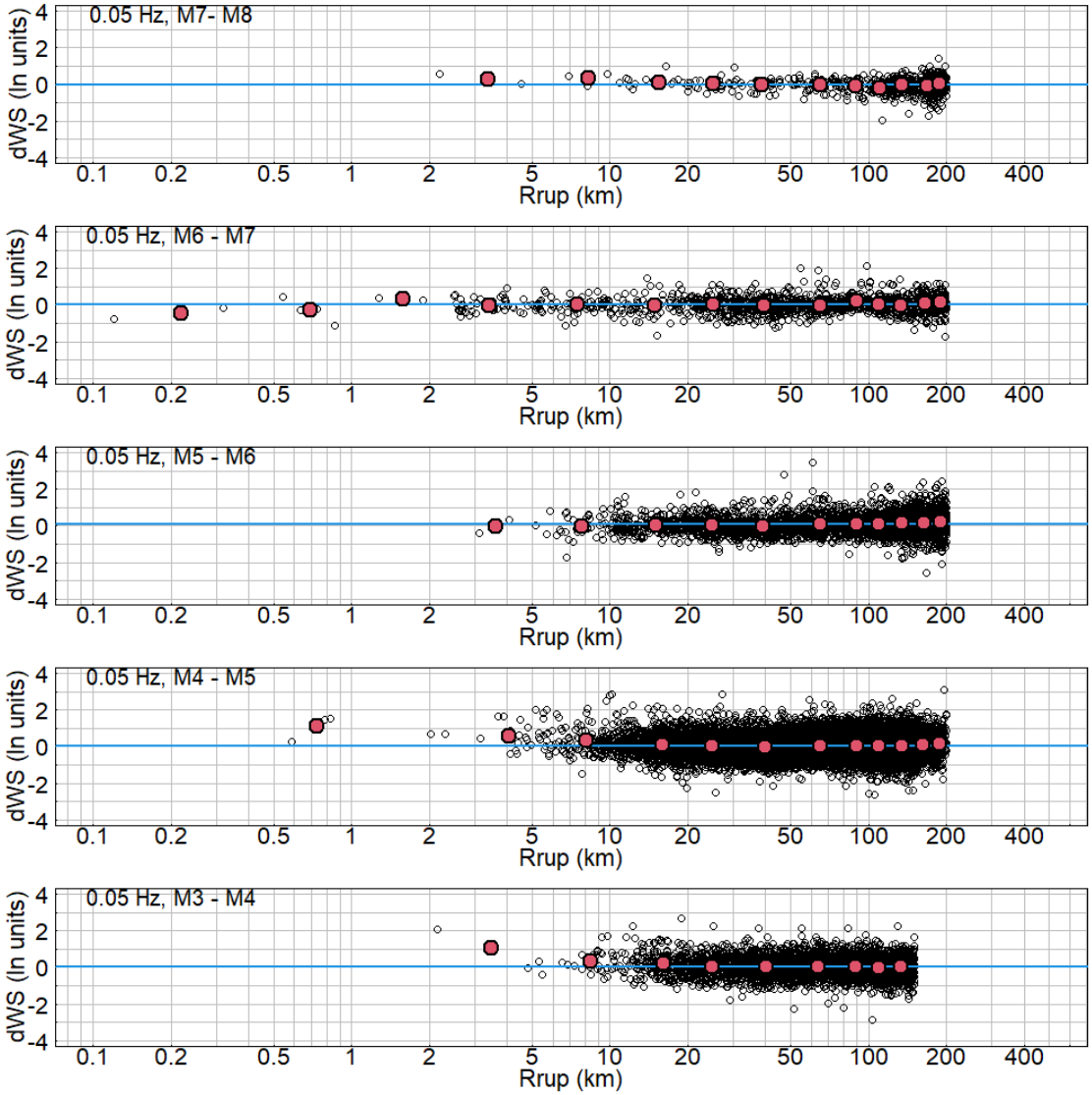


Figure 6.8: δWS **versus** R_{RUP} **in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 0.05 sec.**

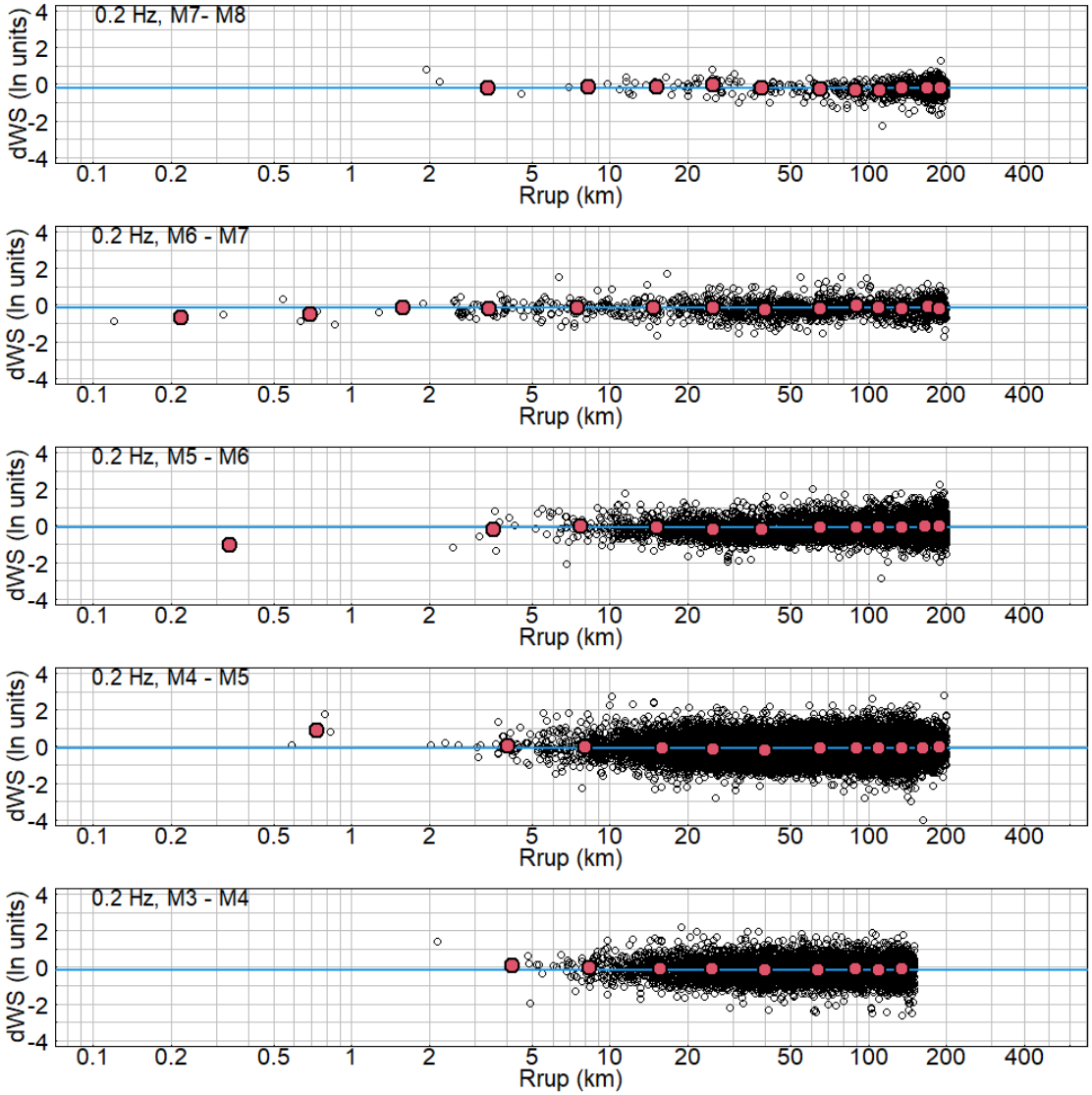


Figure 6.9: δWS **versus** R_{RUP} **in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 0.2 sec.**

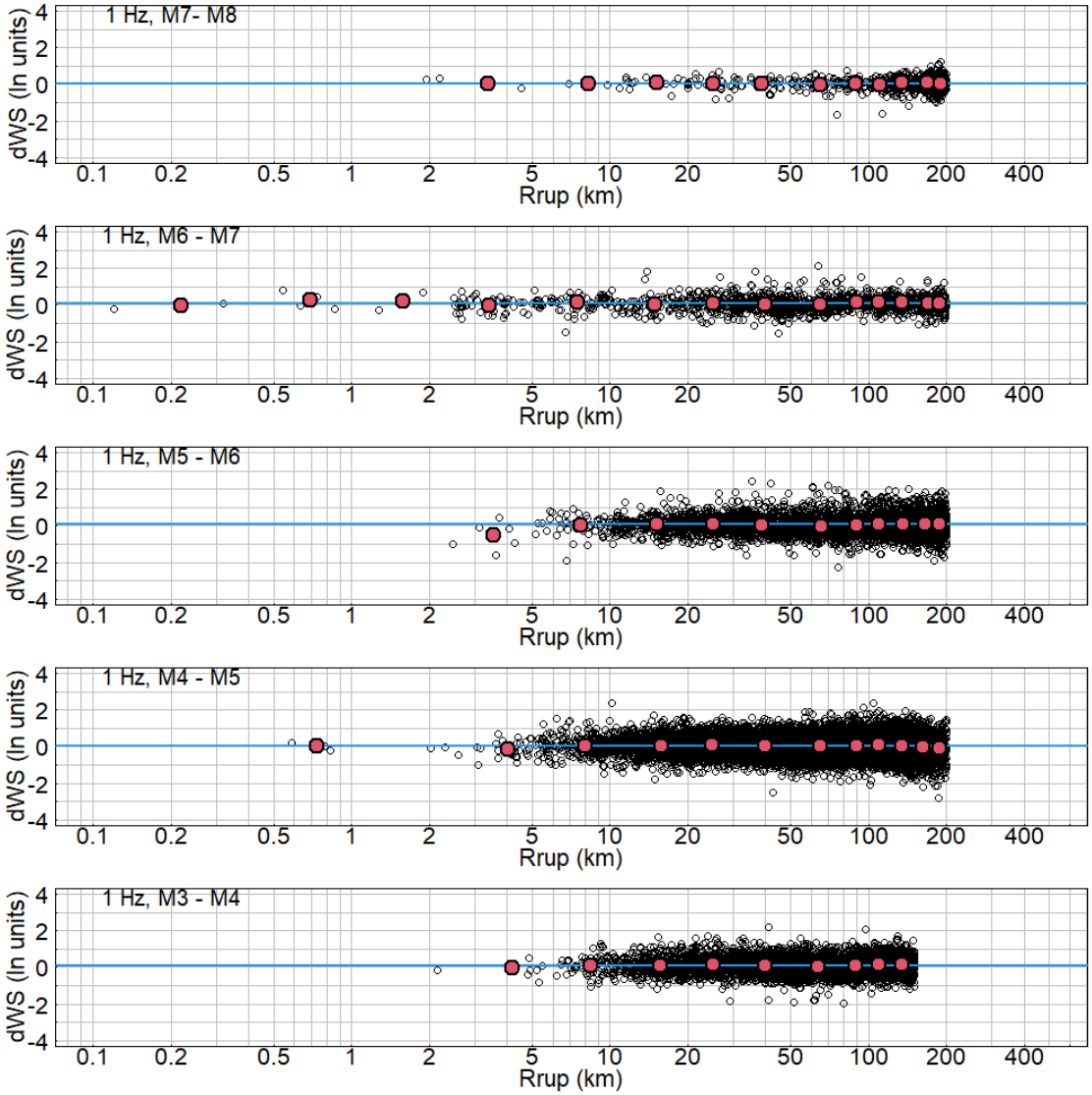


Figure 6.10: δWS **versus** R_{RUP} **in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 1 sec.**

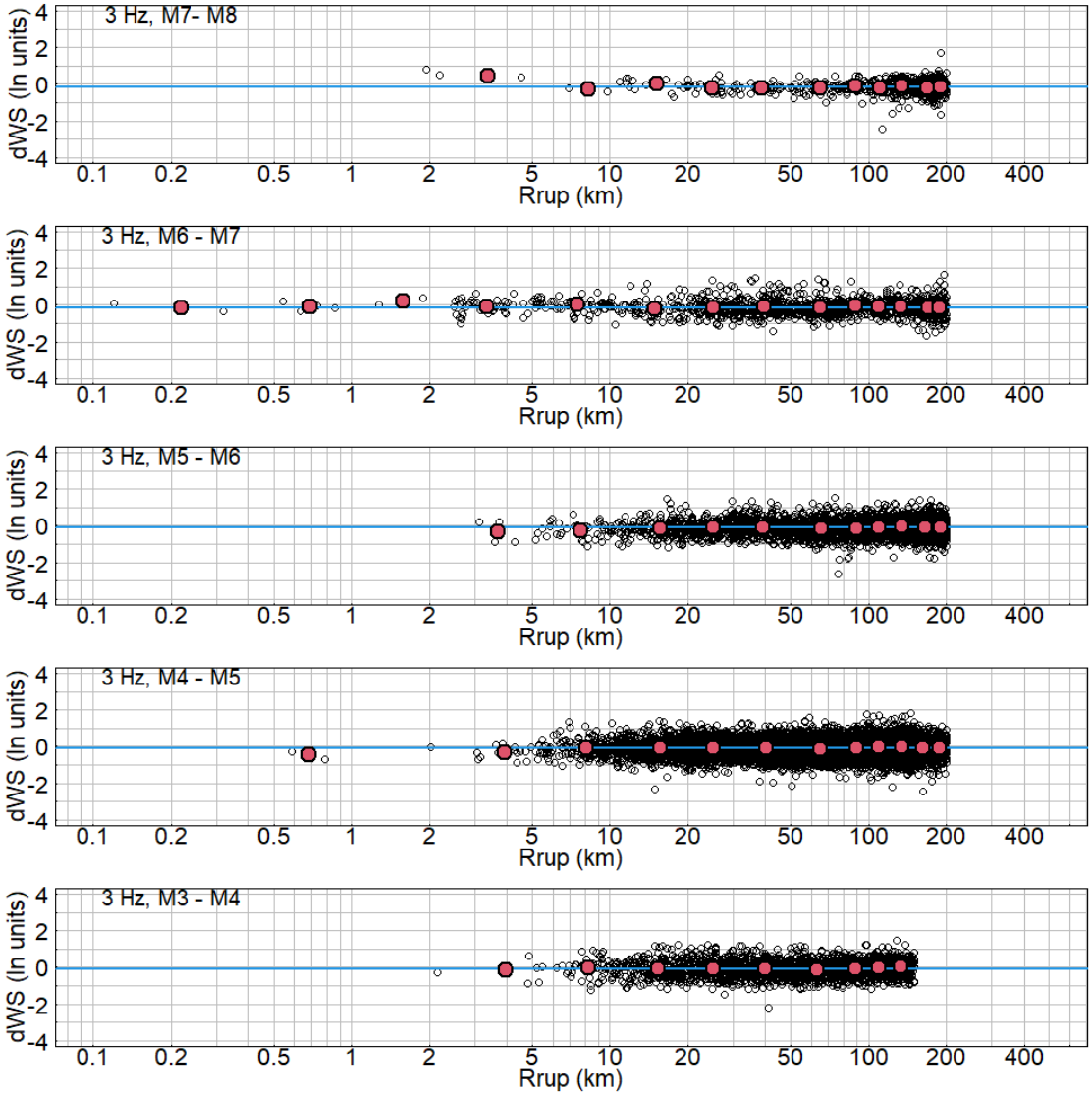


Figure 6.11: δWS **versus** R_{RUP} **in (a) M7-M8, (b) M6-M7, (c) M5-M6, (d) M4-M5, and (e) M3-M4 for 3 sec.**

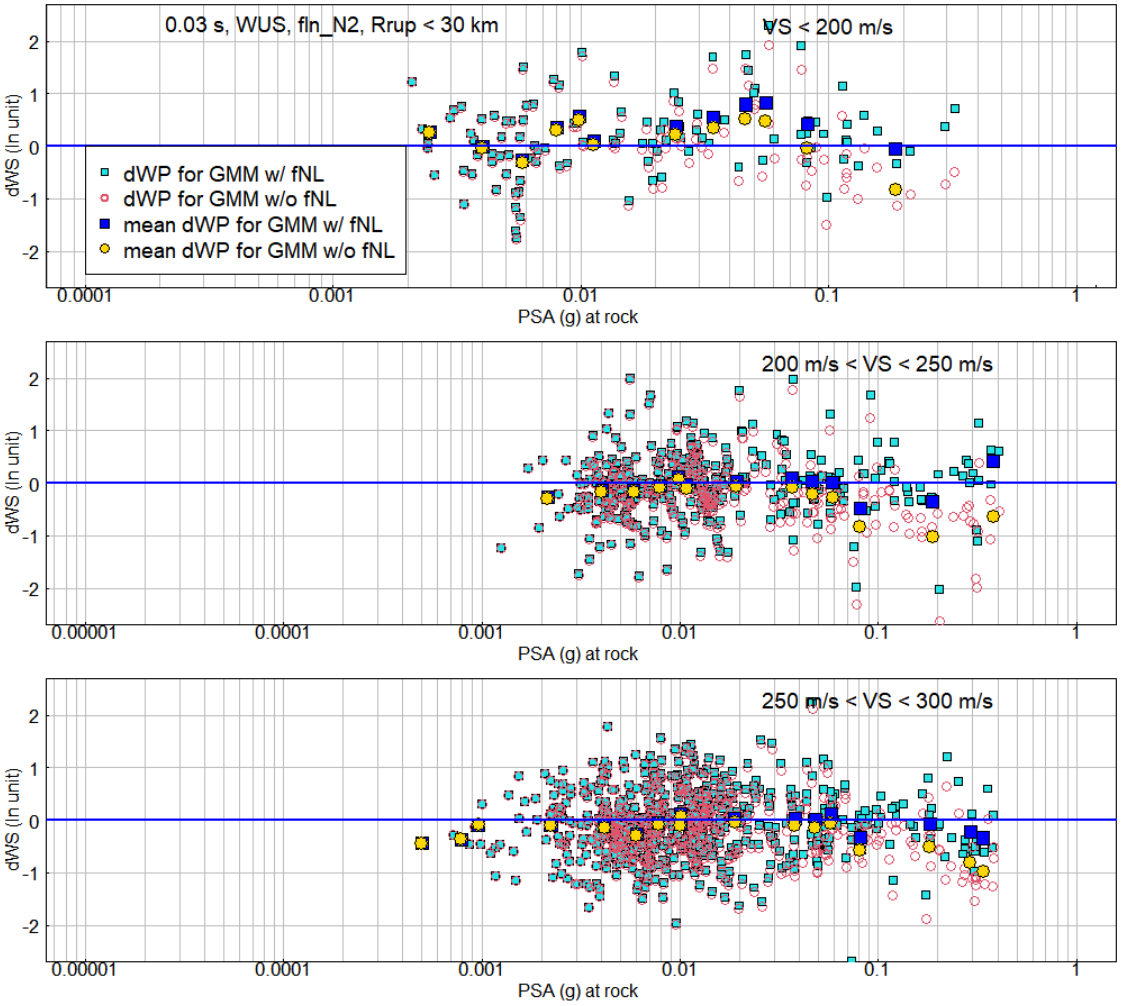


Figure 6.12: δWS **from the GMM with and without fNL for 0.03 sec.**

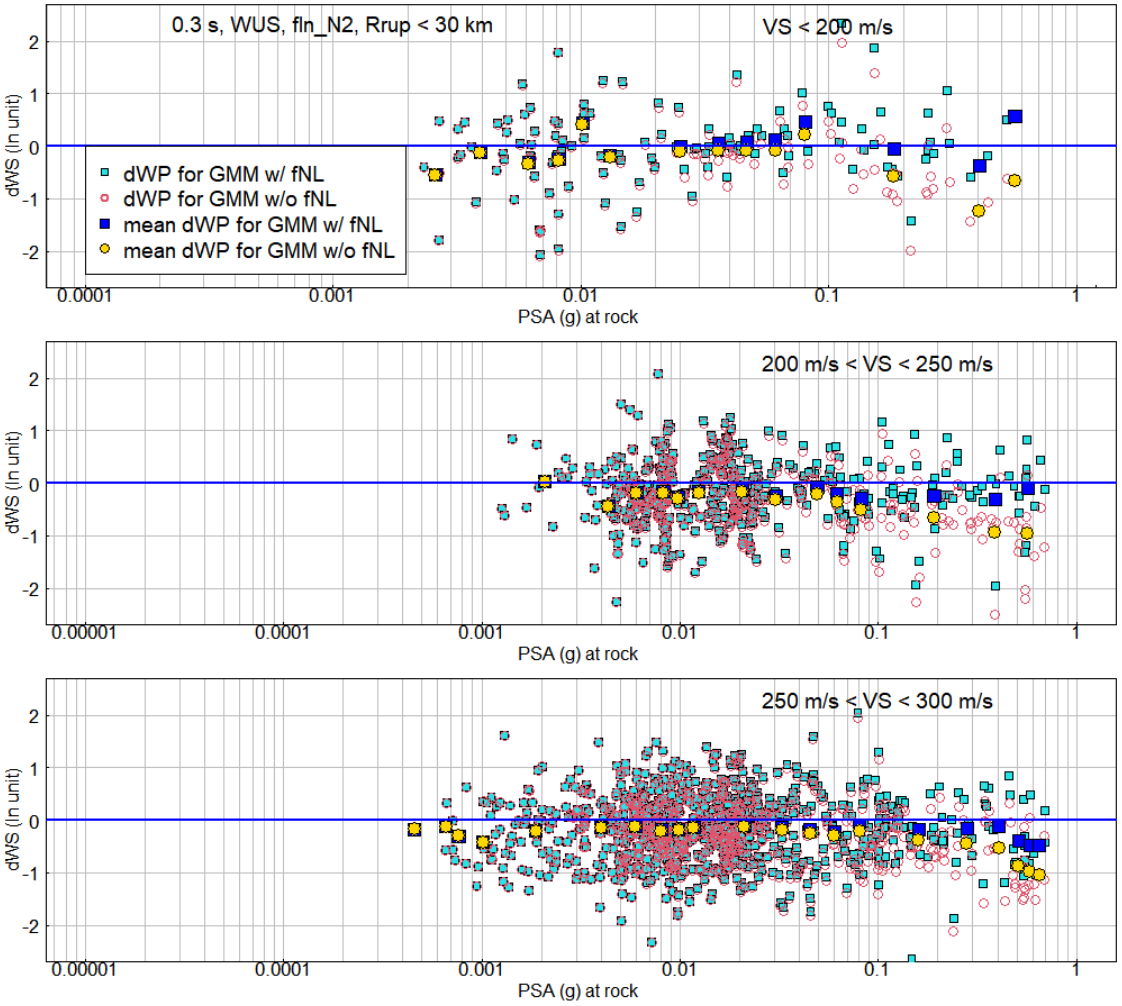


Figure 6.13: δWS **from the GMM with and without fNL for 0.3 sec.**

7 STANDARD DEVIATION MODEL

The total standard deviation consists of three random effects and the within-site residual. Assuming the random effects are uncorrelated, except the two event terms (δB with δP), using simple propagation of errors, the total variance is given by:

$$\sigma^2 = \tau_e^2 + \tau_p^2 \left(\frac{R_{rup} - 50}{100}\right)^2 + \phi_{SS}^2 + \phi_{S2S}^2 + 2\rho(\delta B, \delta P)\, \tau_e\, \tau_p \left(\frac{R_{rup} - 50}{100}\right), \tag{7.1}$$

in which $\rho(\delta B, \delta P)$ is the correlation coefficient between the event-specific constant and the event-specific slope. The period dependence of $\rho(\delta B, \delta P)$ is shown in Figure 7.1. There is no clear trend with period, so an average period-independent value is used for the $\rho(\delta B, \delta P)$ model.

Over 90 percent of the selected ground-motion data have a RotD50 PGA less than 0.05 g. Therefore, the standard deviations estimated from the data represent the variability for linear site response for all four terms. We first develop the models for the standard deviation for linear site response, and then develop the models for the standard deviation with non-linear site effects.

7.1 STANDARD DEVIATION TERMS FOR LINEAR SITE EFFECTS

τ_E, τ_P and ϕ_{SS} values are evaluated for a magnitude dependence. The ϕ_{S2S} values are evaluated for a V_{S30} dependence.

7.1.1 τ_E Model for Linear Site Response

The τ_E for PSA at T=0.1, T=1 sec are shown as a function of magnitude in Figure 7.2. At T=0.1 sec, there is an increase in the τ_E for small-magnitude events, but at T=1 sec, there is no clear magnitude dependence. The magnitude dependence of τ_E is modeled by a trilinear model with breaks at M4 and M6.5:

$$\tau_E(M,T) = \begin{cases} s_{e1}(T) & \text{for } M < 4 \\ s_{e1}(T) + 0.4(s_{e2}(T) - s_{e1}(T))(M-4) & \text{for } M_{b1} \le M \le M_{b2} \\ s_{e2}(T) & \text{for } M > 6.5. \end{cases} \tag{7.2}$$

The period dependence of the s_{e1} and s_{e2} are shown in Figure 7.3. There is no trend with period for the s_{e2} (M>6) coefficients. Therefore, a period-independent average value is used for s_{e2}. The s_{e1} coefficients, (M<4), show a large increase for periods less than 0.3 sec. For periods greater than 0.3 sec, an average value is used. The s_{e1} and s_{e2} coefficients are listed in Tables 7.1 and Tables 7.2.

7.1.2 τ_P Model for Linear Site Response

The τ_P for PSA at T=0.1, T=1 sec are shown as a function of magnitude in Figure 7.4. At T=0.1 sec, there is an increase in the τ_P for small-magnitude events, but at T=1 sec, there is no clear magnitude dependence. Similar to the τ_E model, the magnitude dependence of τ_P is modeled by a trilinear model with breaks at M4 and M6.5:

$$\tau_P(M,T) = \begin{cases} s_{p1}(T) & \text{for } M < 4 \\ [s_{p1}(T) + 0.4(s_{p2}(T) - s_{p1}(f))(M-4)] & \text{for } 4 \le M \le 6.5 \\ s_{p2}(T) & \text{for } M > 6.5 \end{cases} \tag{7.3}$$

The period dependence of the s_{p1} and s_{p2} are shown in Figure 7.5. There is no trend with period for the s_{p2} (M>6) coefficients. Therefore, a period-independent average value is used for s_{p2}.

The s_{p1} coefficients, (M<4), show a strong period dependence with larger τ_P for small-magnitude events at short periods (T< 1 sec) and smaller τ_P for for small-magnitude events at long periods (T> 1 sec). The physical cause for the larger path effects for small-magnitude earthquakes has not been identified, but the trend in the empirical data is clear. The smoothed s_{p1} are shown in Figure 7.5. The s_{p1} and s_{p2} coefficients are listed in Tables 7.1 and Tables 7.2.

7.1.3 ϕ_{S2S} Model for Linear Site Response

The ϕ_{S2S} values for PSA at T=0.1, T=1 sec are shown as a function of V_{S30} in Figure 7.6. At T=0.1 sec, there is an increase in the ϕ_{S2S} for larger V_{S30} values, whereas at T=1 sec, there is a decrease in ϕ_{S2S} for larger V_{S30} values.

As noted in Chapter 6, the increase in the ϕ_{S2S} for larger V_{S30} values is controlled by V_{S30} values between 600 and 700 m/s and is not seen for V_{S30} values greater than 800 m/s (Figure 6.4).

For the EAS model, we model the ϕ_{S2S} dependence the V_{S30}, but without a clear increase in the ϕ_{S2S} for V_{S30} values greater than 800 m/s, we do not include the V_{S30}-dependence of ϕ_{S2S} in the PSA GMM. The black line in Figure 7.7 indicates the V_{S30} independent ϕ_{S2S} model for the PSA GMM.

$$\phi_{S2S}(T) = s_{s1}(T). \tag{7.4}$$

7.1.4 ϕ_{SS} Model for Linear Site Response

The ϕ_{SS} for PSA at T=0.1, T=1 sec are shown as a function of magnitude in Figure 7.4. At T=0.1 sec, there is an increase in the ϕ_{SS} for small-magnitude events for the full period range. The magnitude dependence of ϕ_{SS} is modeled by a trilinear model with breaks at M4 and M6.5:

$$\phi_{SS}(M,T) = \begin{cases} s_{rc1}(T) & \text{for } M < 4 \\ s_{rc1}(T) + 0.4(s_{rc2}(f) - s_{rc1}(T))(M-4) & \text{for } 4 \leq M \leq 6.5 \\ s_{rc2}(T) & \text{for } M > 6.5. \end{cases} \tag{7.5}$$

The period dependence of the s_{rc1} and s_{rc2} are shown in Figure 7.9. There is a period dependence with larger ϕ_{SS} at short periods. The smoothed s_{rc1} and s_{rc2} coefficients are listed in Tables 7.1 and 7.2.

7.1.5 Total Sigma

The total sigma is computed from the four standard deviation terms. The total standard deviation is given by:

$$\begin{aligned} \sigma^2(M, R_{rup}, V_{S30}, T) =& \tau_E^2(M,T) + \tau_P^2(M,f) \left(\frac{R_{RUP} - 50}{100}\right)^2 + \phi_{SS}^2(M,T) + \phi_{S2S}^2(V_{S30},T) \\ &+ 2\rho(\delta B_e, \delta P_e)\, \tau_E\, \tau_P \left(\frac{R_{RUP} - 50}{100}\right). \end{aligned} \tag{7.6}$$

The σ has a distance dependence due to the random effects for the path terms.

7.2 NON-LINEAR SITE EFFECTS ON ALEATORY VARIABILITY

The non-linear site response effects on the standard deviation terms is computed using the results of the analytical modeling of the non-linear effects. In this section, we derive the modifications to the standard deviation terms for linear site response to compute the standard deviation terms with non-linear site effects.

We consider a simple model with the input motion at depth (point B) and the observed ground motion at the surface (point C). The PSA at the surface (point C) is given by:

$$\ln(PSA_C) = GMM_C(M,R,...) + \delta B_C + \delta P_C \left(\frac{R_{RUP} - 50}{50}\right) + \delta S2S_C + \delta WSP_C, \tag{7.7}$$

in which τ_C, $\tau_{P,C}$, $\phi_{S2S,C}$, and $\phi_{WSP,C}$ are the standard deviations of δB_C, δP_C, $\delta S2S_C$, and δWSP_C, respectively. Although the data used to develop the GMM are only on the surface, the PSA at depth (location B) can be written in the same form:

$$\ln(PSA_B) = GMM_B(M, R, ...) + \delta B_B + \delta P_B \left(\frac{R_{RUP} - 50}{50}\right) + \delta S2S_B + \delta WSP_B. \quad (7.8)$$

The ground motion at the surface, $PSA_C(T))$, is the product of the input motion at depth, $PSA_B(T)$, and the site amplification, $Amp(T)$.

$$ln(PSA_C(T)) = ln(PSA_B(T)) + ln(Amp(T, PSA_B(T))). \quad (7.9)$$

Non-linear site effects are a function of the input motion to the soil profile, $PSA_B(T)$, which depends on all four terms in equation (7.8).

Using simple propagation of errors, the standard deviation of $ln(PSA_C(T))$ with non-linear site effects is given by:

$$\begin{aligned} VAR[ln(PSA_C(T))] = &\left(\frac{\partial ln(PSA_C(T))}{\partial ln(PSA_B(T))}\right)^2 VAR[ln(PSA_B(T))] \\ &+ VAR[ln(Amp(T, PSA_B(T)))]. \end{aligned} \quad (7.10)$$

Combining Equations (7.8) and (7.10), the variance of the $ln(PSA_B(T))$ is given by:

$$\begin{aligned} VAR[ln(PSA_B(T))] = &\tau^2_{E,B} + \tau^2_{P,B}\left(\frac{R_{Rup} - 50}{50}\right) + \phi^2_{S2S,B} + \phi^2_{WS,B} \\ &+ 2\rho(\delta B_e, \delta P_e)\, \tau_E\, \tau_P. \end{aligned} \quad (7.11)$$

We don't have direct estimates of the standard deviation terms of the input motion at depth ($\tau_{E,B}$, $\tau_{P,B}$, $\phi_{S2S,B}$, and $\phi_{WSP,B}$), but the surface ground-motion data used to estimate τ and τ_P are dominated by ground motions in the linear range. Therefore, we can approximate $\tau_{E,B}$ and $\tau_{P,B}$ as

$$\tau_{E,B} \approx \tau_{E,C}, \quad (7.12)$$

and

$$\tau_{P,B} \approx \tau_{P,C}. \quad (7.13)$$

The $\phi_{S2S,B}$ term is more difficult to estimate from the surface ground-motion data. We assume that the variance of the PSA at the depth (point B) is the variance of the PSA at the surface minus the variance of linear site amplification:

$$VAR[ln(PSA_B] = \sigma_C^2 - \phi_{amp}^2. \tag{7.14}$$

From evaluations of the variability of amplification from down-hole arrays, $\phi_{amp}(T) \approx 0.3$ ln units independent of frequency (Liou et al., 2024).

For the non-linear site term, we use the results of the analytical modeling of the non-linear site amplification by Ilhan et al. (2025), which uses the ground motion for V_S=800 m/s as outcropping motion for point B. For our application, we ignore the difference between site conditions with V_S=800 m/s and V_{S30}=800 m/s. (Note: the input motions used by Ilhan et al. (2025) were developed for V_{S30}=800 m/s but were input at the half-space with V_S=800 m/s.) With this assumptions, the nonlinear site term, f_{snl}, is modeled by:

$$f_{snl}(V_{S30}, PSA_{800}(T)) = f_2(V_{S30})\, ln\left(\frac{PSA_{800}(T) + f_3}{f_3}\right), \tag{7.15}$$

and the aleatory variability of the non-linear site amplification is given by:

$$VAR[ln(Amp(T, PSA_B(T)))] = \phi_{amp}^2(T) + \phi_{amp,NL}^2, (PSA_{800}, T), \tag{7.16}$$

in which $\phi_{amp,NL}^2(PSA_{800}, T)$ is the increase in the standard deviation of the site amplification due to non-linear site effects

The partial derivative in equation (7.10) is given by:

$$\frac{\partial ln(PSA_C(T))}{\partial ln(PSA_{800}(T))} = 1 + f_2\left(\frac{PSA_{800}(T)}{PSA_{800}(f) + f_3}\right), \tag{7.17}$$

and the variance at point C is given by:

$$\begin{aligned} VAR[ln(PSA_C(T))] = &\left(\frac{\partial ln(PSA_C(T))}{\partial ln(PSA_B(T))}\right)^2 VAR[ln(PSA_B(T))] \\ &+ VAR[ln(Amp(T, PSA_B(T)))]. \end{aligned} \tag{7.18}$$

The non-linear site response is due to the shallow soil material, but it affects all four of the standard deviation terms for the surface motions because the soil responds to the amplitude of the input motion, regardless of which terms contribute to the amplitude. That is, the soil does not distinguish between larger input motions due to the δB at depth or larger input motions due to the δWS at depth.

7.2.1 Sigma Models with Non-linear Site Response

Combining equations (7.18), (7.17), (7.14), and (7.16), the standard deviation of the surface motion with non-linear site effects is given by:

$$\sigma_{C,NL}(T) = \sqrt{\left(1 + \frac{f_2(f)\, PSA_{800}(T)}{PSA_{800}(T) + f_3(T)}\right)^2 \left(\sigma^2_{C,LIN}(T) - \phi^2_{amp}(T)\right) + \phi^2_{amp}(T) + \phi^2_{amp,NL}(T)}. \tag{7.19}$$

Coefficients $f_2(T)$ and $f_3(T)$ are listed in Table 5.3.

Similarly, the standard deviation with non-linear effects for the components of the total sigma are given by:

$$\tau_{E,NL}(T) = \left(1 + \frac{f_2(T)\, PSA_{800}(f)}{PSA_{800}(T) + f_3(T)}\right) \tau_E(T) \tag{7.20}$$

$$\tau_{P,NL}(T) = \left(1 + \frac{f_2(T)\, PSA_{800}(f)}{PSA_{800}(f) + f_3(T)}\right) \tau_P(T) \tag{7.21}$$

$$\phi_{S2S,NL}(T) = \left(1 + \frac{f_2(T)\, PSA_{800}(f)}{PSA_{800}(T) + f_3(T)}\right) \phi_{S2S}(T) \tag{7.22}$$

and

$$\phi_{SS,NL}(T) = \sqrt{\left(1 + \frac{f_2(T)\, PSA_{800}(f)}{PSA_{800}(T) + f_3(T)}\right)^2 \left(\phi^2_{SS}(T) - \phi^2_{amp}(T)\right) + \phi^2_{amp}(T) + \phi^2_{amp,NL}(T)}. \tag{7.23}$$

As mentioned earlier, the ϕ_{amp} for linear site response is modeled as a frequency-independent constant:

$$\phi_{amp}(f) = 0.3 \text{ ln units}. \tag{7.24}$$

7.2.2 $\phi_{amp,NL}$ Model for Non-linear Effects on ϕ_{amp}

For the non-linear range, there can be an increase in the standard deviation of the site amplification for large input motions as compared to the standard deviation for the linear range. The variability in the surface motions from the analytical site response calculations includes the effects of the different V_S profiles, different input motions, and variability of the non-linear effects. The $\phi_{amp,NL}(T)$ term only accounts for variability of the non-linear effects. It is computed by removing the variability in the linear range due to the different V_S profiles and the different input motions from the computed variability in the nonlinear range from the site response modeling.

$$\phi_{amp,NL}(T) = \sigma_2(T) \ln\left(\frac{PSA_{800}(T) + \sigma_3(T)}{\sigma_3(T)}\right). \tag{7.25}$$

The σ_2 and σ_3 terms are given by Ilhan et al. (2025) as a table of values for different periods. For ease of application, we fit these point estimates to piecewise continuous functions.

$$\sigma_2(T) = \begin{cases} \sigma_4(\text{T}) & \text{for } V_{S30} < V_{n1} \\ \sigma_4(T)\left(1 - \frac{ln(V_{S30}/V_{n1})}{ln(V_{n2}/V_{n1})}\right) & \text{for } V_{n1} < V_{S30} < V_{n2} \\ 0 & \text{for } V_{S30} > V_{n2}, \end{cases} \tag{7.26}$$

and

$$\sigma_3(f) = \begin{cases} 0.001 & \text{for f<0.4 Hz} \\ 0.001 + 0.004\ \frac{ln(f/0.4)}{ln(2/0.4)} & \text{for 0.4 Hz} < \text{f} < 2\text{ Hz} \\ 0.005 & \text{for f} > 2\text{ Hz}, \end{cases} \tag{7.27}$$

in which V_{n1}, = 200 m/s, and σ_4 and V_{n2} are period-dependent coefficients given piece-wise continuous functions;

$$\sigma_4(f) = \begin{cases} 0.1 & \text{for T<0.055 sec} \\ 0.1 + 0.0575ln(T/0.05) & \text{for 0.05 s} < \text{T} < 0.1\text{ s} \\ 0.14 + 0.0369ln(T/0.1) & \text{for 0.05 s} < \text{T} < 0.1\text{ s} \\ 0.20 - 0.0152ln(T/0.5) & \text{for 0.05 s} < \text{T} < 0.1\text{ s} \\ 0.193 - 0.1100ln(T/0.75) & \text{for 0.05 s} < \text{T} < 0.1\text{ s} \\ 0.085 - 0.0598ln(T/2) & \text{for 0.05 s} < \text{T} < 0.1\text{ s} \\ 0.044 - 0.0397ln(T/4) & \text{for 0.05 s} < \text{T} < 0.1\text{ s} \\ 0.016 & \text{for T} > 8\text{ s} \end{cases} \tag{7.28}$$

and

$$V_{n2}(T) = \begin{cases} 1100\ \text{m/s} & \text{for T<0.01 sec} \\ 1100 + 373\ \text{ln(T/0.01)} & \text{for 0.01} < \text{T /le 0.05 sec} \\ 1700 - 585\ \text{ln(T/0.05)} & \text{for 0.05} < \text{T /le 0.02 sec} \\ 1000 - 186\ \text{ln(T/0.2)} & \text{for 0.2} < \text{T /le 1.0 sec} \\ 700 - 432\ \text{ln(T)} & \text{for 1} < \text{T /le 2 sec} \\ 400 & \text{for 2} < \text{T} \end{cases} \tag{7.29}$$

The period dependence of the total sigma with and without non-linear site effects are compared in Figure 7.10 for V_{S30} of 250 and 400 m/s for M7 and R_{RUP}=10 km. For large-magnitude events at short distances, the non-linear site effects reduce the total sigma at short periods on soil sites by about 25% for V_{S30}=250 m/s and by about 10% for V_{S30}=400 m/s.

Table 7.1: Coefficients for the standard deviation model based on the GMM without $\mathbf{T_{Amp1.5}}$

Period(s)	s_{e1}	s_{e2}	s_{p1}	s_{p2}	ϕ_{s2s}	s_{r1}	s_{r2}	$\rho 1_{\delta B \delta P}$	$\rho 2_{\delta B \delta P}$
0.01	0.564	0.285	0.506	0.289	0.404	0.533	0.402	0.050	0.189
0.02	0.571	0.285	0.506	0.289	0.406	0.534	0.402	0.050	0.189
0.025	0.576	0.285	0.506	0.289	0.408	0.540	0.402	0.050	0.189
0.03	0.589	0.285	0.506	0.289	0.411	0.545	0.402	0.050	0.189
0.05	0.611	0.285	0.506	0.289	0.430	0.546	0.402	0.050	0.189
0.075	0.612	0.285	0.506	0.289	0.446	0.542	0.402	0.050	0.189
0.1	0.600	0.285	0.506	0.289	0.449	0.552	0.387	0.050	0.189
0.15	0.563	0.285	0.506	0.289	0.456	0.543	0.361	0.050	0.189
0.2	0.519	0.285	0.506	0.289	0.446	0.540	0.361	0.050	0.189
0.25	0.492	0.285	0.506	0.289	0.436	0.535	0.363	0.050	0.189
0.3	0.478	0.295	0.506	0.289	0.428	0.525	0.366	0.050	0.189
0.4	0.450	0.311	0.506	0.289	0.431	0.511	0.357	0.050	0.189
0.5	0.444	0.320	0.497	0.289	0.432	0.500	0.358	0.050	0.189
0.75	0.455	0.349	0.465	0.289	0.433	0.472	0.340	0.050	0.189
1	0.456	0.371	0.428	0.289	0.442	0.450	0.331	0.050	0.189
1.5	0.465	0.401	0.378	0.269	0.437	0.444	0.327	0.050	0.189
2	0.471	0.402	0.349	0.250	0.441	0.429	0.328	0.050	0.189
2.5	0.471	0.402	0.323	0.237	0.432	0.419	0.320	0.050	0.189
3	0.465	0.400	0.304	0.223	0.427	0.407	0.321	0.050	0.189
5	0.426	0.400	0.251	0.189	0.427	0.394	0.303	0.050	0.189
6	0.415	0.400	0.240	0.178	0.427	0.382	0.303	0.050	0.189
7.5	0.406	0.400	0.240	0.178	0.427	0.372	0.303	0.050	0.189
10	0.385	0.400	0.240	0.178	0.427	0.369	0.303	0.050	0.189

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Table 7.2: Coefficients for the standard deviation model based on the GMM with $\mathbf{T_{Amp1.5}}$.

Period(s)	s_{e1}	s_{e2}	s_{p1}	s_{p2}	ϕ_{s2s}	s_{r1}	s_{r2}	$\rho 1_{\delta B \delta P}$	$\rho 2_{\delta B \delta P}$
0.01	0.558	0.224	0.550	0.378	0.409	0.517	0.364	0.046	0.145
0.02	0.563	0.212	0.553	0.378	0.411	0.517	0.366	0.046	0.145
0.025	0.566	0.208	0.556	0.380	0.413	0.522	0.370	0.046	0.145
0.03	0.575	0.199	0.560	0.384	0.416	0.526	0.377	0.046	0.145
0.05	0.591	0.209	0.556	0.389	0.432	0.524	0.380	0.046	0.145
0.075	0.596	0.253	0.546	0.360	0.446	0.520	0.370	0.046	0.145
0.1	0.589	0.268	0.545	0.328	0.454	0.530	0.358	0.046	0.145
0.15	0.555	0.270	0.515	0.328	0.452	0.530	0.355	0.046	0.145
0.2	0.517	0.270	0.522	0.328	0.446	0.530	0.351	0.046	0.145
0.25	0.492	0.270	0.506	0.328	0.436	0.535	0.363	0.046	0.145
0.3	0.478	0.270	0.506	0.328	0.428	0.525	0.366	0.046	0.145
0.4	0.450	0.270	0.506	0.328	0.431	0.511	0.357	0.046	0.145
0.5	0.444	0.310	0.497	0.328	0.432	0.500	0.358	0.046	0.145
0.75	0.455	0.349	0.465	0.328	0.433	0.472	0.340	0.046	0.145
1	0.456	0.371	0.428	0.328	0.442	0.450	0.331	0.046	0.145
1.5	0.465	0.401	0.378	0.269	0.437	0.444	0.327	0.046	0.145
2	0.471	0.402	0.349	0.250	0.441	0.429	0.328	0.046	0.145
2.5	0.471	0.402	0.323	0.237	0.432	0.419	0.320	0.046	0.145
3	0.465	0.400	0.304	0.223	0.427	0.407	0.321	0.046	0.145
5	0.426	0.400	0.251	0.189	0.427	0.394	0.303	0.046	0.145
6	0.415	0.400	0.240	0.178	0.427	0.382	0.303	0.046	0.145
7.5	0.406	0.400	0.240	0.178	0.427	0.372	0.303	0.046	0.145
10	0.385	0.400	0.240	0.178	0.427	0.369	0.303	0.046	0.145

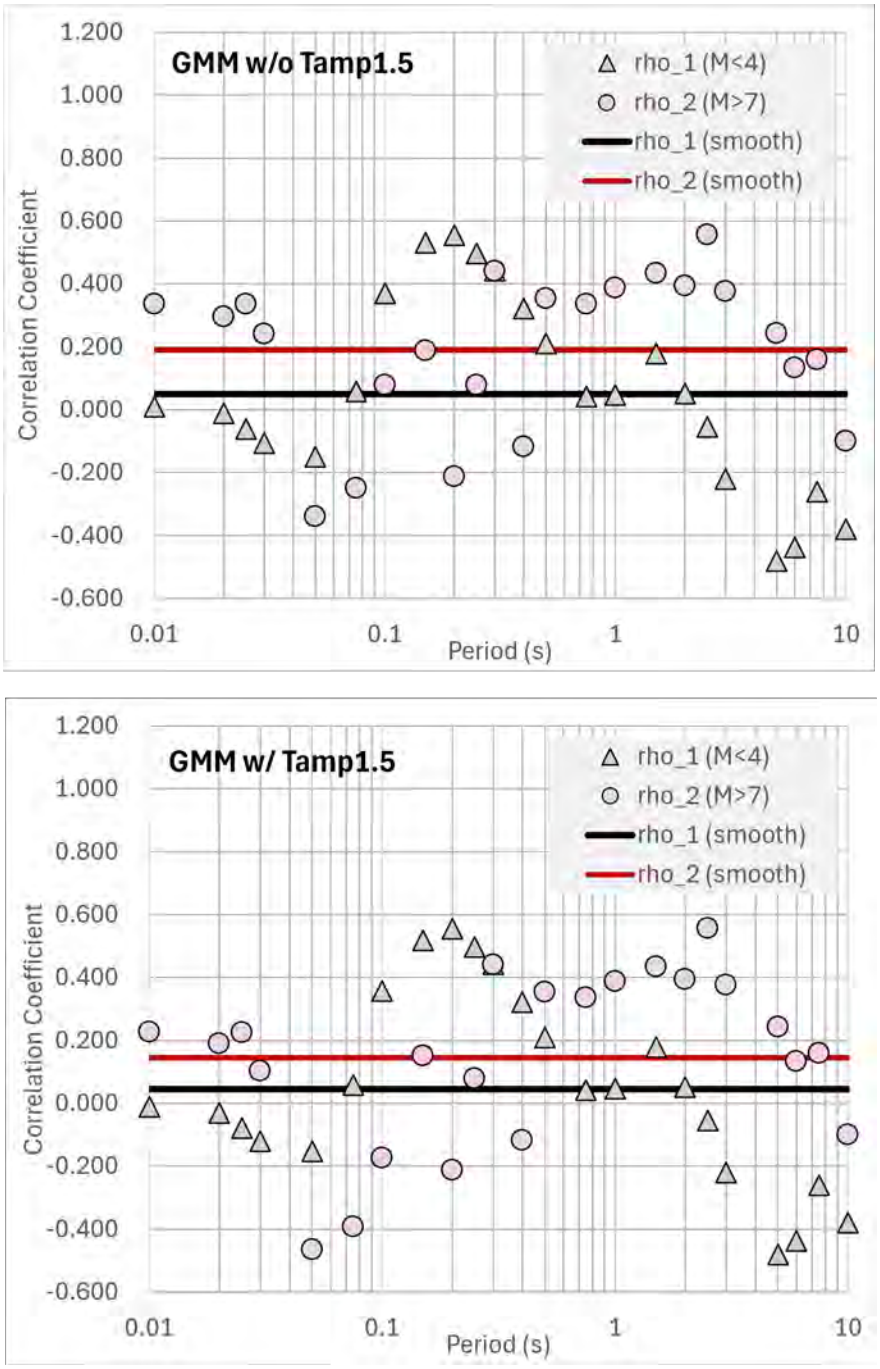


Figure 7.1: Period dependence of the correlation coefficient for the δB_e and δP_e. The period-independent average is used.

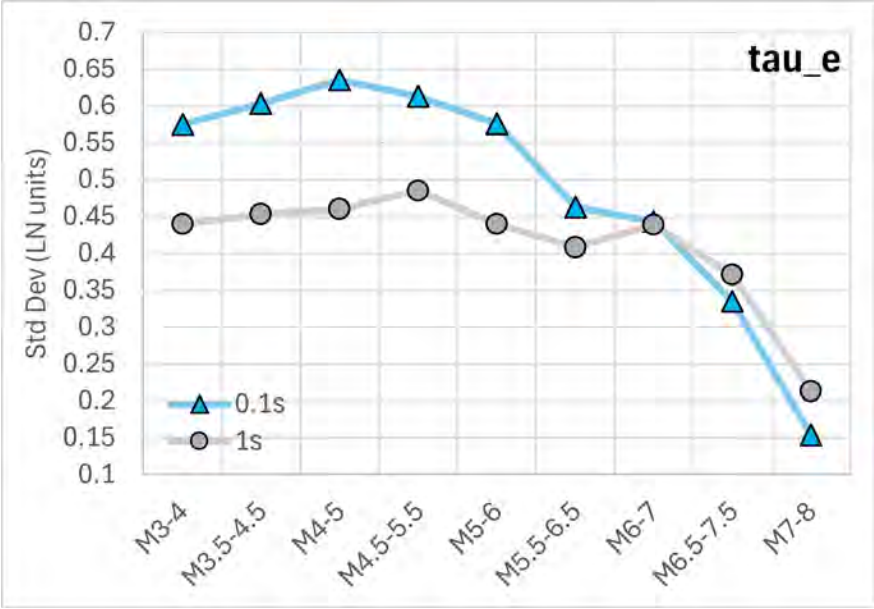


Figure 7.2: Magnitude dependence of the τ_E for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec.

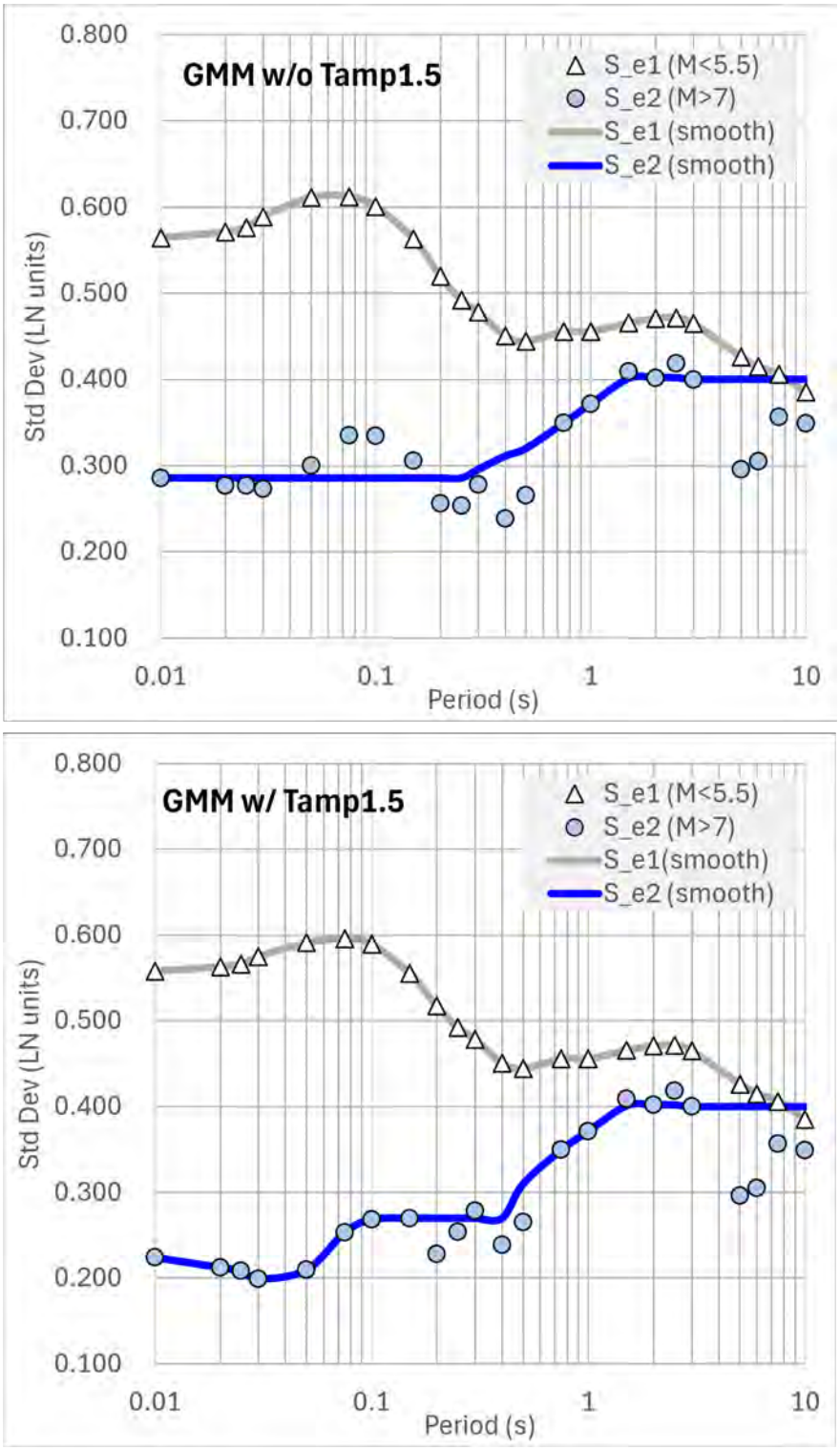


Figure 7.3: Coefficients of τ_E model for the GMM without and with $T_{Amp1.5}$.

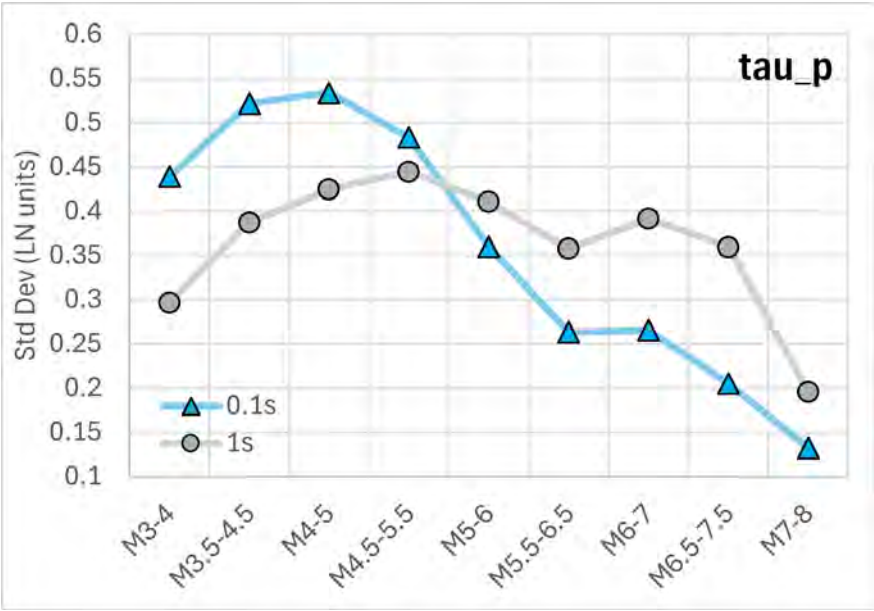


Figure 7.4: Magnitude dependence of the τ_P for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec.

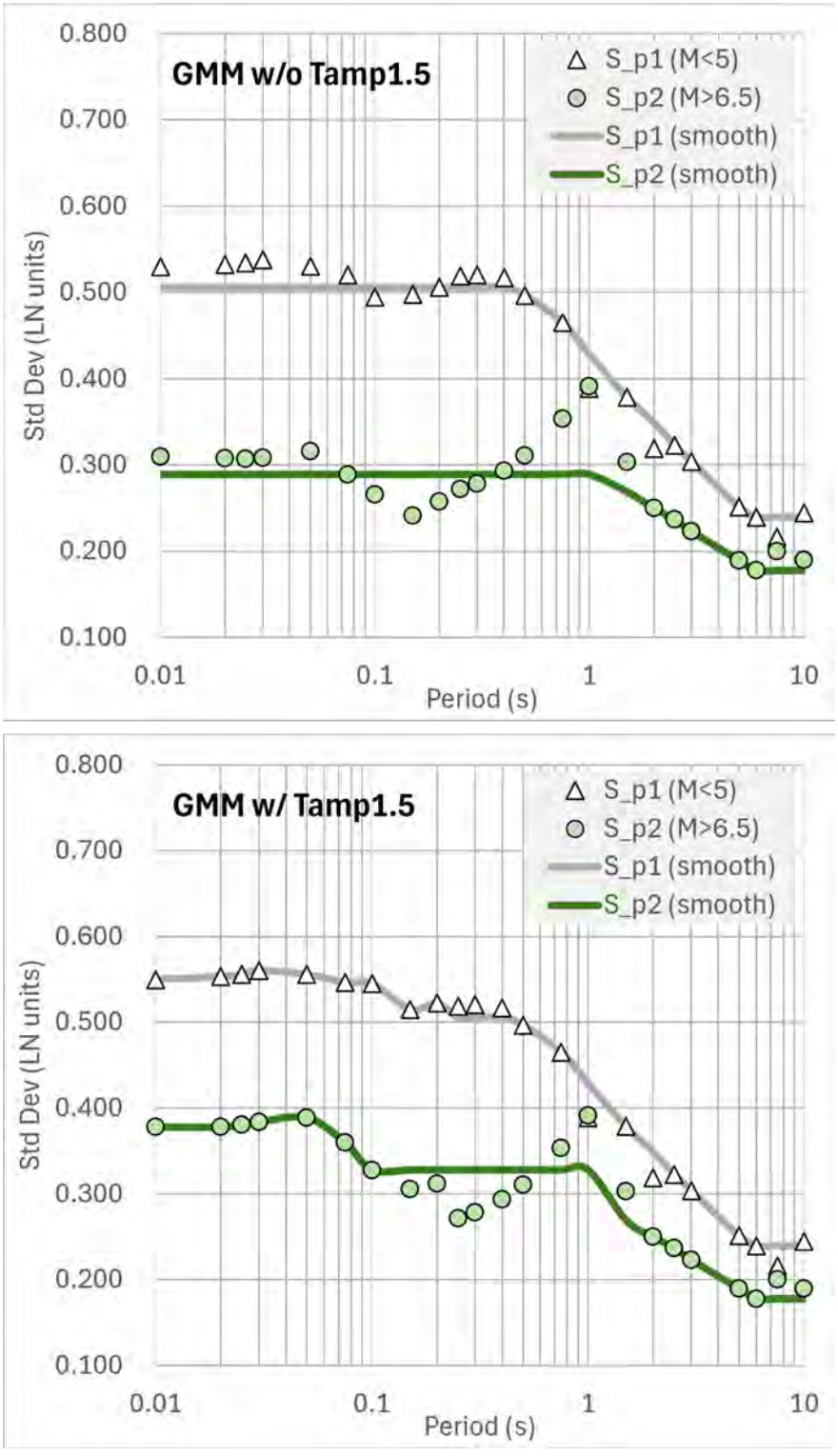


Figure 7.5: Coefficients of τ_P model for the GMM without and with $T_{Amp1.5}$.

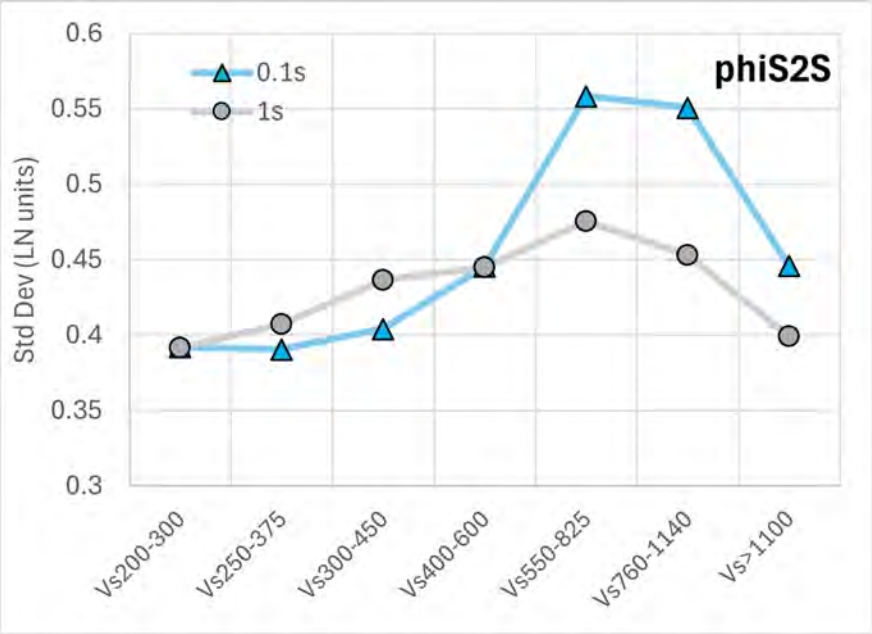


Figure 7.6: V_{S30} **dependence of the** ϕ_{S2S} **for the GMM without** $T_{amp1.5}$ **for T=0.1 and T=1 sec.**

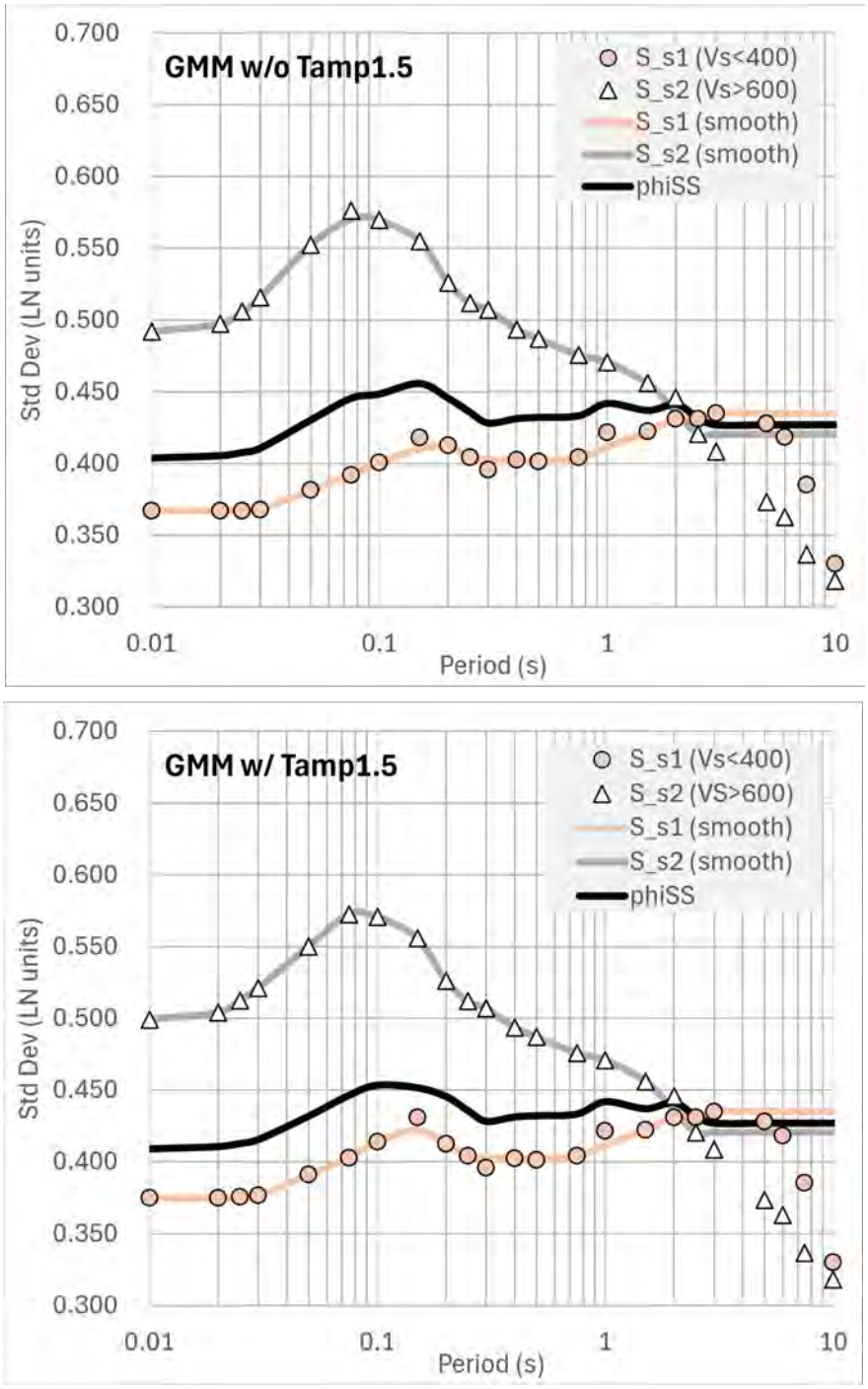


Figure 7.7: Coefficients of ϕ_{S2S} model for the GMM without and with $T_{Amp1.5}$.

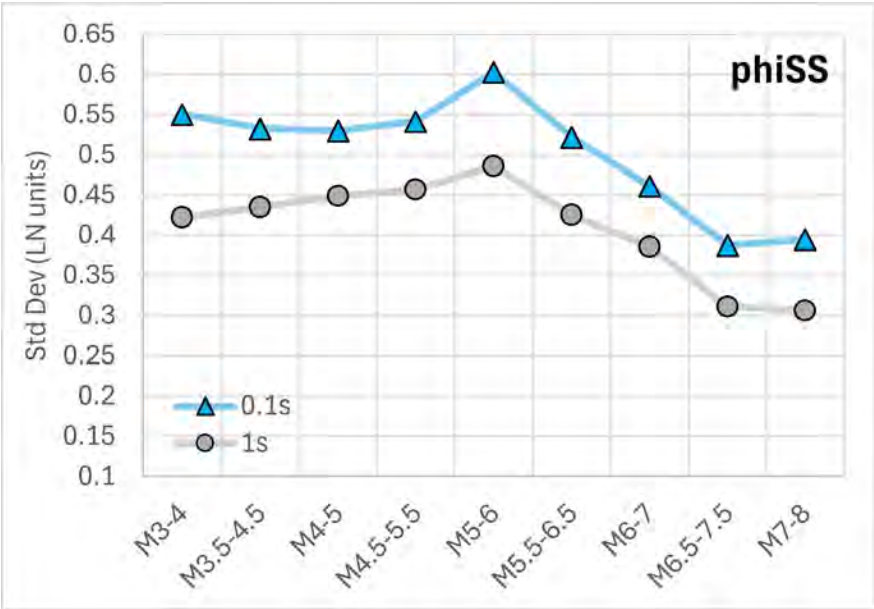


Figure 7.8: Magnitude dependence of the ϕ_{SS} for the GMM without $T_{amp1.5}$ for T=0.1 and T=1 sec.

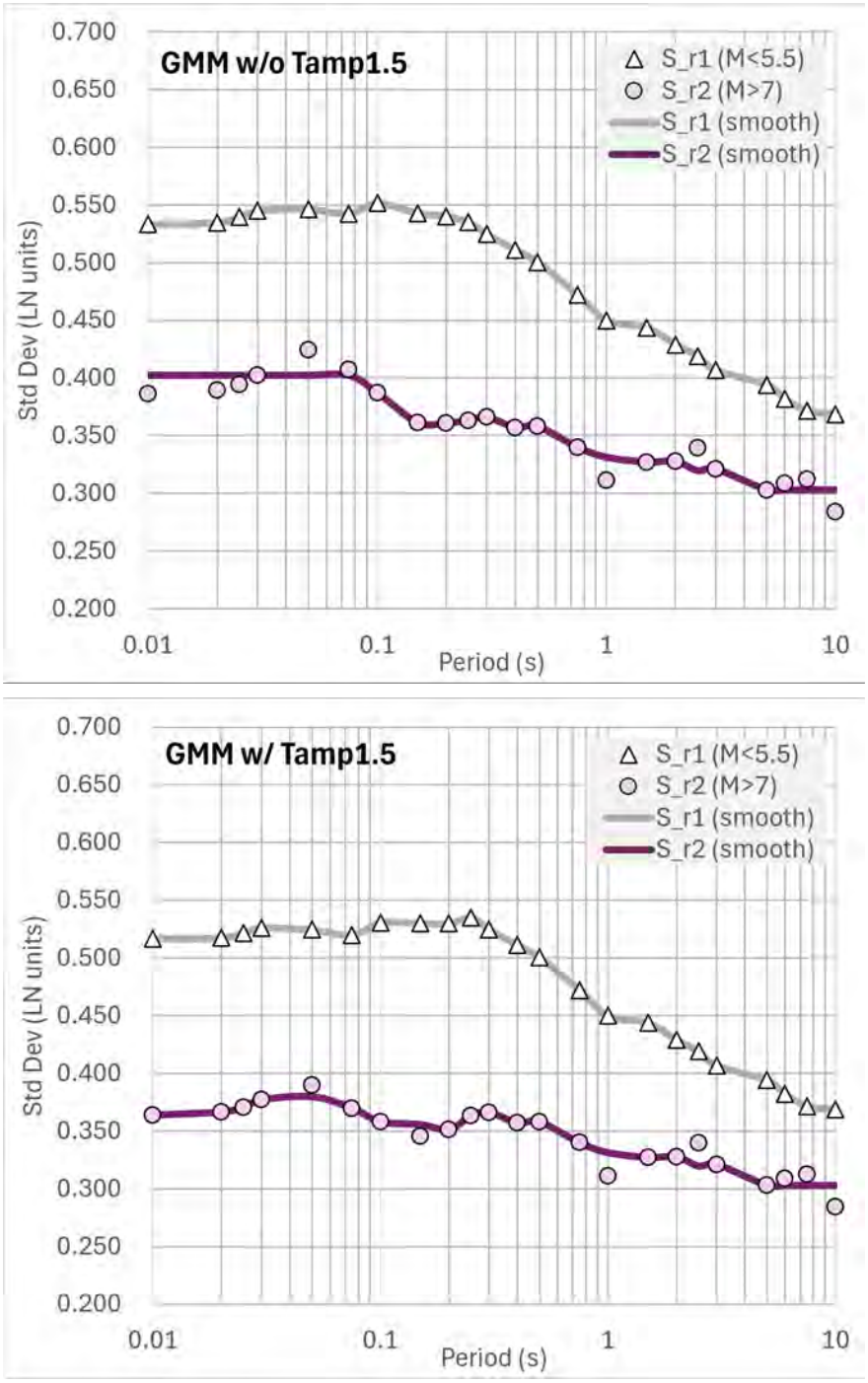
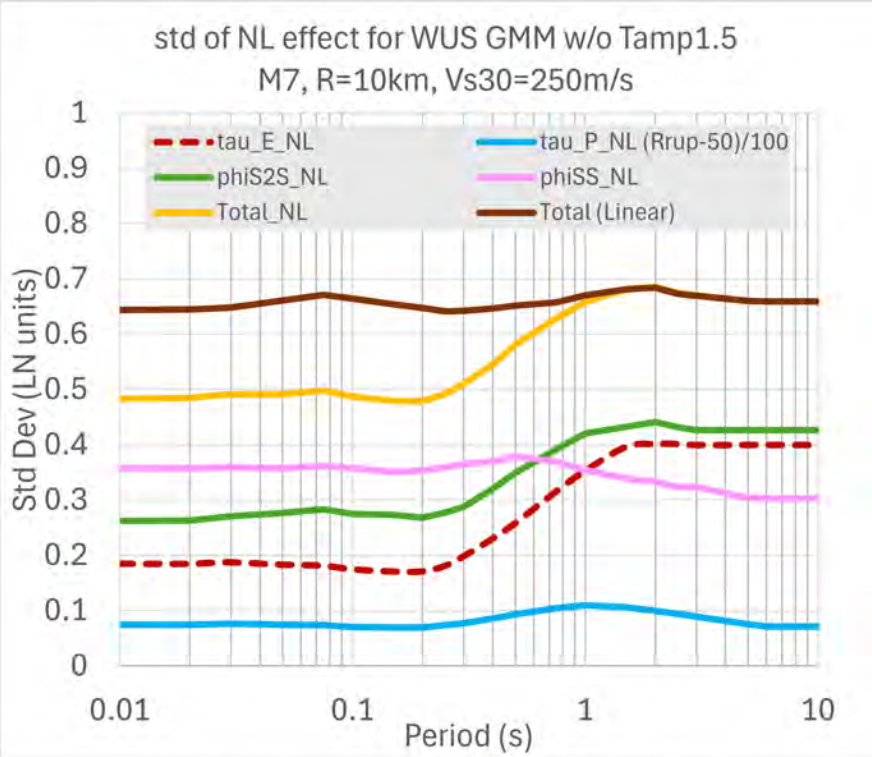


Figure 7.9: Coefficients of ϕ_{SS} model for the GMM without and with Tamp1.5.



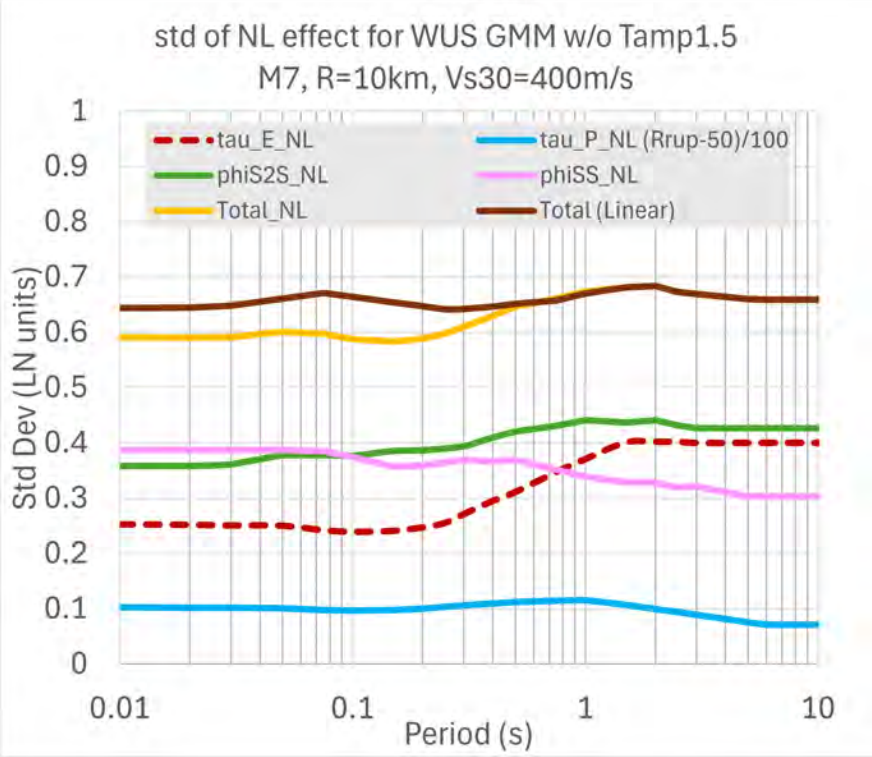


Figure 7.10: Comparison of the total sigma for soil sites (V_{S30}=250 m/s and 400 m/s) for M7 and R_{RUP}=10 km.

8 MODEL COMPARISONS

For the NGA projects, a key feature is that the medians and standard deviations from the different GMMs are compared before the GMMs are finalized. At the time of the preparation of this report, the GMMs from the other NGA-W3 developer teams were not available for comparisons. Without the other NGA-W3 GMMs for comparison, the median and standard deviation for the new GMM are compared to the ASK14 GMM.

8.1 MEDIAN

The distance scaling of the median PSA from vertical strike-slip earthquakes for the new GMM and the ASK14 GMM is compared for five different magnitudes (M7.5, M6.5, M5.5, M4.5, M3.5) in Figure 8.1 and 8.2 for periods of 0.05, 0.2, 1.0, and 3.0 sec, for a soil site condition (V_{S30} = 400 m/s) and a rock site condition (V_{S30} = 760 m/s), respectively. At large magnitudes (M>7), the distance scaling is similar at the long periods, whereas at short periods, there is weaker saturation in the new GMM, leading the larger median short-period PSA values for large at rupture distances less than 10 km. For the 4.5 $<$ M $<$ 5.5, the new GMM leads to lower median PSA values for distances greater than 5 km.

The magnitude scaling for five different distances (R_{RUP} = 1, 5, 20, 50, 100 km) is compared in Figures 8.3 and 8.4 for the same two site conditions (V_{S30} = 400 m/s and 760 m/s) for T = 0.05, 0.2, 1, 3 sec. Overall, the magnitude scaling is similar for the new GMM and the ASK14 GMM. The largest differences in the magnitude scaling are at the short periods with the new GMM having steeper magnitude scaling between M4 and M6.5.

The median response spectra for a vertical strike-slip event at rupture distances of 15 km and 80 km, are shown in Figures 8.5 and 8.6, respectively. The main differences are in the short-period range for the small and moderate magnitudes (M<6.5), with the new GMM PSA values 20-40% smaller than the ASK14 GMM, and in the long period range for the large magnitudes, with the new GMM PSA values about 20% larger than the ASK14 GMM values.

8.2 STANDARD DEVIATION

The period dependence of the components of the standard deviation is compared to the ASK14 GMM standard deviations for M7, V_{S30}=400 m/s and R_{RUP} = 20 km and 100 km in Figures 8.7

and 8.8. The total sigma for the new GMM is similar to the ASK14 sigma for short periods (T$<$ 0.15 sec) and is about 10% lower than the ASK14 sigma at long periods. The reduction of the total sigma at long periods is due to the reduction in the ϕ.

Similar comparisons are shown for M=4.5 events in Figures 8.9 and 8.9 for R_{RUP} of 20 km and 100 km respectively. At short periods (T$<$0.2 sec), the total sigma for the new GMM is up to 10% larger than for the ASK14 GMM for the model without $T_{Amp1.5}$. The increase at short periods is reduced to about 3% if the $T_{Amp1.5}$ is included in the GMM due to the better performance of the short-period GMM at large distances. At long periods (T$>$0.2 sec), the total sigma from the new GMM and the ASK14 GMM are similar.

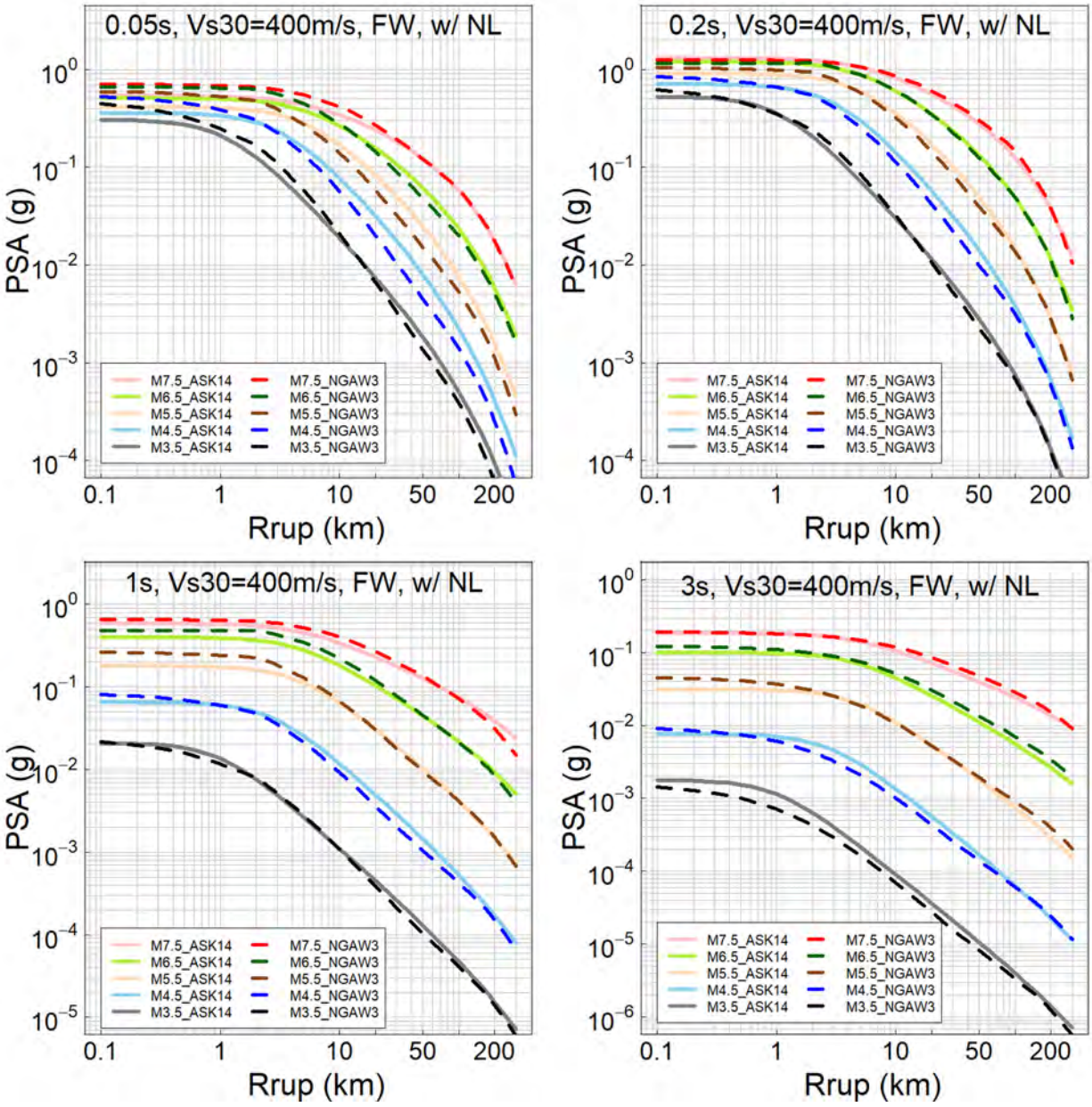


Figure 8.1: Distance scaling of the median PSA of the PSA GMM with V_{S30} = 400 m/s.

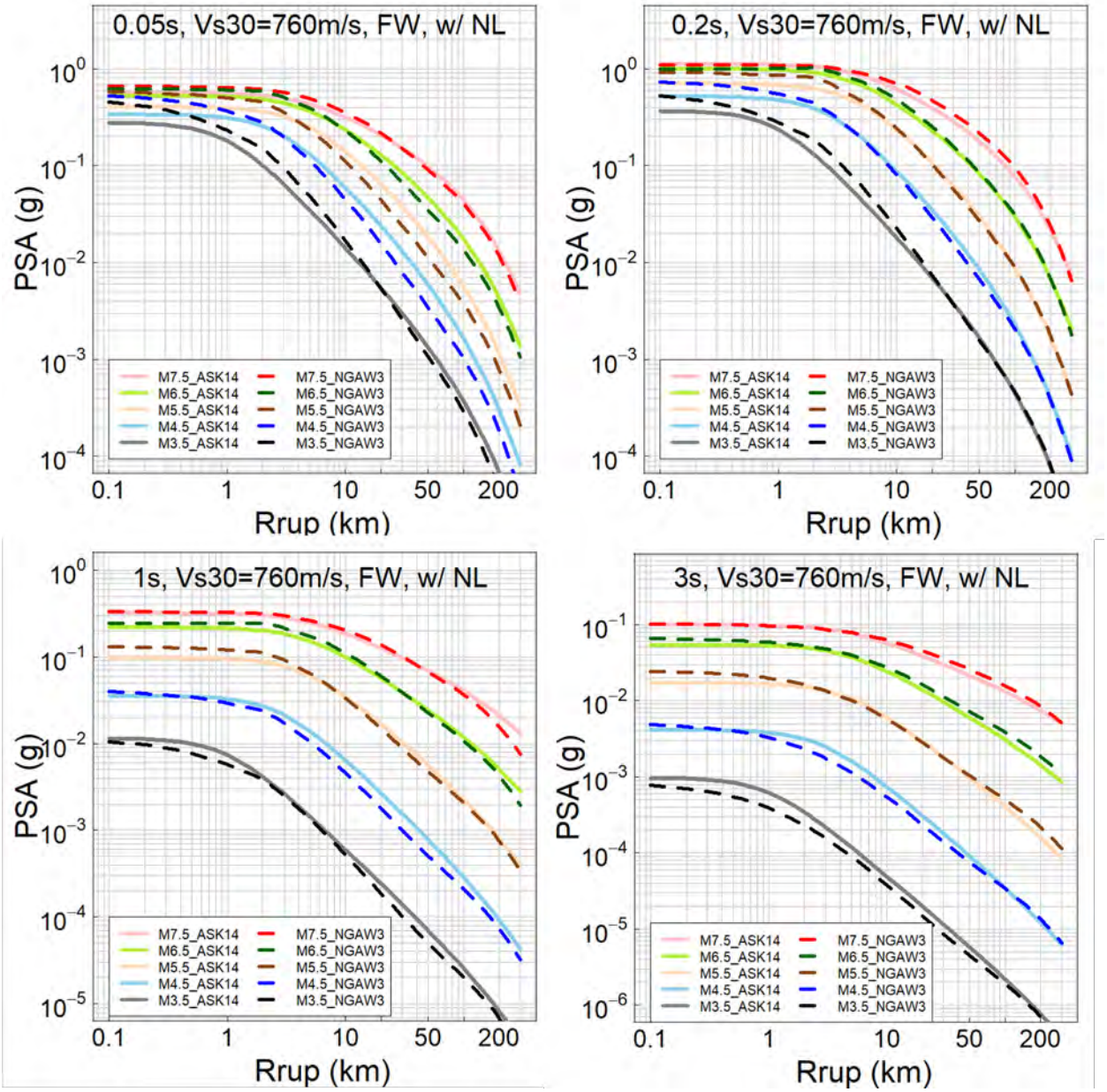


Figure 8.2: Distance scaling of the median PSA of the PSA GMM with V_{S30} = 760 m/s.

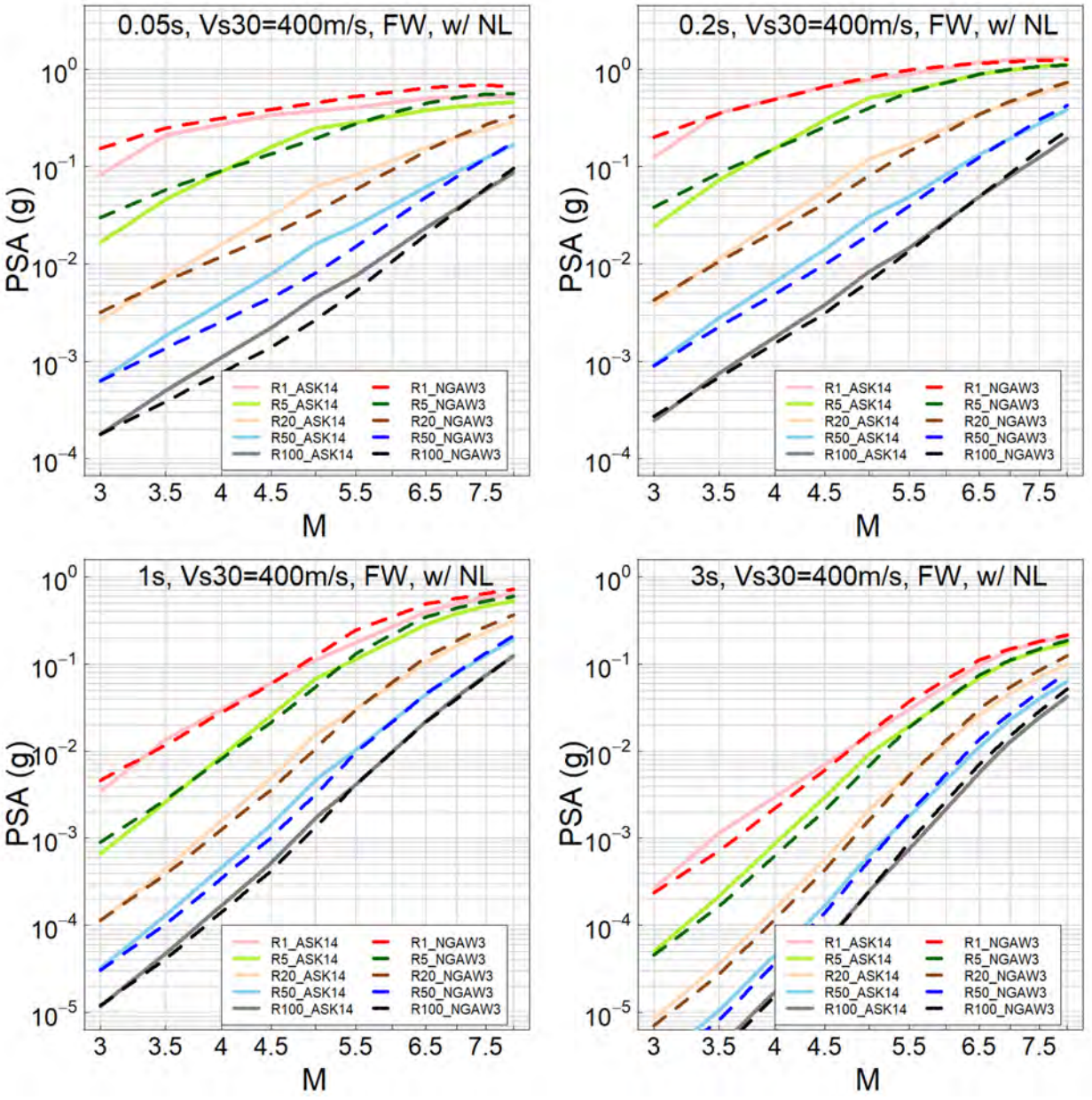


Figure 8.3: Magnitude scaling of the median PSA of the PSA GMM with V_{S30} = 400 m/s.

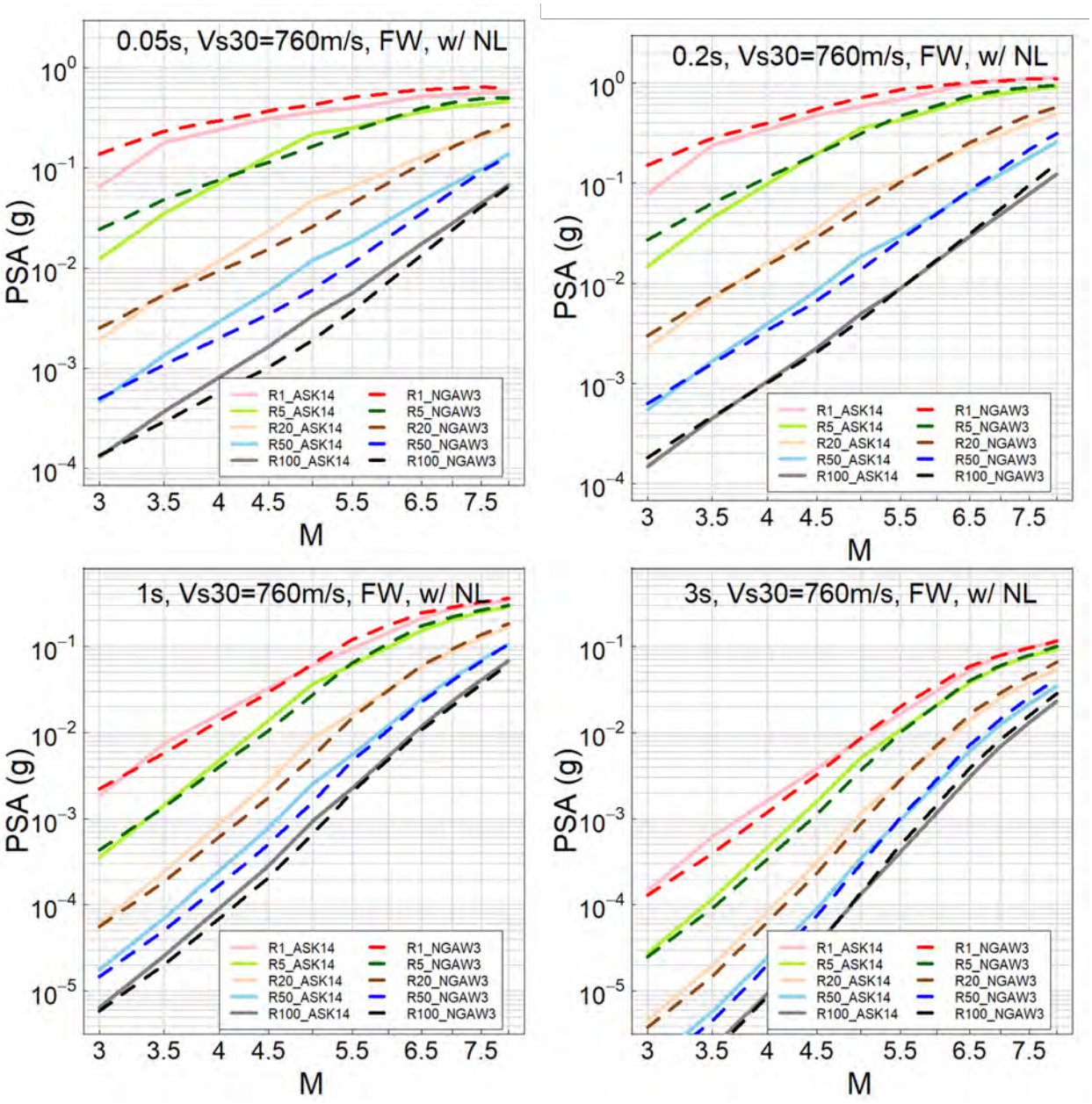


Figure 8.4: Magnitude scaling of the median PSA of the PSA GMM with V_{S30} = 760 m/s.

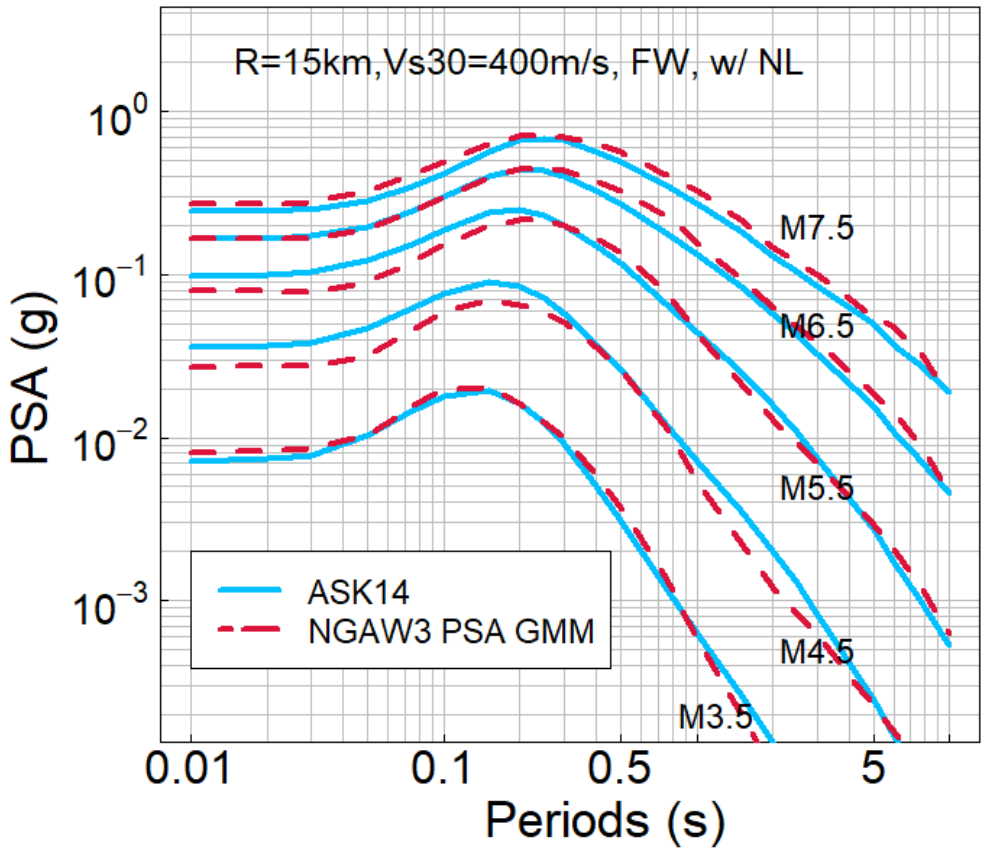


Figure 8.5: Median PSA spectrum of the NGA-W3 PSA model and ASK14 for a strike-slip event and with the non-linear site effect for R_{RUP}=15 km.

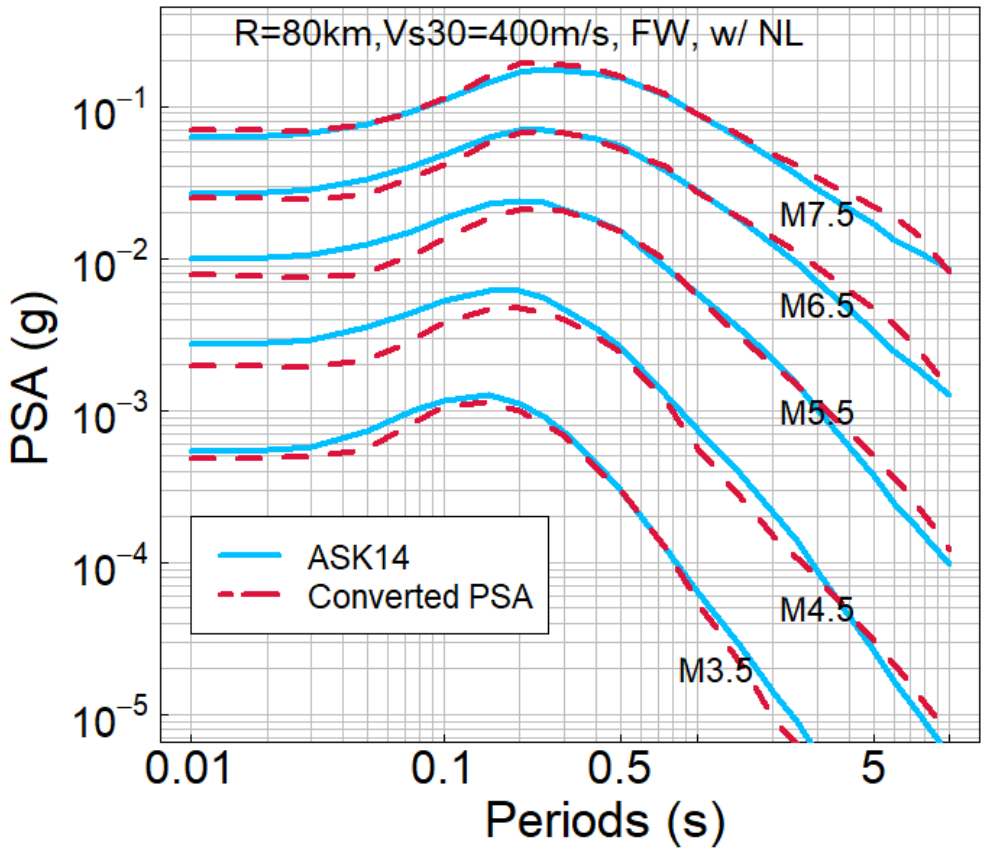


Figure 8.6: Median PSA spectrum of the NGA-W3 PSA model and ASK14 for a strike-slip event and with the non-linear site effect for R_{RUP}=80 km.

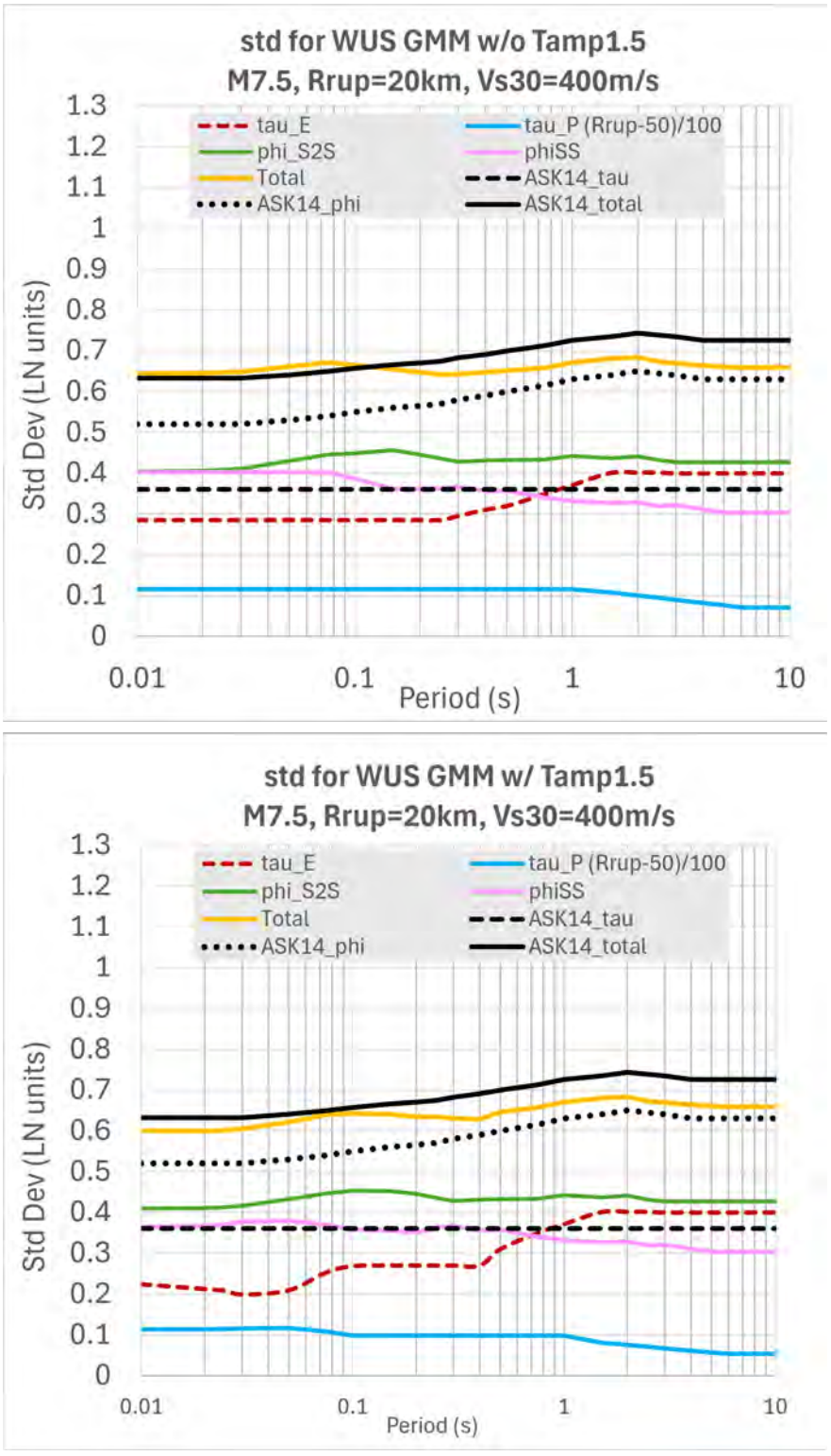


Figure 8.7: The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with $T_{Amp1.5}$) are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M7.5, V_{S30}=400 m/s and R_{RUP} = 20 km.

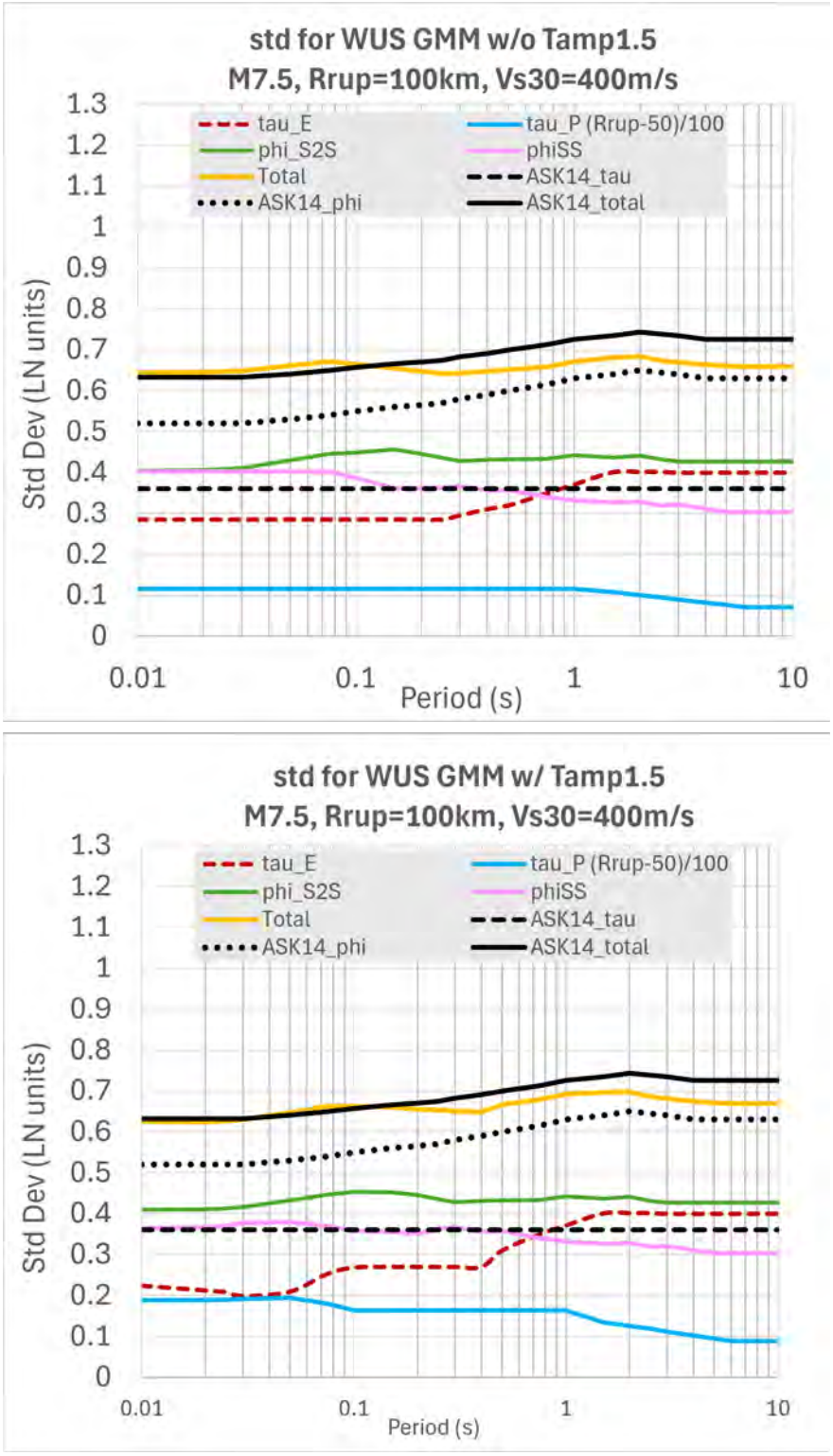


Figure 8.8: The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with Tamp1.5) are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M7.5, V_{S30}=400 m/s and R_{RUP} = 100 km.

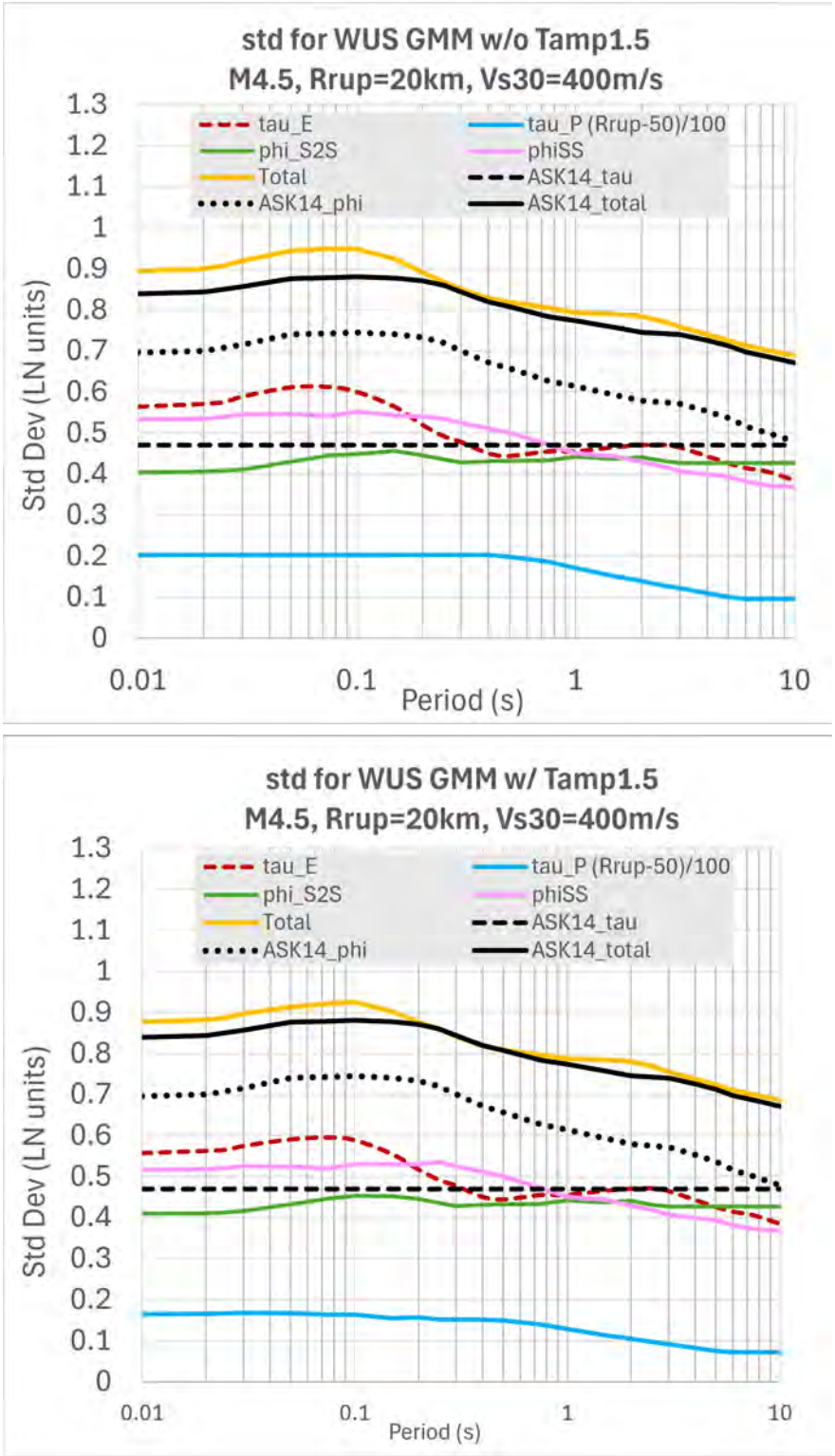


Figure 8.9: The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with $T_{Amp1.5}$) are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M4.5, V_{S30}=400 m/s and R_{RUP} = 20 km.

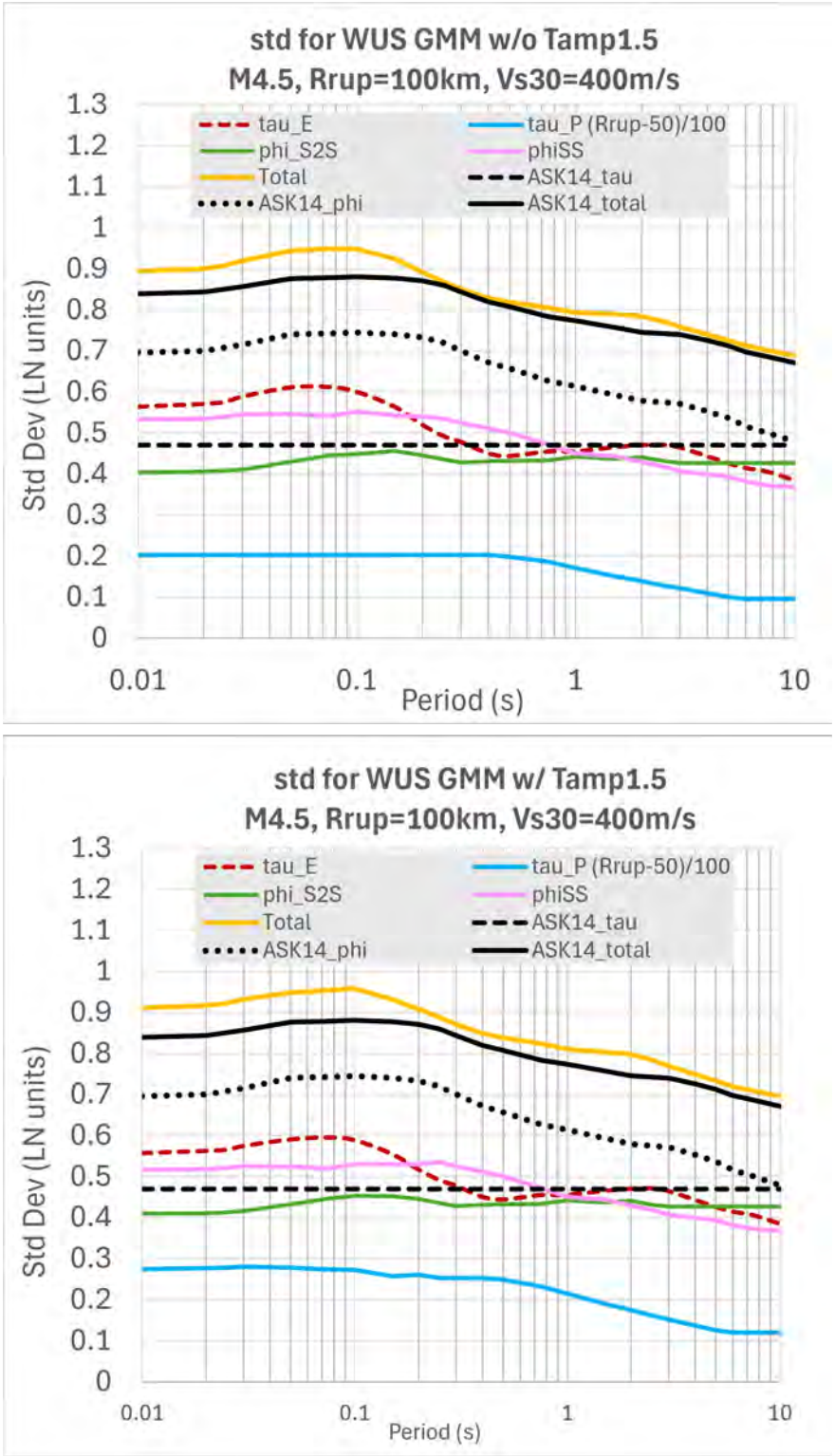


Figure 8.10: The period dependence τ, τ_P, ϕ_{S2S}, ϕ_{SS}, and σ_T for this study (upper: GMM w/o Tamp1.5, bottom: GMM with $T_{Amp1.5}$ are compared with the corresponding the τ, ϕ and σ_T of ASK14 for M4.5, V_{S30}=400 m/s and R_{RUP} = 100 km.

9 MODEL APPLICATION GUIDELINES

The PSA GMM developed in this report is intended for application to the WUS. It is a regionalized GMM, but it is ergodic within the WUS region. The GMM is applicable to the following ranges: M3-M8, $R_{RUP} < 300$ km, and 200 m/s $< V_{S30} <$ 1000 m/s.

For hard rock sites with $V_{S30} >$ 1000 m/s, the GMM uses $V_{S30} =$ 1000 m/s and is likely conservative. For hard-rock sites, we recommend using the GMM for $V_{S30} = 760m/s$ and applying the hard-rock factors from Al Atik et al. (2022).

For soil sites with $V_{S30} <$ 200 m/s, the GMM shows too strong non-linearity, which leads to an underestimation of the short-period ground motions for large input motions. We recommend conducting a site-specific site response analysis for soft sites with sites with $V_{S30} <$ 200 m/s.

9.1 MISSING PARAMETERS

The GMM includes a new parameter $\delta_{Tamp1.5}$. To use this parameter requires an estimate of the $T_{Amp1.5}$ for the site, which can be from small-magnitude events recorded at the site on on similar site conditions near the site. This site-specific $T_{Amp1.5}$ is used with the Largent et al. (2025) GMM for $T_{Amp1.5}$ to compute the $\delta_{Tamp1.5}$. If a site-specific estimate of $T_{amp1.5}$ is not available, then the $\delta_{Tamp1.5}$ should be set to zero and the standard deviation models without $\delta_{Tamp1.5}$ should be used.

9.2 USE OF THE PSA GMM

The evaluation of this version of our new GMM should be completed before the GMM is used in seismic hazard applications. The evaluation should include comparisons with GMMs from the other NGA-W3 developers once they become available as well as trial applications in probabilistic seismic hazard analyses. It may take up to a year of trial applications and comparisons before a new GMM is ready for use on projects.

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